Waveform Generators

7

7.1 INTRODUCTION

The op-amps are widely used in circuits for generating various waveforms. Most of the analog and digital equipments require one or more periodic waveforms for timing, control and other functions. The commonly used sinusoidal, square and triangular waveform generations are other forms of evolution in the design of operational amplifiers. Their applications have made the design of oscillators possible, which are capable of generating repetitive waveforms of fixed frequency and amplitude without the need of any other signal. The terms oscillator and function generator or waveform generator represent the circuits employed for generating such waveforms. This chapter discusses sinusoidal oscillators, and non-sinusoidal oscillators including multivibrators and applications of timer IC 555. The monolithic ICL8038 function generator and its applications are also dealt with.

7.2 SINE-WAVE GENERATORS

The sine-wave is one of the most fundamental waveforms. The generation of sine-wave is a challenging task if ideal waveform characteristics are expected. In the sine-wave oscillators using op-amps, the required phase-shift of 180° in the feedback loop from output to input is obtained by using either L and C or R and C components.

7.2.1 Conditions for Oscillations

An oscillator is basically a feedback amplifier, in which a fraction of the output is fed back to the input with the use of a feedback circuit. The block diagram of an oscillator is shown in Fig. 7.1.

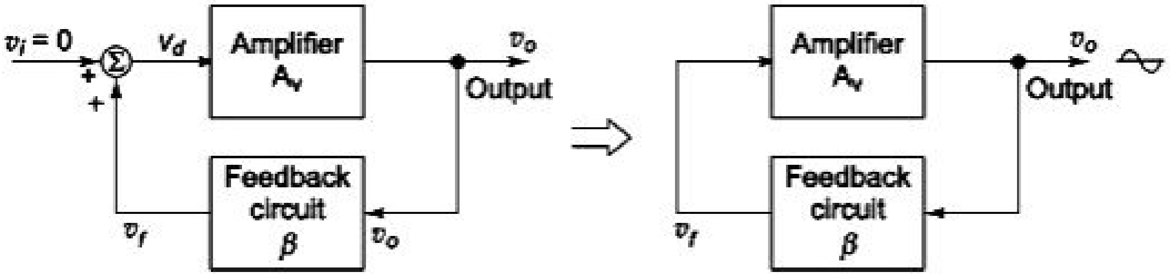


Fig. 7.1 *Block diagram of an oscillator*

As illustrated in Fig. 7.1,

$$v_d = v_f + v_i$$

$$v_o = A_v\, v_d = A_v(v_f + v_i)$$

$$v_f = \beta v_o = \beta A_v(v_f + v_i)$$

or

$$v_o = A_v(\beta v_o + v_i)$$

Therefore,

$$\frac{v_o}{v_i} = \frac{A_v}{1 - A_v\beta}$$

For an oscillator, $v_i = 0$, and for sustained oscillation $v_o \neq 0$. Therefore,

$$|A_v\,\beta| = 1 \tag{7.1}$$

and

$$\angle A_v\,\beta = 0° \text{ or } 360° \tag{7.2}$$

Therefore, the two basic requirements for sustained oscillations are:

(i) the magnitude of the loop gain, $A_v\,\beta$, must be unity and

(ii) the total phase-shift of the loop gain, $A_v\,\beta$, must be equal to 0° or 360°

Many different types of oscillators are available which are characterised by the types of components used in the feedback network, e.g. RC oscillator, LC oscillator and crystal oscillator.

7.2.2 *LC* Oscillators

A general form of *LC* oscillator is shown in Fig. 7.2(a). Here the amplifier using op-amp is considered ideal and it has a non-zero output resistance R_o. Referring to its equivalent shown in Fig. 7.2(b), we get $\beta = -\frac{V_f}{AV_f'}$.

Applying the voltage divider equation to Fig. 7.2 (b) gives

$$V_f = \frac{Z_1}{Z_1 + Z_3} V_o$$

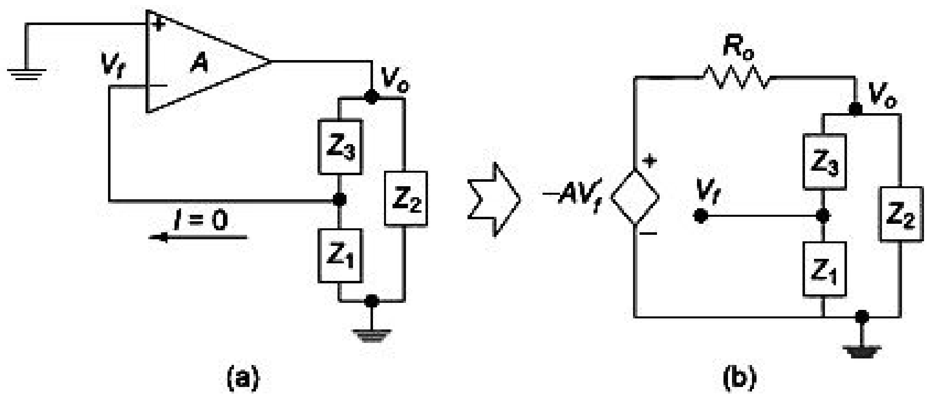


Fig. 7.2 *(a) LC Oscillator circuit using an ideal op-amp with output impedance R_o (b) Its equivalent circuit*

From Fig. 7.2(b), we get

$$V_o = -\frac{Z}{Z + R_o} AV_f'$$

where R_o is the output impedance and $Z = Z_2 \| (Z_1 + Z_3)$.

That is,
$$\frac{1}{AV_f'} = -\frac{1}{V_o}\frac{Z}{Z+R_o} = -\frac{1}{V_o}\frac{Z_2(Z_1+Z_3)}{Z_2(Z_1+Z_3)+R_o(Z_1+Z_2+Z_3)}$$

Solving for β, we get
$$\beta = -\frac{V_f}{AV_f'} = -\frac{Z_1Z_2}{R_o(Z_1+Z_2+Z_3)+Z_2(Z_1+Z_3)} \tag{7.3}$$

We know that for the LC tunable oscillators, the three impedances ($Z_1+Z_2+Z_3$) are purely reactive, i.e. the real part is equal to zero as given by
$$Z_i = jX_i, X_i > 0 \qquad \text{for } i = 1, 2, 3$$

Then, Eq. (7.3) becomes
$$\beta = \frac{X_1X_2}{jR_o(X_1+X_2+X_3)+X_2(X_1+X_3)}$$

For β to be real
$$X_1+X_2+X_3 = 0, \tag{7.4}$$

and
$$\beta(\omega_o) = -\frac{X_1}{X_1+X_3}$$

where ω_o is the oscillation frequency. Using the two previous equations, we get
$$\beta(\omega_o) = \frac{X_1}{X_2}.$$

Since $\beta(\omega_o)$ must be positive, X_1 and X_2 must have the same sign. This means that they are of the same kind of reactance, namely, two capacitors or two inductors. From the condition that the imaginary part equals zero, we find that if X_1 and X_2 are capacitors, X_3 must be an inductor and vice versa. Therefore, based on the choice of reactive elements, the LC oscillators are identified as Colpitts and Hartley oscillators.

Hartley oscillator The circuit diagram of Hartley oscillator is shown in Fig. 7.3. The op-amp is connected to operate in inverting mode. An LC network consisting of inductive reactances X_1 and X_2 and a capacitive reactance X_3 forms the feedback network and the phase-shift through the feedback network is 180°. The reactances are:
$$X_1 = j\omega L_1$$
$$X_2 = j\omega L_2$$

and
$$X_3 = -\frac{j}{\omega C}$$

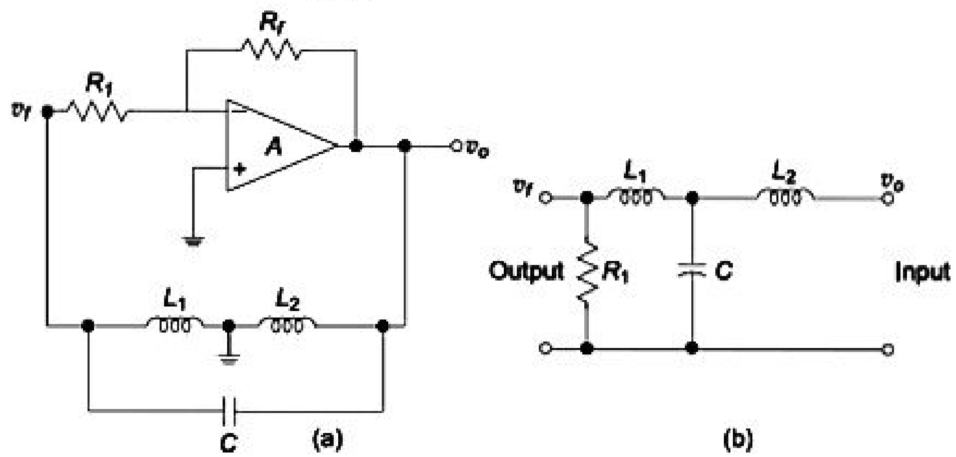


Fig. 7.3 *Hartley oscillator (a) Circuit diagram and (b) Feedback network*

Using Eq. (7.4), we get the angular frequency of oscillation and the gain as

$$\omega_o = \sqrt{\frac{1}{C(L_1 + L_2)}} \quad \text{and} \quad \beta(\omega_o) = \frac{L_1}{L_2}$$

The frequency of oscillation is $f_o = \dfrac{1}{2\pi\sqrt{CL_T}}$ (7.5)

where $L_T = L_1 + L_2$ or $L_T = L_1 + L_2 + 2M$

and M is the mutual inductance of coils L_1 and L_2.

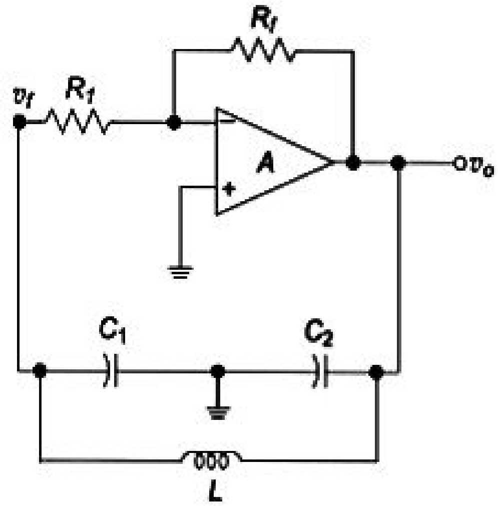


Fig. 7.4 *Colpitts oscillator*

Colpitts oscillator The circuit diagram of Colpitts oscillator is shown in Fig. 7.4. The feedback signal is connected to (–) input terminal, such that the op-amp is working as an inverting amplifier. The feedback circuit consisting of two capacitive reactances X_1 and X_2, and the inductive element X_3 provides 180° phase-shift. The oscillation occurs when the feedback β is real. The reactances are:

$$X_1 = \frac{1}{j\omega C_1} = -\frac{j}{\omega C_1}$$

$$X_2 = \frac{1}{j\omega C_2} = -\frac{j}{\omega C_2}$$

and

$$X_3 = j\omega L$$

Using Eq. (7.4), we get the angular frequency of oscillation ω_o, and the gain $\beta(\omega_o)$ as

$$\omega_o = \sqrt{\frac{1}{L\left(\dfrac{C_1C_2}{C_1 + C_2}\right)}}$$

and

$$\beta(\omega_o) = \frac{C_2}{C_1}$$

Thus, the frequency of oscillation for the Colpitts oscillator is

$$f_o = \frac{1}{2\pi\sqrt{LC_T}} \tag{7.6}$$

where

$$C_T = \frac{C_1C_2}{C_1 + C_2}.$$

The feedback factor β at the oscillation frequency is -1.

7.2.3 *RC* Oscillators

All the oscillators using tuned *LC* circuits operate well at high frequencies. At low frequencies, as the inductors and capacitors required for the timing circuit would be very bulky, *RC* oscillators are found to be more suitable. Two important *RC* oscillators are (i) *RC* phase shift oscillator and (ii) Wien Bridge oscillator.

RC phase-shift oscillator The *RC* phase-shift oscillator consisting of an op-amp serving as the amplifier stage and the cascaded RC network acting as the feedback circuit is shown in Fig. 7.5. The op-amp is used in inverting configuration and it produces 180° phase-shift at the output. The cascaded RC networks connected in the feedback network path provide an additional phase-shift of 180°. Therefore, the total phase-shift around the loop is 360° (or 0°). When the gain of the amplifier is sufficiently large and the phase-shift of the cascaded RC network is exactly 180°, the circuit will oscillate at the frequency determined by the RC feedback network.

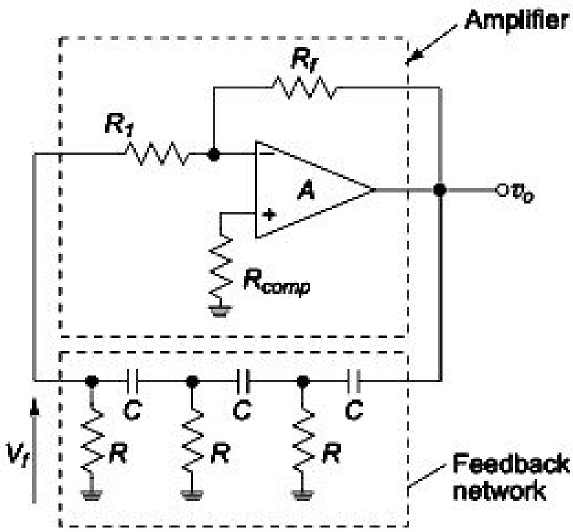


Fig. 7.5 *RC phase-shift oscillator*

Frequency of oscillation Consider the *RC* feedback circuit shown in Fig. 7.6. Applying Kirchhoff's current law at node $V_1(s)$, we have

$$I_1(s) = I_2(s) + I_3(s)$$

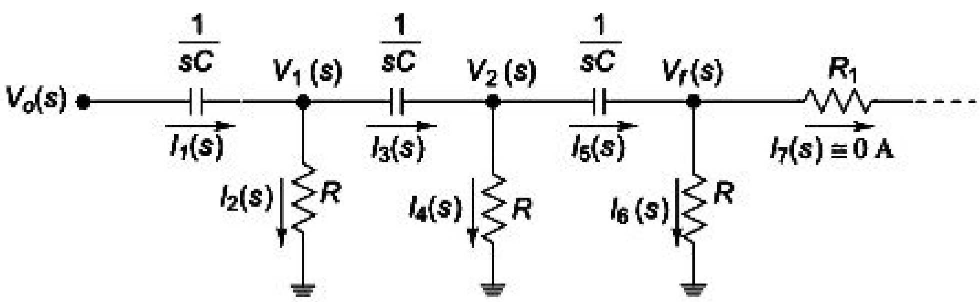


Fig. 7.6 *RC network of the phase-shift oscillator in s-domain*

That is,
$$\frac{V_o(s)-V_1(s)}{(1/sC)} = \frac{V_1(s)}{R} + \frac{V_1(s)-V_2(s)}{(1/sC)}$$

Thus,
$$V_1(s) = \frac{[V_o(s)+V_2(s)]\,RCs}{2RCs+1} \tag{7.7}$$

Similarly, at node $V_2(s)$,

$$I_3(s) = I_4(s) + I_5(s)$$

That is,
$$\frac{V_1(s)-V_2(s)}{(1/sC)} = \frac{V_2(s)}{R} + \frac{V_2(s)-V_f(s)}{(1/sC)}$$

Solving for $V_1(s)$, we get

$$V_1(s) = \frac{(2RCs+1)V_2(s)}{RCs} - V_f(s) \tag{7.8}$$

When $R_1 >> R$, $I_5 \approx I_6$ and $V_f(s) = V_2(s)\dfrac{R}{\left(R+\dfrac{1}{sC}\right)}$

Therefore,
$$V_2(s) = \frac{(RCs+1)}{RCs} V_f(s) \tag{7.9}$$

Substituting Eq. (7.9) in Eq. (7.8) results in

$$\frac{RCsV_o(s)}{2RCs+1} + \frac{(RCs+1)V_f(s)}{2RCs+1} = \frac{(2RCs+1)(RCs+1)V_f(s)}{(RCs)^2} - V_f(s)$$

Simplifying for $\frac{V_f(s)}{V_o(s)}$, we get

$$\beta = \frac{V_f(s)}{V_o(s)} = \frac{R^3C^3s^3}{R^3C^3s^3 + 6R^2C^2s^2 + 5RCs + 1} \tag{7.10}$$

The voltage gain of the op-amp A_v is

$$A_v = \frac{V_o(s)}{V_f(s)} = -\frac{R_f}{R_1} \tag{7.11}$$

We know that for any oscillator, the necessary condition for the oscillation is $A_v\,\beta = 1$. Therefore, using Eqs. (7.10) and (7.11), we obtain

$$A_v\,\beta = -\frac{R_f}{R_1}\frac{R^3C^3s^3}{R^3C^3s^3 + 6R^2C^2s^2 + 5RCs + 1} = 1$$

Substituting $s = j\omega$ in the above equation, we obtain

$$\left(-\frac{R_f}{R_1}\right)(-j\omega^3 R^3 C^3) = (-j\omega^3 R^3 C^3) - 6\omega^2 R^2 C^2 + j5\,\omega RC + 1$$

Equating the real parts, we get

$$0 = -6\omega_o^2 R^2 C^2 + 1$$

$$\omega_o^2 = \frac{1}{6R^2C^2} \tag{7.12}$$

or
$$f_o = \frac{1}{2\pi\sqrt{6}RC}$$

Equating the imaginary parts, we get

$$\left(-\frac{R_f}{R_1}\right)(-j\omega_o^3 R^3 C^3) = -j\omega_o^3 R^3 C^3 + 5j\omega_o RC$$

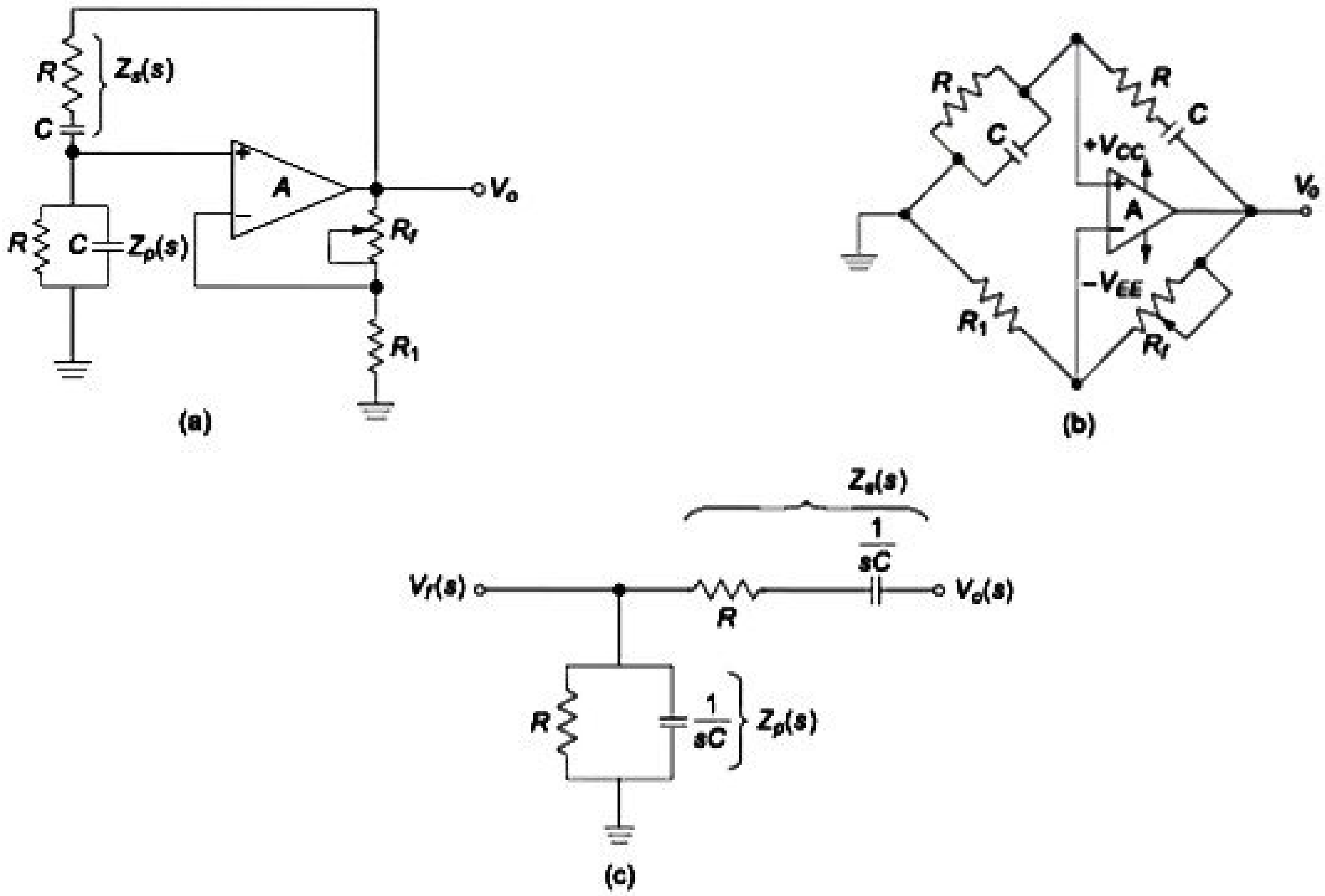
That is,
$$\frac{-R_f}{R_1} = 1 - \frac{5}{\omega_o^2 R^2 C^2}$$

Substituting for ω_o^2 from Eq. (7.12),

$$\left|\frac{R_f}{R_1}\right| = |1 - 30| = 29 \text{ or } R_f = 29\,R_1 \tag{7.13}$$

Therefore, the gain of the amplifier should be at least 29, and the total phase-shift around the loop should be exactly 360°.

Wien Bridge oscillator Wien Bridge oscillator is the most commonly used audio frequency oscillator due to its inherent simplicity and stability. Figure 7.7(a) shows the Wien Bridge oscillator using op-amp. Since the op-amp is connected to operate in non-inverting mode, it produces no phase-shift at the output. The Wien Bridge circuit is connected between the input and output terminals of the amplifier. The bridge consists of a series RC network (shown as $Z_s(s)$) forming one arm of the bridge circuit, a parallel RC network (shown as $Z_p(s)$) forming the second arm, input resistance R_1 and feedback resistance R_f forming the third and fourth arms of the bridge circuit respectively as shown in Fig. 7.7(b). The feedback circuit of the Wien Bridge oscillator is shown in Fig. 7.7(c).



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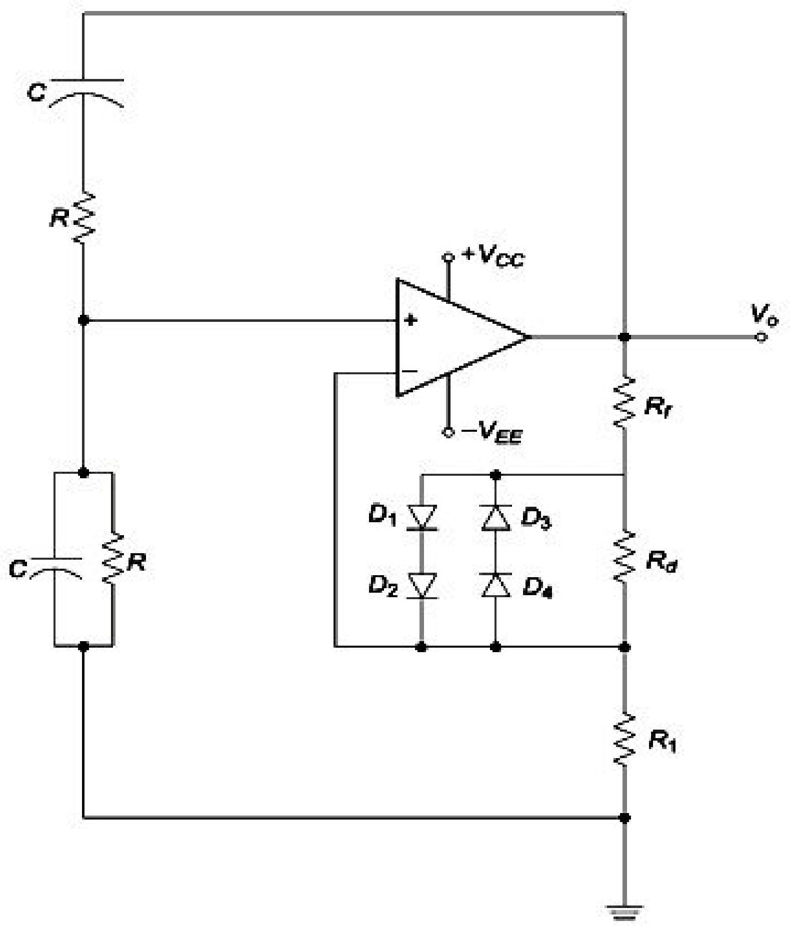


(d)

Fig. 7.7 *Wien Bridge oscillator: (a) Circuit diagram (b) Equivalent circuit (c) Feedback circuit of the Wien Bridge oscillator in s-domain and (d) Output amplitude stabilisation*

It is known that the total phase-shift around the circuit must be 0° or 360° for oscillations to occur. It is achieved when the bridge is balanced, i.e. at resonance. Thus the frequency of oscillation is the resonant frequency of the balanced Wien Bridge.

Frequency of oscillation Considering the circuit shown in Fig. 7.7(c) in the s-domain,

$$V_f(s) = \frac{Z_P(s)}{Z_p(s) + Z_s(s)} V_o(s)$$

where $Z_P(s) = R \,||\, \frac{1}{sC} = \frac{R}{(RCs+1)}$ and $Z_s(s) = R + \frac{1}{sC} = \frac{(RCs+1)}{sC}$

Therefore, $$\beta = \frac{V_f(s)}{V_o(s)} = \frac{Z_p(s)}{Z_p(s) + Z_s(s)} = \frac{RCs}{(RCs+1)^2 + RCs}$$

$$= \frac{RCs}{R^2C^2s^2 + 3RCs + 1} \tag{7.14}$$

The voltage gain A_v of the op-amp is $A_v = \frac{V_o(s)}{V_f(s)} = 1 + \frac{R_f}{R_1}$ (7.15)

Using Eqs. (7.14) and (7.15) for satisfying the condition of oscillation $|A_v \beta| = 1$, we get

$$\left(1+\frac{R_f}{R_1}\right)\frac{RCs}{R^2C^2s^2+3RCs+1}=1$$

Substituting $s = j\omega_o$ in the above equation, we get

$$\left(1+\frac{R_f}{R_1}\right) j\omega_o RC = (-\omega_o^2 R^2C^2 + 3j\omega_o RC + 1)$$

Equating the real parts, we have

$$\omega_o^2 R^2C^2 = 1$$

Therefore,

$$f_o = \frac{1}{2\pi RC} \tag{7.16}$$

Equating the imaginary parts, we get

$$\left(1+\frac{R_f}{R_1}\right) j\omega_o RC = 3j\omega_o RC$$

or

$$A_v = 1 + \frac{R_f}{R_1} = 3 \text{ and } R_f = 2R_1 \tag{7.17}$$

The Wien Bridge RC oscillators can be used in the range of 5 Hz to about 1 MHz, e.g. in low frequency applications like audio generators. It is not suitable for high frequency applications above 1MHz, due to the fact that the phase-shift through the amplifier and the phase-shift of the *lead-lag* circuit jointly cause resonance to occur at different frequencies other than the specified resonant frequency.

To maintain a stable amplitude for the oscillations, the resistor of Fig. 7.7(a) is normally replaced by a non-linear resistor. This adjusts the loop gain based on the existing amplitude of oscillations. When the amplitude increases, it causes a rise in the current through R_1 which in turn rises the value of R_1. It actually means an increased amount of negative feedback results in a corresponding reduction in loop gain and oscillation amplitude.

It is observed that only the response of the non-linear resistance R_1 determines the stability of oscillations of the circuit shown in Fig. 7.7(d). The stabilisation of output amplitude is realised by controlling the amplifier gain through the diodes D_1 to D_4.

The series combinations of $D_1 - D_2$ and $D_3 - D_4$ are connected in opposite directions across resistor R_d, introduced between the two resistive arms formed by R_f and R_1 of Fig. 7.7(a). When the output oscillation amplitude rises large enough to forward bias either the $D_1 - D_2$ or $D_3 - D_4$ combination, resistor R_d is short circuited and the amplifier gain gets controlled. In turn, this will lower the output amplitude. When the output level falls, the diodes would turn-off, thereby increasing the effective R_f and the resulting gain. Hence, sustained oscillations can be realized.

7.2.4 Quadrature Oscillator

Quadrature signals are those two signals of same frequency but separated by a phase shift of 90° from each other. The quadrature oscillator circuit shown in Fig. 7.8 generates two sinusoidal signals that are in *quadrature* with each other or with 90° phase difference. The quadrature oscillator circuit requires dual op-amp formed by A_1 and A_2 and three RC combinations. The two amplifiers are connected in cascade to form a feedback loop. The op-amp A_1 operates as a non-inverting integrator. The op-amp A_2 acts as an inverting integrator. The output of A_2 is connected to voltage divider network consisting of $R_1 - C_1$ combination and the voltage across C_1 is fedback to the non-inverting input of A_1.

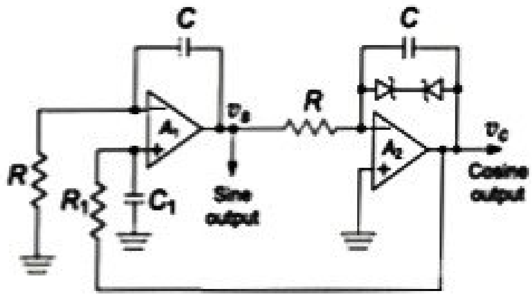


Fig. 7.8 *Quadrature oscillator*

A total phase shift of 360° is needed around the loop. The op-amp A_2 acts as an inverting integrator, contributing for 270° of phase shift i.e. 180° due to inverter arrangement and 90° due to integrator. The remaining 90° phase shift is realised in the voltage divider R_1C_1 and op-amp A_1.

The feedback loop is represented by the set of differential equations $RC\dfrac{dv_s}{dt} = v_c$ and $RC\dfrac{dv_c}{dt} = -v_s$. Then, assuming $R_1C_1 = RC$, the frequency of sinusoidal oscillations is given by $f_o = \dfrac{1}{2\pi RC}$. In practice, the resistance R_1 is chosen slightly larger than R. This ensures sufficient feedback for oscillation. The amplitude of oscillations is stabilised by the use of Zener diodes connected back to back.

7.3 MULTIVIBRATORS

Multivibrators are regenerative circuits, which are mainly used in timing applications. Based on their operational characteristics, they can be classified into three categories, namely,

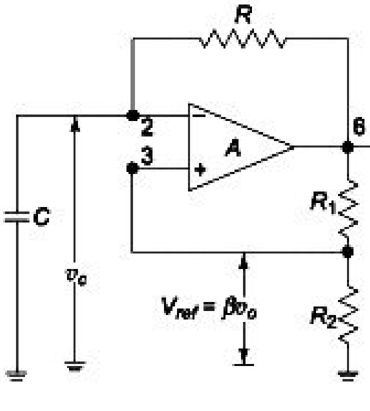
(i) Astable multivibrator
(ii) Monostable multivibrator
(iii) Bistable multivibrator

The astable multivibrator toggles between one state and the other without the influence of any other external control signal. It is also called a *free-running multivibrator*. The monostable multivibrator or *one-shot* requires an external signal called a *trigger* to force the circuit into a quasi-stable state for a particular time duration or delay. A suitable timing network determines the time delay and it returns to the stable state at the end of the delay time.

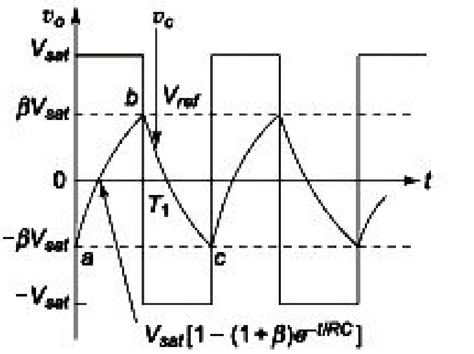
7.3.1 Astable (Free-running) Multivibrator

An astable multivibrator is a square-wave generator. Figure 7.9(a) shows the circuit of an astable multivibrator with the output of op-amp fedback to the (+) input terminal. The resistors R_1 and R_2 form a voltage divider network, and a fraction $\beta = \dfrac{R_2}{R_1 + R_2}$ of the output is fed back to the input. The output can take values of $+\beta V_{sat}$ or $-\beta V_{sat}$. The voltage $\pm\beta V_{sat}$ acts as V_{ref} at the (+) input terminal. The output is also connected to the (–) input terminal through an integrating low-pass RC network. When the voltage v_c across capacitor C just exceeds V_{ref}, switching takes place resulting in a square-wave output.

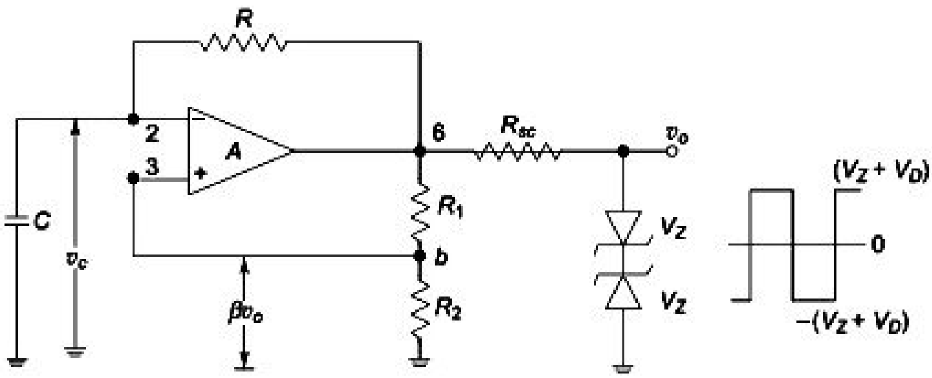
To understand the operation of the circuit, let us consider that initially the output is at $+V_{sat}$ as shown in Fig. 7.9(b).



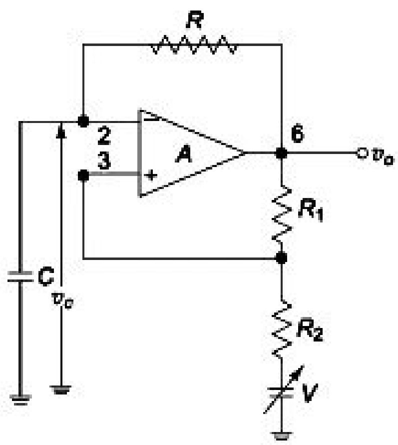
(a) Circuit diagram



(b) Waveforms at output and capacitor terminals



(c) Use of Zener diodes for different peak-to-peak voltages



(d) Asymmetric square wave generator

Fig. 7.9 *Square-wave generator using op-amp: (a) Circuit diagram (b) Waveforms at output and capacitor terminals (c) Use of Zener diodes for different peak-to-peak voltages (d) Asymmetric square-wave generator*

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The capacitor C with its voltage shown as v_c starts charging through resistor R towards $+V_{sat}$. The voltage at (+) input terminal is held at $+\beta V_{sat}$ as indicated by the use of $R_1 - R_2$ potential divider network. The charging of C continues until the voltage v_c at the (–) input terminal is just greater than the voltage at the (+) input terminal, $+\beta V_{sat}$. When this happens as shown at point b of Fig. 7.9(b), the output is switched down to $-V_{sat}$. The voltage $+\beta v_o$ across the capacitor now starts discharging through resistance R and charging towards $-V_{sat}$. The capacitor voltage v_c now becomes increasingly more and more negative and at point c just exceeds $-\beta V_{sat}$. The output now switches back to $+V_{sat}$, and the cycle repeats.

Summarising,

(i) when $v_o = +V_{sat}$, C charges from $-\beta V_{sat}$ to $+\beta V_{sat}$ and switches v_o to $-V_{sat}$ and

(ii) when $v_o = -V_{sat}$, C charges from $+\beta V_{sat}$ to $-\beta V_{sat}$ and switches v_o to $+V_{sat}$.

Period and frequency of oscillation The frequency of the free running multivibrator is determined by the charging and discharging time of the capacitor between the voltage levels $-\beta V_{sat}$ and $+\beta V_{sat}$ and vice versa. The voltage across the capacitor as a function of time can be represented as

$$v_c(t) = V_{fin} + \left(V_{ini} - V_{fin}\right)e^{\frac{-t}{RC}}$$

where V_{fin} is the final value of the voltage and V_{ini} is the initial voltage. Considering the charging of the capacitor from point a towards $+V_{sat}$,

$$v_c(t) = +V_{sat} + (-\beta V_{sat} - V_{sat})e^{\frac{-t}{RC}}$$

$$= V_{sat} - V_{sat}(1+\beta)e^{\frac{-t}{RC}} \tag{7.18}$$

At $t = T_1$, the voltage across the capacitor reaches $+\beta V_{sat}$ and switches at point b. Therefore, capacitor voltage v_c at time T_1 is

$$\beta V_{sat} = V_{sat}\left(1 - (1+\beta)e^{\frac{-T_1}{RC}}\right)$$

That is,
$$(1-\beta) = (1+\beta)e^{\frac{T_1}{RC}}$$

and
$$T_1 = RC\ln\frac{1+\beta}{1-\beta} = RC\ln\frac{R_1+2R_2}{R_1}$$

As shown in Fig. 7.9(b), the total time period is given by

$$T = 2T_1 = 2RC\ln\frac{1+\beta}{1-\beta} \tag{7.19}$$

That is,
$$T = 2\,RC\ln\left(\frac{R_1+2R_2}{R_1}\right) \tag{7.20}$$

and the output is a symmetrical waveform.

Hence, the frequency of oscillation is
$$f_o = \frac{1}{T} = \frac{1}{2\,RC\ln\left(\frac{1+\beta}{1-\beta}\right)} \tag{7.21}$$

Considering $R_1 = R_2$, we have $\beta = \dfrac{R}{2R} = 0.5$, $T = 2RC \ln 3$ and

$$f_o = \frac{1}{2RC \ln 3} = \frac{1}{2.2RC} \tag{7.22}$$

Equation (7.19) shows that the period is directly proportional to the time constant, RC. Thus, varying either R or C changes the period correspondingly. Therefore, providing a tunable resistance R paves the way for a continuously tunable square-wave generator. The output peak amplitudes can be varied by the use of Zener diodes connected back to back as shown in Fig. 7.9(c).

The output voltage is then regulated to $\pm(V_z + V_D)$ where V_z is the Zener voltage. Then the peak-to-peak output voltage is given by $v_{o\,(\text{peak-to-peak})} = 2(V_z + V_D)$. To generate an asymmetric square-wave, a variable voltage source V can be introduced as shown in Fig. 7.9(d).

For designing a square wave oscillator, the op-amp based astable multivibrator, shown in Fig. 7.9(a) may be used with the assumption of $R_1 = R_2 = 10\ \text{k}\Omega$. Referring to Eq. (7.22), we have

$$f_o = \frac{1}{2.2RC}$$

Assuming $\quad C = 0.1\mu F$, we get $R = \dfrac{1}{2.2Cf_o} = \dfrac{1}{2.2 \times 0.1 \times 10^{-6} \times 1 \times 10^3} = 4.545\ \text{k}\Omega$

7.3.2 Monostable Multivibrator

The circuit diagram of a monostable multivibrator, also called a *one-shot* multivibrator is shown in Fig. 7.10(a). This has a stable state and a quasi-stable state. Single output pulse of adjustable time duration in response to a triggering signal can be generated using the monostable multivibrator. The time duration for the output pulse is achieved by connecting the required external components to the op-amp.

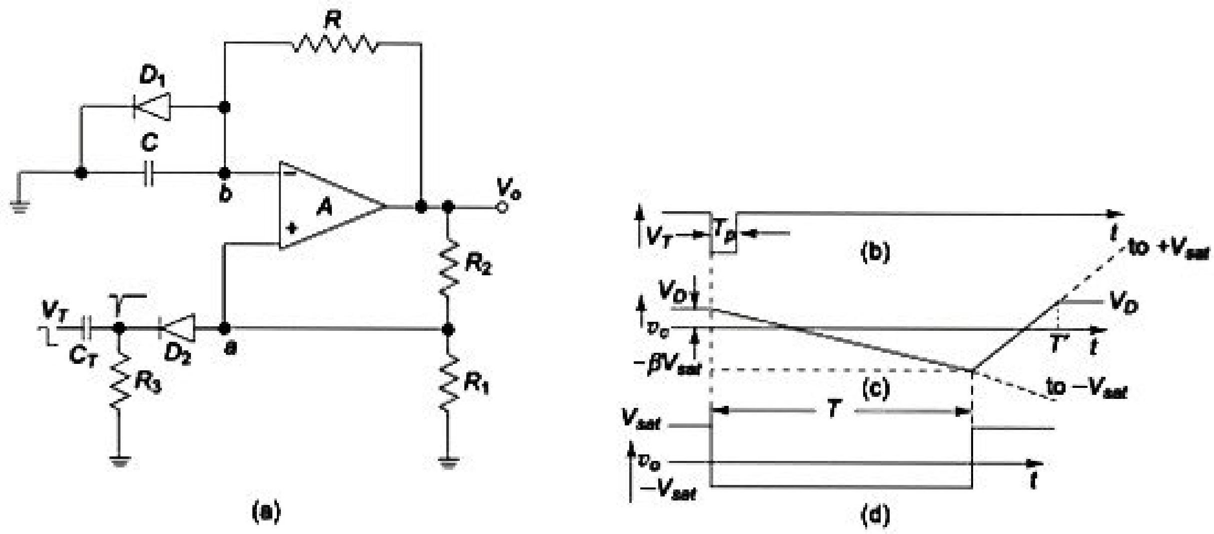


Fig. 7.10 *Monostable multivibrator: (a) Circuit diagram (b) Negative polarity triggering signal, (c) Voltage across the capacitor (d) Output voltage waveform*

When the output V_o is at positive saturation, the diode D_1 clamps the capacitor (C) voltage to V_D (0.7 V). This is the stable state of the circuit. In this state, the inverting terminal b is clamped to ground by diode D_1. This results in preventing the inverting input terminal going more positive than V_D (0.7 V). This makes terminal a also positive by the same voltage V_D. A negative-going narrow trigger pulse is passed through differentiator R_3C_T and diode D_2, and connected to the non-inverting input terminal of op-amp. R_3 is made much larger than R_1, so that its loading effect may be minimised. The diode D_2 prevents positive spikes arriving from the triggering circuit.

Let us assume that in the stable state, the output V_o of op-amp is at $+V_{sat}$, the voltage at inverting terminal is V_D and the voltage at (+) terminal through the potential divider $R_1 - R_2$ is $+\beta V_{sat}$, where $\beta = \dfrac{R_1}{R_1 + R_2}$. When a negative trigger signal V_T is applied to the (+) terminal through the trigger line, the effective signal is less than 0.7 V. That is, voltage at (+) terminal is $[\beta V_{sat} + (-V_T)] < 0.7$ V. Then, the op-amp output switches from $+V_{sat}$ to $-V_{sat}$. The diode D_1 is now reverse-biased and the capacitor C starts charging exponentially to $-V_{sat}$ through the resistance R in the closed loop. In this condition, the voltage at positive input terminal is $-\beta V_{sat}$.

While charging exponentially, just as the capacitor voltage v_c becomes slightly more than $-\beta V_{sat}$, the voltage at (–) terminal becomes more negative than that at (+) terminal. Then the op-amp output switches back to $+V_{sat}$. The capacitor C now starts charging towards $+V_{sat}$ through the resistance R. This continues only until the voltage at (–) terminal becomes V_D(0.7 V) and then it clamps the capacitor C to V_D. The waveforms of the negative polarity triggering signal, voltage across the capacitor and output of op-amp are shown in Fig. 7.10 (b), (c) and (d) respectively.

To determine the pulse width T For a low-pass RC circuit, the general solution is

$$v_o = V_{fin} + (V_{ini} - V_{fin})e^{-t/RC}$$

where V_{ini} is the initial voltage value and V_{fin} is the final voltage value. For the circuit explained above, $V_{fin} = -V_{sat}$ and $V_{ini} = V_D$ (diode forward voltage)

The output v_c is then given by,

$$v_c = -V_{sat} + (V_D + V_{sat})e^{-t/RC} \tag{7.23}$$

At the end of time $t = T$ as shown in Fig. 7.9(c), $v_c = -\beta V_{sat}$. Thus,

$$-\beta V_{sat} = -V_{sat} + (V_D + V_{sat})e^{-T/RC}$$

Simplifying for the pulse width, we get

$$T = \mathrm{RC}\ \ln\left(\frac{1 + V_D/V_{sat}}{1 - \beta}\right)$$

where
$$\beta = \frac{R_2}{R_1 + R_2}. \tag{7.24}$$

When $V_{sat} >> V_D$(0.7 V) and $R_1 = R_2$ with $\beta = 0.5$,

$$T = 0.693\ RC \tag{7.25}$$

Understandably, the trigger pulse width should be less than the pulse width T.

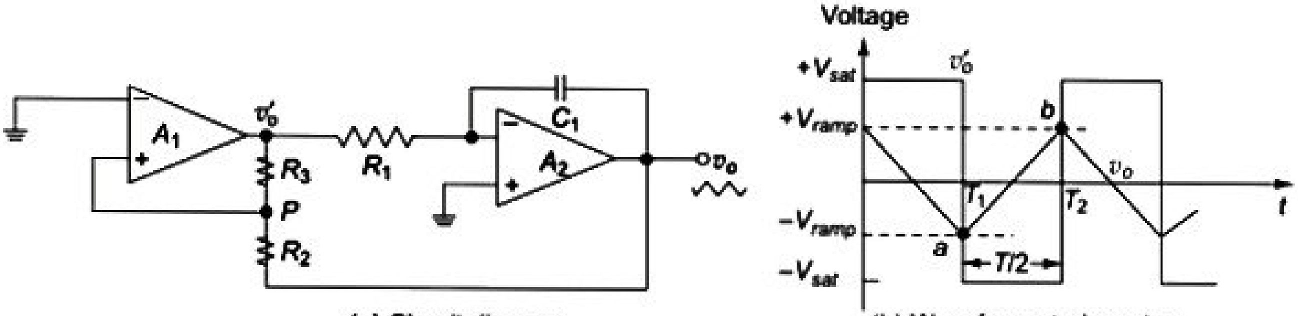
The monostable multivibrator circuit can generate a fast transition after a calculated time T equal to the pulse width in response to the application of the input trigger pulse. Therefore, it can be used as a time-delay circuit. The rectangular waveform can be used as a gating signal in counters and analog-to-digital converters.

7.4 TRIANGULAR WAVE GENERATOR

Figure 7.11(a) shows the circuit of a triangular wave generator. It consists of two op-amps and several passive components. The op-amp A_1 forms a non-inverting comparator with hysteresis, which is a Schmitt Trigger. The op-amp A_2 forms an integrator which integrates the output obtained from the Schmitt trigger. The op-amp A_1 is a two level comparator whose outputs are determined by $\pm V_{sat}$. The square-wave output from A_1 is applied to the (–) input terminal of the op-amp A_2. The output of A_2 is a triangular wave and it is fed back as an input to the comparator A_1 through a voltage divider network formed by R_2 and R_3.

Loop analysis Let us consider that the output v'_o of comparator A_1 is $+V_{sat}$ initially. The integrator integrates $+V_{sat}$ and produces a negative going ramp at its output as shown in Fig. 7.11(b). Hence, the voltages at the two ends of the voltage divider formed by $R_2 - R_3$ are $+V_{sat}$ at the output of A_1 and $-V_{ramp}$ at the output of A_2. At $t = T_1$, when the negative going ramp reaches a value of $-V_{ramp}$, represented as point a in Fig. 7.11(b), the effective value at the point P becomes slightly less than 0 V. This switches the op-amp A_1 to its negative saturation level $-V_{sat}$.

With the output of A_1 at $-V_{sat}$, the op-amp A_2 starts integrating and increases its output in the positive direction. At $t = T_2$, shown as point b in Fig. 7.11(b), the voltage at P becomes just more than 0 V. This switches the output of op-amp A_1 from $-V_{sat}$ to $+V_{sat}$. This cycle repeats itself, and generates a triangular waveform. The frequency of the waveform is determined by the RC value of the integrator formed by op-amp A_2 and the saturation voltage levels $\pm V_{sat}$ of comparator op-amp A_1.



(a) Circuit diagram (b) Waveform at v'_o and v_o

Fig. 7.11 *Triangular waveform generator: (a) Circuit diagram (b) Waveforms at v'_o and v_o*

To determine the amplitude and frequency of the triangular waveform When the comparator output is at $+V_{sat}$, the effective voltage at the point P is

$$-V_{ramp} + \frac{R_2}{R_2 + R_3}[+V_{sat} - (-V_{ramp})] = 0$$

Simplifying, we get

$$-V_{ramp} = \frac{-R_2}{R_3}(+V_{sat})$$

Similarly, at $t = T_2$, when the output of A_1 switches from $-V_{sat}$ to $+V_{sat}$,

$$V_{ramp} = \frac{-R_2}{R_3}(-V_{sat}) = \frac{R_2}{R_3}V_{sat} \tag{7.26}$$

Thus the peak-to-peak amplitude of the triangular wave is

$$v_{o(pp)} = +V_{ramp} - (-V_{ramp}) = 2\frac{R_2}{R_3}V_{sat} \tag{7.27}$$

The time taken for the output of A_2 to switch from $-V_{ramp}$ to $+V_{ramp}$ is half of the time period, i.e. $T/2$. From the basic integrator output equation,

$$v_o = -\frac{1}{R_1C_1}\int_0^{T/2}(-V_{sat})dt = \frac{V_{sat}}{R_1C_1}\left(\frac{T}{2}\right)$$

Then,

$$T = 2\frac{R_1C_1\,v_{o(pp)}}{V_{sat}}$$

Substituting the value of $v_{o(pp)}$ from Eq. (7.27) in the above equation, we get

$$T = \frac{4R_1C_1R_2}{R_3} \tag{7.28}$$

Therefore, the frequency of oscillation is

$$f_o = \frac{1}{T} = \frac{R_3}{4R_1C_1R_2} \tag{7.29}$$

A triangular waveform generator can also be constructed by a simple alternate arrangement of a square-wave generator connected to an integrator as shown in Fig. 7.12(a).

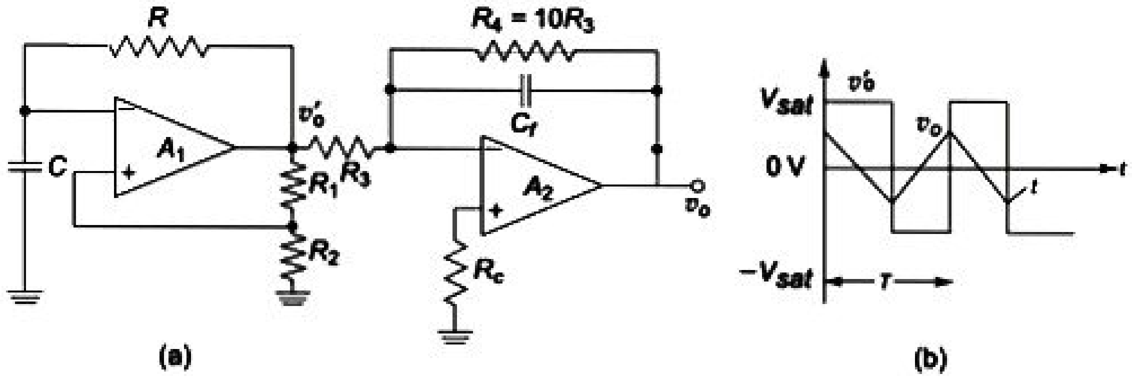


Fig. 7.12 *Alternate triangular waveform generator circuit: (a) Circuit diagram (b) Waveforms at the output of op-amps*

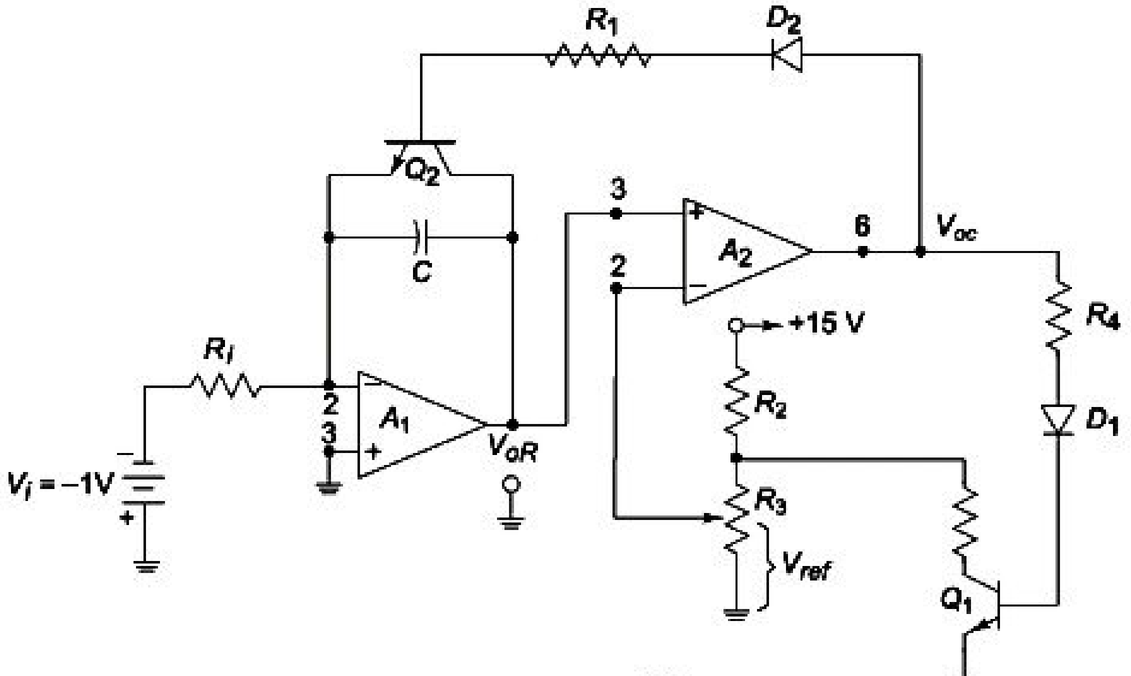
Let us assume that the voltage v_o' is high at $+V_{sat}$. This forces a current of $+V_{sat}/R_3$ through capacitor C_f of the integrator, producing a negative ramp at the output of the integrator. When v_o' is low at voltage $-V_{sat}$, the output of integrator ramps up linearly. This cycle repeats itself and hence the frequency of the triangular wave is the same as that of the square-wave. Hence, the value of resistor R connected in the square-wave generator part of the circuit determines the frequency of the triangular wave. The amplitude of the triangular wave decreases with an increase in frequency value. This is due to the fact that the capacitive reactance decreases at high frequencies and increases at low frequencies. The square waveform and triangular waveform of the circuit are shown in Fig. 7.12(b).

A stable triangular wave can be obtained by maintaining $5R_3C_2 > T/2$, where T is the period of the square-wave input. A resistance R_4 is normally connected across C_f to avoid saturation problems occurring at low frequencies.

7.5 SAWTOOTH WAVE GENERATOR

The sawtooth wave refers to a waveform with its rise time being many times longer than the corresponding fall time or fall time very longer as compared to the rise time. Triangular wave generator can be modified to produce a sawtooth waveform.

The sawtooth wave generator circuit is shown in Fig. 7.13(a).The op-amp A_1 functions as a ramp generator and the op-amp A_2 function as a comparator. The input reference signal V_i of value less than zero is connected to the inverting input of the op-amp. Since V_i is negative, the output of op-amp A_1 can only ramp up. The rate of raise is given by $\frac{V_{oR}}{t} = \frac{V_i}{R_i C}$. The ramp voltage V_{oR} is monitored by the comparator A_2. The output V_{oR} is connected to the non-inverting terminal of op-amp A_2, and a reference voltage set by a potentiometer V_{ref} is connected to the inverting input of comparator. When the capacitor C charges, and when the voltage V_{oR} is below V_{ref}, the output of comparator is negative. Then the transistors Q_1 and Q_2 do not conduct. The diodes D_1 and D_2 protect the transistors from excessive reverse bias voltages.



(a)

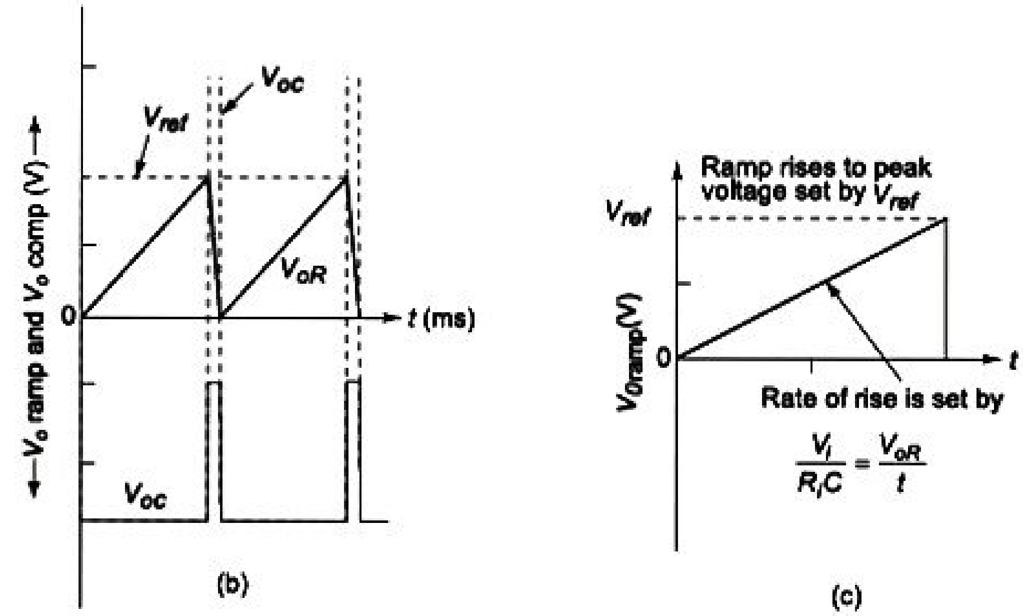


Fig. 7.13 *Sawtooth wave generator: (a) Circuit Diagram (b) Sawtooth wave and comparator outputs (c) Sawtooth wave design*

When V_{oR} rises and just exceeds V_{ref}, the output V_{oc} of comparator goes to positive saturation. This action forward biases the transistor Q_2 into saturation. The saturated transistor then acts as a switch across the capacitor C, which discharges quickly making the output V_{oc} come down essentially to 0 V. The positive saturation V_{oc} of A_2 also makes the transistor Q_1 ON, and the V_{ref} or negative input of op-amp A_2 drops to 0 V. As the capacitor C discharges rapidly making V_{oR} zero volts, it drops below V_{ref}. This causes the comparator output to become negatively saturated. This action switches the transistor Q_2 OFF and C begins charging linearly and the cycle repeats.

Frequency of oscillation The charging time T_c is obtained from period $T = \frac{V_{ref}}{V_i / R_iC}$. The frequency is the reciprocal of the time period. Therefore,

$$f = \frac{1}{T} = \left(\frac{1}{R_iC}\right)\frac{V_i}{V_{ref}}$$

The sawtooth output waveform is shown in Fig. 7.13(b). Figure 7.13(c) shows the sawtooth wave designing method. The circuit can also be operated as a current-controlled sawtooth wave generator by applying an external control current sink I_s at the (–) input terminal of op-amp A_1 of Fig. 7.13(a). Then,

$$f = \frac{I_s}{CV_{ref}}.$$

Here, the op-amp A_1 can preferably be a FET input op-amp with low bias current and good slew rate.

7.6 ICL8038 FUNCTION GENERATOR

The ICL8038 waveform generator is a monolithic integrated circuit which is capable of producing sine, square, triangular, sawtooth and pulse waveforms of high accuracy with a minimum number of external components. It is fabricated through advanced monolithic fabrication technology employing Schottky barrier diodes and thin film resistors. The output voltage is stable over a wide range of temperature and operating supply voltage values.

7.6.1 Functional Block Diagram of ICL8038

The main part of waveform generator is a VCO, that generates the triangular and square-waves. Sine-wave is generated from triangular wave by passing the later through an on-chip wave shaper. Sawtooth and pulse waveforms are obtained by the use of highly asymmetric duty cycle for the oscillator. The most common VCO circuit configurations are *grounded capacitor* and emitter coupled types. The *grounded capacitor* arrangement is used in the waveform generator ICL8038. The principle of operation of a *grounded capacitor* type VCO is shown in Fig. 7.14(a). When switch S is in position 1, the capacitor C charges at a rate fixed by the current source i_H. When voltage v_{TR} across capacitor reaches the upper threshold of Schmitt trigger, it changes state and flips the switch S to position 2. The capacitor C now discharges through the current sink i_L. When v_{TR} reaches the lower threshold level of Schmitt trigger, it triggers flipping the switch S to position 1. This sequence produces the waveforms shown in Fig. 7.14(b). The current source i_H and current sink i_L can be made programmable through the control voltage V_i.

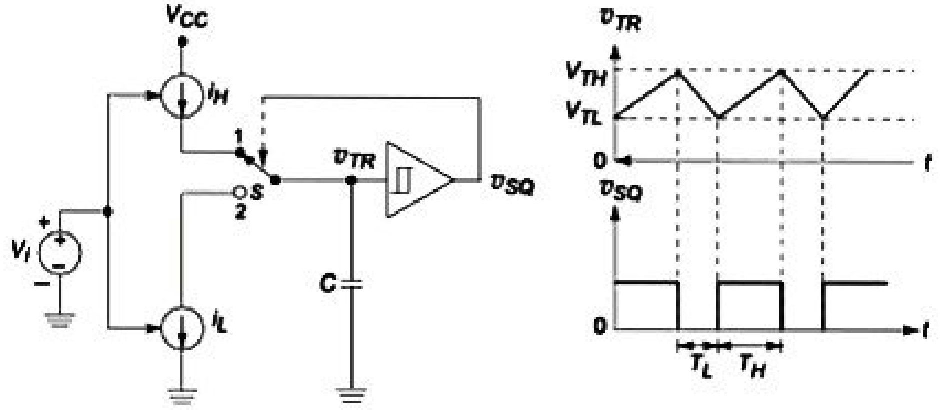


Fig. 7.14 *Grounded-capacitor type of VCO: (a) Circuit arrangement (b) Waveforms*

The circuit diagram of ICL8038 is shown in Fig. 7.15. The transistors Q_1 and Q_2 are two programmable current sources. The magnitudes of current are set by R_A and R_B. The emitter follower formed by the transistor Q_3 drives the base of Q_1 and Q_2. As shown in Fig. 7.15, $i_A = \frac{V_i}{R_A}$ and $i_B = \frac{V_i}{R_B}$. The current i_A is fed to C directly and current i_B is sent to the current mirror formed by the transistors Q_4, Q_5, and Q_6. Hence the transistors Q_5 and Q_6 produce a total sink current of $2i_B$ due to their parallel arrangement.

Schmitt trigger circuit is formed by the comparators CMP_1 and CMP_2 in combination with the flip-flop. The threshold voltage levels are $V_{TL} = \frac{1}{3} V_{CC}$ and $V_{TH} = \frac{2}{3} V_{CC}$. This is achieved by the potential divider arrangement consisting of three resistances of equal value R.

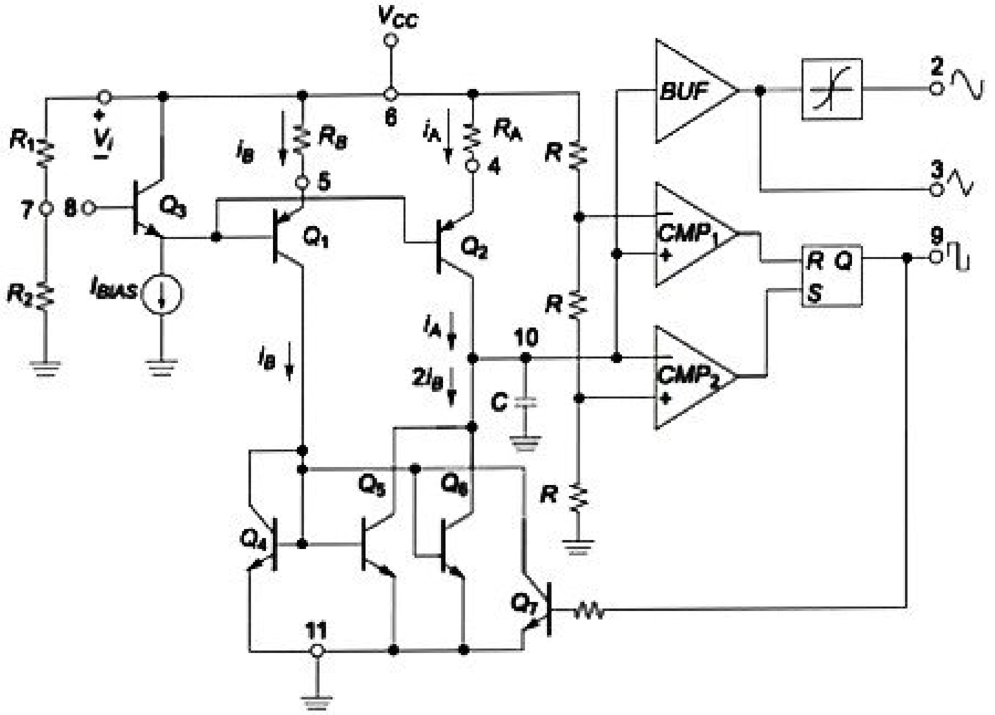


Fig. 7.15 *Functional circuit diagram of ICL8038 waveform generator*

Assume that the output Q of the flip-flop is initially high. This saturates transistor Q_7 and it pulls the bases of Q_5 and Q_6 low, shutting the current sink OFF. Therefore, C charges up at a rate determined by $i_H = i_A$. When the capacitor voltage reaches V_{TH}, the CMP_1 triggers, resetting the output Q. This turns Q_7 OFF and this action enables the current mirror. The current through C is now $i_c = 2i_B - i_A$. When $2i_B > i_A$, the capacitor continues to discharge and once V_{TL} is reached, CMP_2 triggers and it sets the flip-flop, thus repeating the cycle.

The frequency f_o is given by

$$f_o = 3\left(1 - \frac{R_B}{2R_A}\right)\frac{V_i}{R_A C V_{CC}}$$

and duty cycle $\quad D = \left(1 - \frac{R_B}{2R_A}\right)100\%.$

When $R_A = R_B = R$,

$$f_o = KV_i, \quad \text{where } K = \frac{1.5}{RCV_{CC}} \tag{7.30}$$

The sine-wave is obtained by use of a wave shaper which converts the triangular wave to a sine-wave. The buffer isolates the waveform developed across C. The pin diagram of ICL8038 is shown in Fig. 7.16.

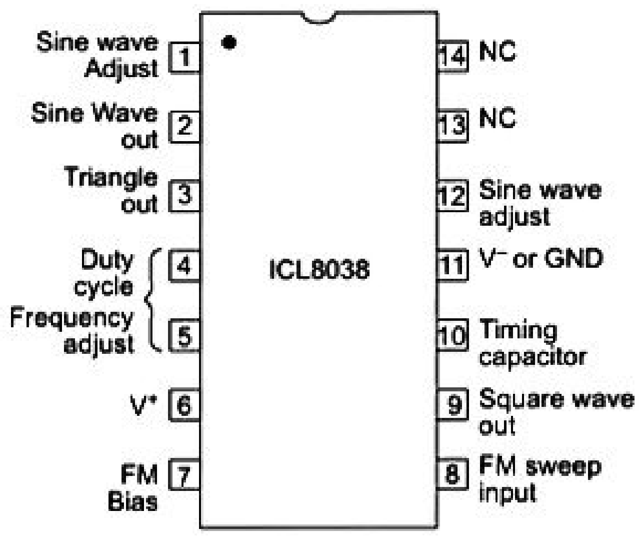


Fig. 7.16 *Pin diagram of ICL8038*

The important specifications of ICL8038 are as follows:

Supply voltage	-	± 18V or 36V
Power dissipation	-	750 mW
Input voltage	-	max of supply voltage level
Input current (at pins 4 and 5)	-	25 mA
Output sink current (pins 3 and 9)	-	25 mA
Storage temperature range	-	65°C to +125°C
Low frequency drift with temperature	-	50 ppm/°C
Distortion	-	1 % (for sine-wave output)
Linearity	-	0.1 % (for triangular wave output)
Operating frequency range	-	0.001 Hz to 300 kHz.
Duty cycle	-	2% to 98%

7.6.2 Basic ICL8038 Application

The basic connection diagram for obtaining 50% duty cycle operation is shown in Fig. 7.17. The control voltage V_i is derived from the supply voltage V_{CC} through the internal voltage divider R_1 and R_2 (Refer to Fig. 7.15). Hence from Eq. (7.30), $V_i = \frac{1}{5} V_{CC}$ and $f_o = \frac{0.3}{RC}$. Here, the duty cycle $D = 50\%$.

It can be noted that f_o is independent of supply voltage V_{CC}. The selection of external components R and C enables oscillation of any desired frequency. The frequency range is 0.001 Hz to 1 MHz and the thermal drift of f_o is typically 50 ppm/°C. The resistance R_{PU} acts as the pull-up resistor for the square-wave output. The waveform generator can be operated either from a single power supply (10 to 30V) or a dual power supply (±5 to ±15V).

With a single power supply, the peak-to-peak amplitudes of square, triangular and sinusoidal waves are V_{CC}, $0.33V_{CC}$ and $0.22V_{CC}$ respectively. The amplitudes of the waveforms are centered at $V_{CC}/2$. Automatic frequency sweep can also be achieved by varying the voltage at pin 8.

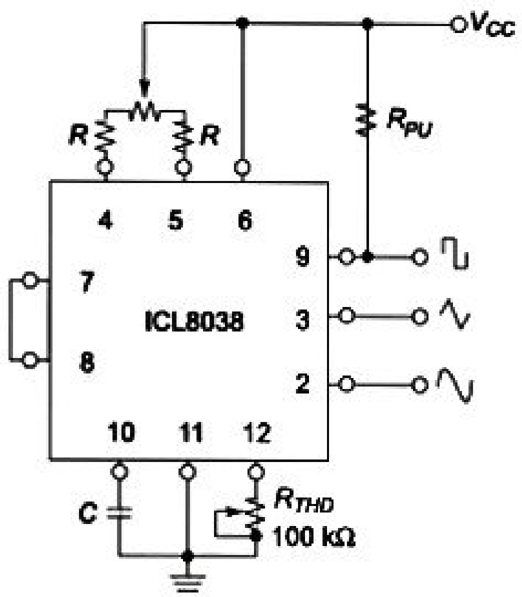


Fig. 7.17 *Basic ICL8038 connection diagram for 50% duty cycle, fixed frequency operation*

7.6.3 Frequency Modulation using ICL8038

The ICL8038 can be used for frequency modulation. The modulating signal voltage is applied at pin 8. The frequency of the waveform generator is then a direct function of the voltage at terminal 8. Frequency modulation is performed by varying the voltage at pin 8. Figures 7.18(a) and (b) show two arrangements of achieving FM modulation using ICL8038.

Figure 7.18(a) shows the circuit arrangement for frequency modulating signals of small deviations, e.g. ±10%. The modulating signal is then directly connected to pin 8 through a dc decoupling capacitor. The resistance R is used to increase the input impedance by the value of R.

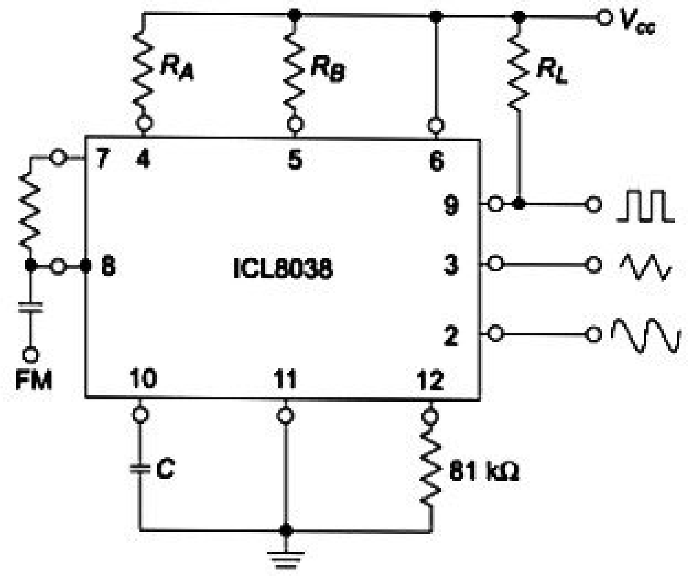


Fig. 7.18 *(a) Frequency modulation for small FM deviations*

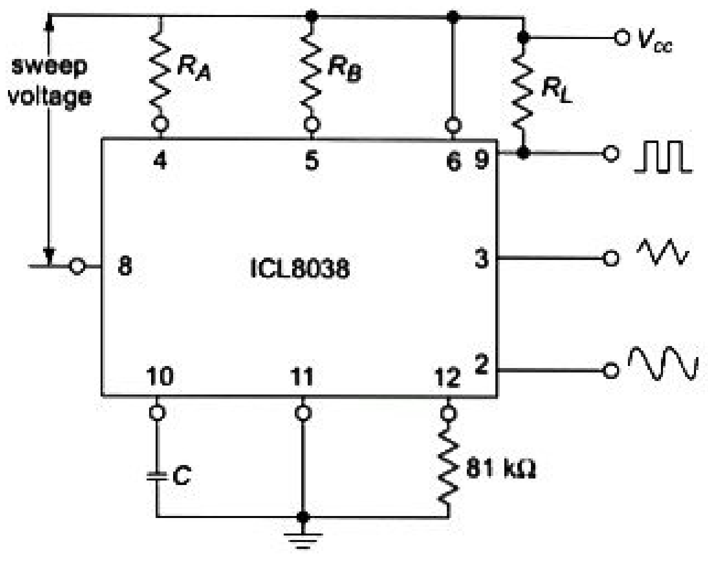


Fig. 7.18 *(b) Frequency modulation for larger FM deviations*

Figure 7.18(b) is used for larger deviations of FM or for frequency sweeping. The sweep voltage or the modulating signal is applied between the positive supply terminal and pin 8. Hence, the effective bias for the current source is created by the modulating signal. A very large sweep (e.g. 1000:1) can be created in this arrangement and value of f is zero at $V_{sweep} = 0$. It is to be noted that the charging current is a function of the modulating sweep signal and the trigger thresholds are functions of the supply voltage. The potential at pin 8 may span from $+V_{CC}$ down to $\left(\frac{1}{3}V_{CC} - 2V\right)$.

7.7 TIMER IC 555

The 555 integrated circuit timer was first introduced by Signetics Corporation as Type SE555/NE555. It is available in 8-pin circular style TO-99 Can, 8-pin mini-DIP and 14-pin DIP as shown in Fig. 7.19. The

555 IC is widely popular and various manufacturers provide the IC. The IC 556 contains two 555 timers in a 14-pin DIP package and Exar's XR-2240 contains a 555 timer with a programmable binary counter in a single 16-pin package.

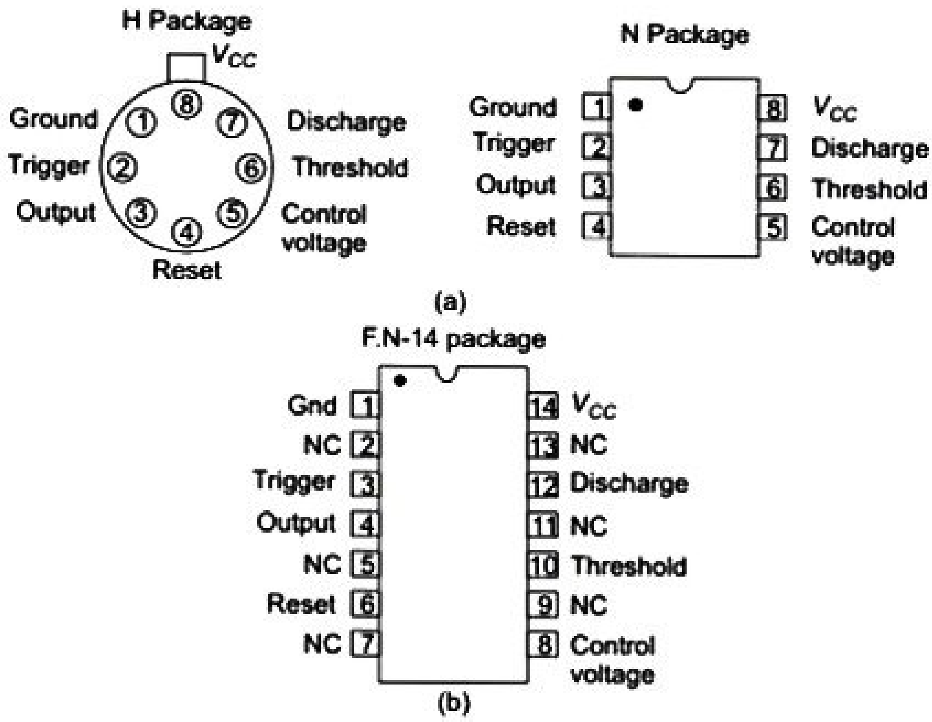


Fig. 7.19 *Pin configurations of IC 555 Timer*

The 555 timer can be operated with a dc supply voltage ranging from +5V to +18V. This feature makes the IC compatible to TTL/CMOS logic circuits and op-amp based circuits. The IC 555 timer is very versatile and its applications include oscillator, pulse generator, square and ramp wave generator, *one-shot* multivibrator, safety alarm and timer circuits, traffic light controllers, etc. The 555 timer can provide time delay, ranging from microseconds to hours.

7.7.1 General Description of the IC 555

Figure 7.20 shows the functional block diagram of 555 IC timer. The positive dc power supply terminal is connected to pin 8(V_{CC}) and negative terminal is connected to pin 1(*Gnd*). The ground pin acts as a common ground for all voltage references while using the IC. The *output* (pin 3) can assume a HIGH level (typically 0.5V less than V_{CC}) or a LOW level (approximately 0.1V).

Two comparators, namely, upper comparator (UC) and lower comparator (LC) are used in the circuit. Three 5 kΩ internal resistors provide a potential divider arrangement. It provides a voltage of $(2/3)V_{CC}$ to the (–) terminal of the upper comparator and $(1/3)V_{CC}$ to the (+) input terminal of the lower comparator. A *control* voltage input terminal (pin 5) accepts a modulation control input voltage applied externally. Pin 5 is connected to ground through a bypassing capacitor of 0.1 μF. It bypasses the noise or ripple from the supply. The (+) input terminal of the UC is called the *threshold terminal* (pin 6) and the (–) input terminal of the LC is the *trigger terminal* (pin 2). The operation of the IC can be summarised as shown in Table 7.1.

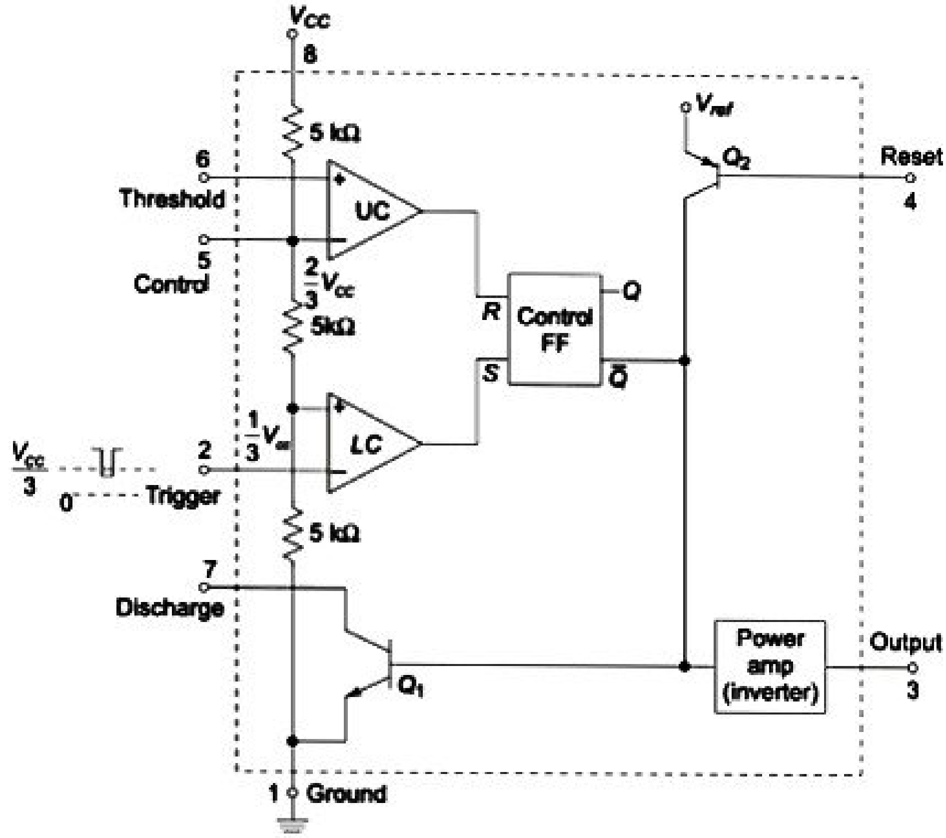


Fig. 7.20 *Functional block diagram of IC 555 Timer*

Table 7.1 *States of operation of IC 555*

Sl. No.	*Trigger (pin 2)*	*Threshold (pin 6)*	*Output state (pin 3)*	*Discharge state (pin 7)*
1	Below $(1/3)V_{CC}$	Below $(2/3)V_{CC}$	High	Open
2	Below $(1/3)V_{CC}$	Above $(2/3)V_{CC}$	Last state remains	Last state remains
3	Above $(1/3)V_{CC}$	Below $(2/3)V_{CC}$	Last state remains	Last state remains
4	Above $(1/3)V_{CC}$	Above $(2/3)V_{CC}$	Low	Ground

The standby (stable) state makes the output $\bar{Q}$ of flip-flop (FF) HIGH. This makes the output of inverting power amplifier LOW. When a negative going trigger pulse is applied to pin 2, as the negative edge of the trigger passes through $\frac{1}{3}(V_{CC})$, the output of the lower comparator becomes HIGH and it sets the control FF making $Q = 1$ and $\bar{Q} = 0$. When the threshold voltage at pin 6 exceeds $\frac{2}{3}(V_{CC})$, the output of upper comparator goes HIGH. This action resets the control FF with $Q = 0$ and $\bar{Q} = 1$.

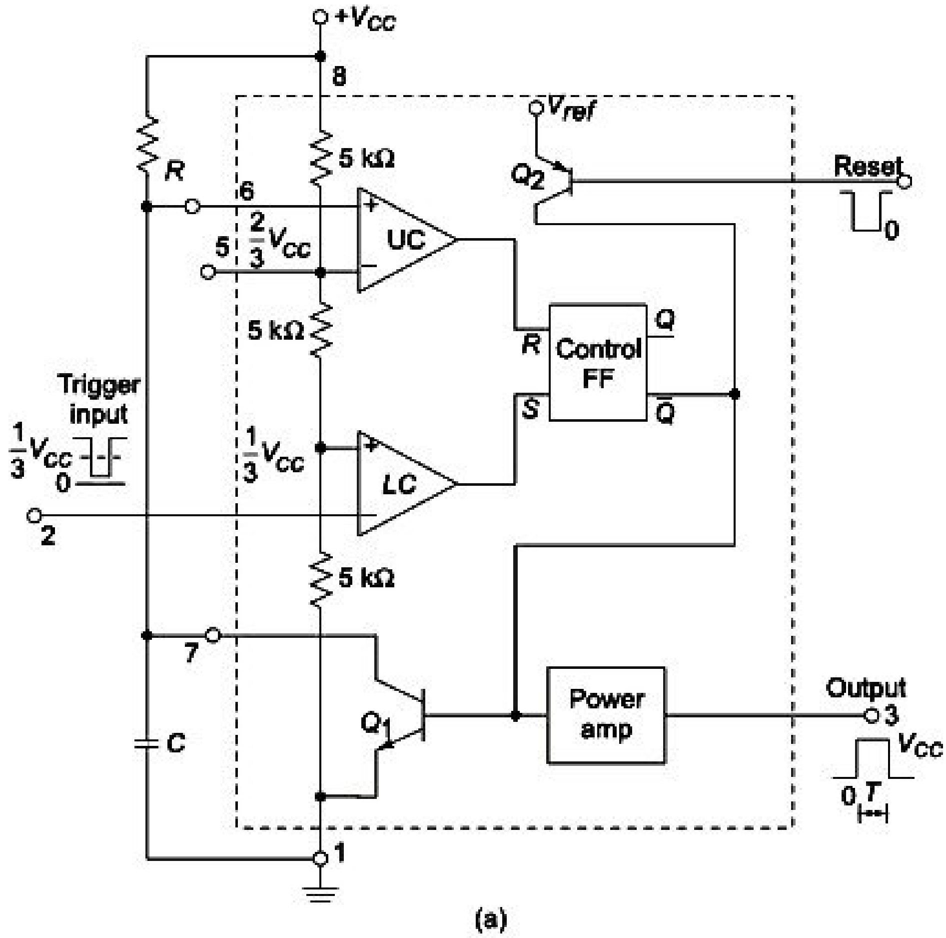
The *reset* terminal (pin 4) allows the resetting of the timer by grounding the pin 4 or reducing its voltage level below 0.4 V. This makes the *output* (pin 3) low overriding the operation of lower comparator. When not used, the *reset* terminal is connected to V_{CC}. Transistor Q_2 isolates the reset input from the *FF* and transistor Q_1. The reference voltage V_{ref} is made available internally from V_{CC}. Transistor Q_1 acts as a *discharge* transistor. When *output* (pin 3) is high, Q_1 is OFF making the *discharge* terminal (pin 7) *open*. When the output is low, Q_1 is forward-biased to ON condition. Then, the *Discharge* terminal appears as a short circuit to ground.

7.7.2 Monostable Operation of Timer IC 555

Monostable multivibrator has one stable state and one quasi-stable state. It is also known as *monoshot* or *one-shot* multivibrator or *uni-vibrator*. It remains in its stable state until an input pulse triggers it into its quasi-stable state. It stays in quasi-stable state for a time duration determined by an RC timing circuit. The output returns to its original stable state automatically at the end of the time and stays there until the next trigger pulse is applied. Therefore, a monostable multivibrator cannot generate square-waves on its own like an astable multivibrator. Only external trigger pulses will cause it to generate the rectangular pulses.

The functional block diagram and connection diagram of a monostable multivibrator using 555 timer is shown in Fig. 7.21(a) and (b) respectively. In the standby mode, the control flip-flop FF holds Q_1 ON, thus clamping the external timing capacitor C to ground. The output (pin 3) during this time is at ground potential, or LOW. The three 5 kΩ internal resistors act as voltage dividers providing bias voltages of $(2/3)V_{CC}$ and $(1/3)V_{CC}$ respectively. Since these two voltages fix the necessary comparator threshold voltages, they aid in determining the timing interval.

The lower comparator (LC) is biased at $(1/3)V_{CC}$ and it remains in the standby state as long as the *trigger* (pin 2) input is held above $(1/3)V_{CC}$. When triggered by a negative going pulse, the output of the lower comparator goes HIGH setting the flip-flop FF with $Q = 1$ and $\bar{Q} = 0$. This turns the transistor Q_1 OFF, and the output goes HIGH (approximately equal to V_{CC}). Since the timing capacitor is now unclamped, the voltage across it now rises exponentially through R towards V_{CC} with a time constant RC. After a period of time, the capacitor voltage will equal $(2/3)V_{CC}$ and the *upper* comparator (UC) resets the internal flip-flop. This makes $\bar{Q} = 1$ and the transistor Q_1 is ON. This in turn discharges the capacitor rapidly to ground potential. As a consequence, the output now returns to the standby state or ground.



(a)

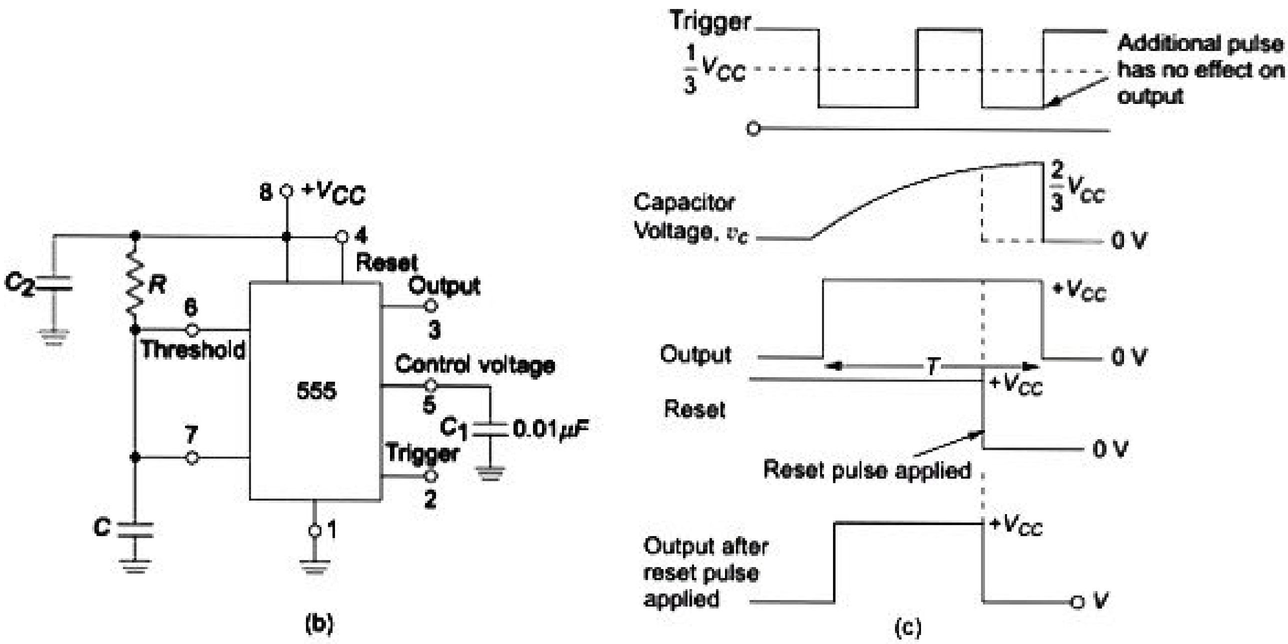


Fig. 7.21 *Monostable multivibrator: (a) Functional diagram (b) Connection diagram (c) Timing pulses*

The timing sequence of 555 monostable multivibrator is shown in Fig. 7.21(c). The circuit triggers only on a negative going pulse, when the level is less than $(1/3)V_{CC}$. Once triggered, the output will remain HIGH until the set time has elapsed, even if it is triggered again during this interval which is indicated in Fig. 7.21(c). Since the external capacitor voltage charges exponentially from 0 to $(2/3)V_{CC}$, the voltage across the capacitor v_c is given by

$$v_c = V_{CC}(1 - e^{-t/RC}) \tag{7.31}$$

At time $t = T$,

$$v_c = (2/3)V_{CC}$$

That is,

$$\frac{2}{3}V_{CC} = V_{CC}(1 - e^{-T/RC})$$

$$\text{Therefore, } T = -RC \ln\left(\frac{1}{3}\right) = 1.1\ RC\ (seconds) \tag{7.32}$$

Figure 7.22 shows a graph of various combinations of R and C necessary to produce a given time delay. Since the charging rate and comparator thresholds are both directly proportional to the supply voltage, the timing interval given by Eq. (7.32) is independent of the supply voltage.

For proper monostable operation with the 555 timer, the negative-going trigger pulse width should be kept shorter compared to the desired output pulse width. The values of external timing resistor and capacitor can be determined either from Eq. (7.32) or from the graph given in Fig. 7.22.

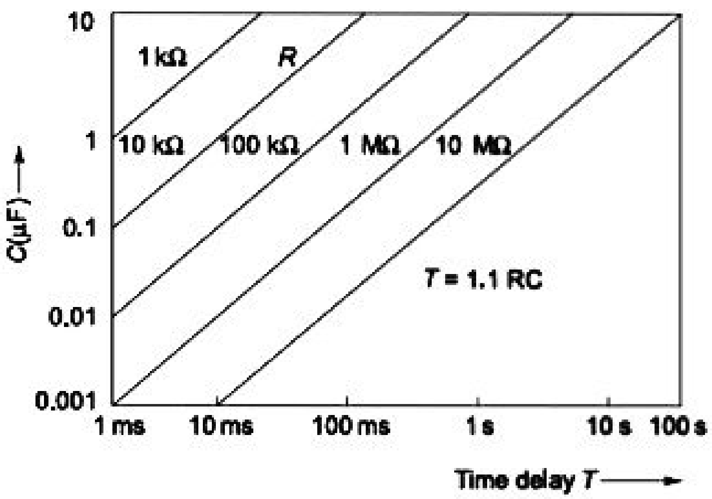


Fig. 7.22 *Graph of RC combinations for different time delays*

To prevent mistriggering on positive pulse edges, an R_1–C_1 combination with a diode D can be connected between pin 2 and pin 8 as shown in Fig. 7.23. The value of R_1 and C_1 should be such that R_1C_1 is smaller than the required pulse width. During the positive edge of the trigger pulse, the diode D gets forward-biased and it limits the amplitude of the positive spike to 0.7 V, which is less than $(1/3)V_{CC}$ to trigger the lower comparator LC ON.

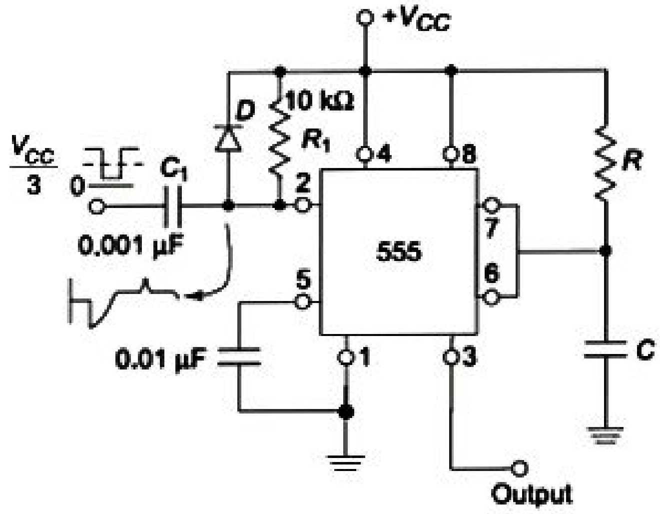


Fig. 7.23 *Waveshaping circuit to avoid positive pulse triggering*

Example 7.7

Design a monostable multivibrator using 555 timer for a pulse period of 1 ms.

Solution

The period of pulse is $t = 1.1\ RC$

Letting $C = 0.1\ \mu F$,

$$1 \times 10^{-3} = 1.1 \times 10^{-7} R$$

Therefore, $R = 8.2\ k\Omega$

7.7.3 Applications of Monostable Multivibrator

The important applications of monostable multivibrator are: (i) Ramp generation, (ii) Frequency division and (iii) Pulse-width modulation.

Ramp generation Linear ramp signal can be generated using the circuit shown in Fig. 7.24(a). The timing capacitor C is charged by constant current source instead of charging by a resistor. The transistor Q_L forms a constant current source. Assuming that i is the current supplied by a constant current source, the capacitor voltage v_c is given by

$$v_c = \frac{1}{C}\int_0^t i\,dt \tag{7.33}$$

Referring to Fig. 7.24(a) and employing KVL,

$$\frac{R_1}{R_1 + R_2}V_{CC} - V_{BE} = (\beta + 1)I_B R_E \tag{7.34}$$

$$\beta I_B R_E = I_C R_E = iR_E$$

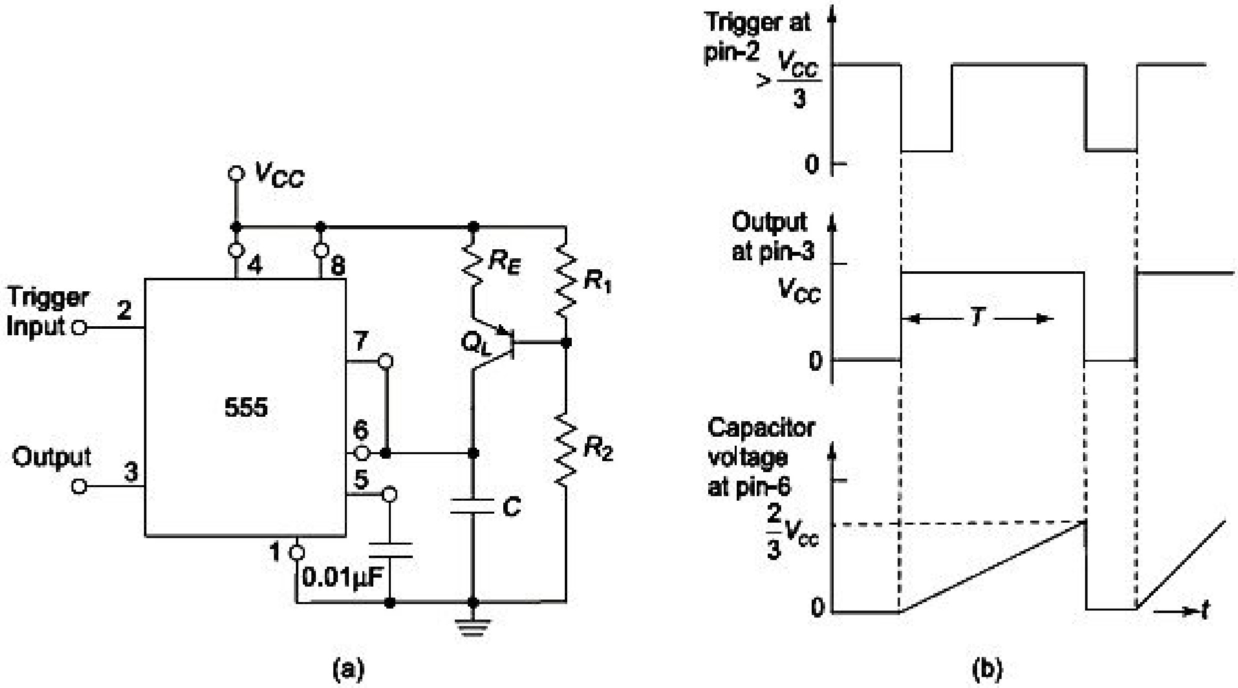


Fig. 7.24 *Ramp Generator: (a) Circuit diagram (b) Waveforms*

where I_B and I_C are the base and collector currents of transistor Q_L and β is the current gain in CE mode.

From Eq. (7.34),

$$i = \frac{R_1 V_{CC} - V_{BE}(R_1 + R_2)}{R_E(R_1 + R_2)} \tag{7.35}$$

Substituting Eq. (7.35) in Eq. (7.33), we get

$$v_c = \frac{1}{C}\int_0^t \frac{R_1 V_{CC} - V_{BE}(R_1 + R_2)}{R_E(R_1 + R_2)}\,dt$$

Therefore,

$$v_c = \frac{R_1 V_{CC} - V_{BE}(R_1 + R_2)}{CR_E(R_1 + R_2)} \times t \tag{7.36}$$

At the instant of time $t = T$, v_c becomes $(2/3)V_{CC}$. Then,

$$(2/3)V_{CC} = \frac{R_1 V_{CC} - V_{BE}(R_1 + R_2)}{CR_E(R_1 + R_2)} \times T$$

That is, the time period T of linear ramp generator is given by

$$T = \frac{(2/3)V_{CC}CR_E(R_1 + R_2)}{R_1 V_{CC} - V_{BE}(R_1 + R_2)} \tag{7.37}$$

When the capacitor voltage v_c just goes above $(2/3)V_{CC}$ the upper comparator triggers, thereby resetting the flip-flop. This discharges the capacitor C, and the output remains at zero until another trigger pulse is applied. The various waveforms of the circuit are shown in Fig. 7.24(b).

Frequency divider The monostable multivibrator, when continuously triggered by a square-wave signal can be used as a frequency divider, if the timing interval T of the monostable multivibrator is designed to be longer than the period of the triggering square-wave signal. The one-shot is triggered by the first negative edge of the square-wave input. During the second negative edge of the square-wave, the output of monostable multivibrator remains HIGH. However, during the third negative edge of the square-wave signal, the mono-shot once again triggers ON. In this manner, the output pulse can be made any integral fraction of the input triggering square-wave signal. The waveform of the input square-wave and output of monoshot are shown in Fig. 7.25 for a frequency *divided-by-two* operation.

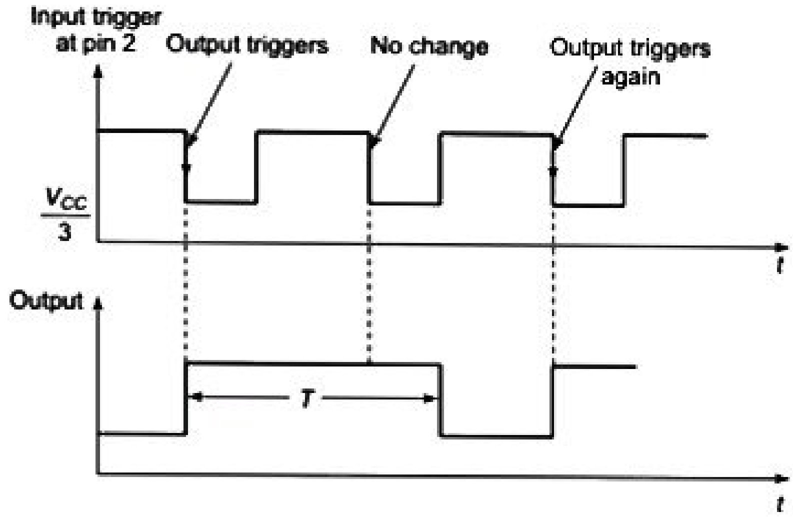


Fig. 7.25 *The input and output waveforms of frequency divider circuit*

Pulse-width modulation The monostable multivibrator, when applied with a *modulating control* input signal at pin 5 can act as a pulse width modulator. The circuit of the mono-shot with modulating input signal at pin 5 is shown in Fig. 7.26(a). The series of trigger pulses at pin 2 generates a series of output pulses. The duration of the output pulses are determined by the triggering of the upper comparator, which in turn depends on the modulating signal input at pin 5. This is due to the fact that, the modulating signal is superimposed upon the voltage $(2/3)V_{CC}$ obtained through voltage divider circuit. The threshold level of the upper comparator thus changes and the output pulse modulation occurs. The modulating control input signal and the output waveforms are shown in Fig. 7.26(b).

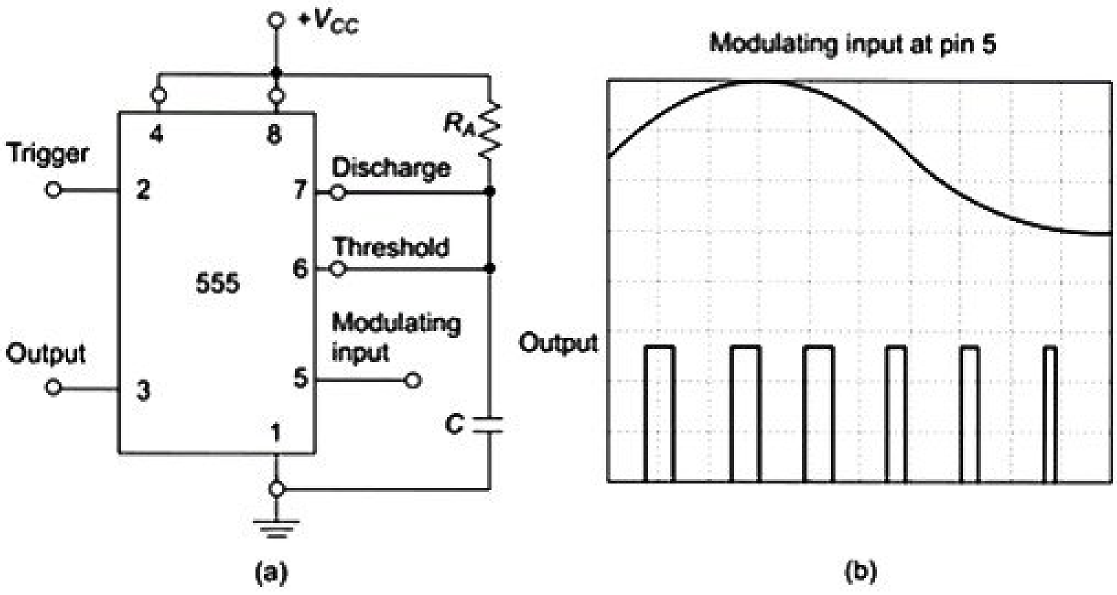


Fig. 7.26 *Pulse width modulator using monostable multivibrator: (a) Circuit diagram (b) Input and output waveforms*

It can be seen that, the frequency of the output remains the same, with the duty cycle of the output varying in response to the modulating input.

7.7.4 Astable Operation of the Timer IC 555

The functional diagram of the IC 555 connected for astable operation is shown in Fig. 7.27. The device connection diagram with external components is shown in Fig. 7.28. Resistors R_A and R_B form the timing resistors. The *discharge* (pin 7) terminal is connected to the junction of R_A and R_B. *Threshold* (pin 6) and *trigger* (pin 2) terminals are connected to the v_c terminal, and *control* (pin 5) terminal is by-passed to ground through a 0.01 μF capacitor.

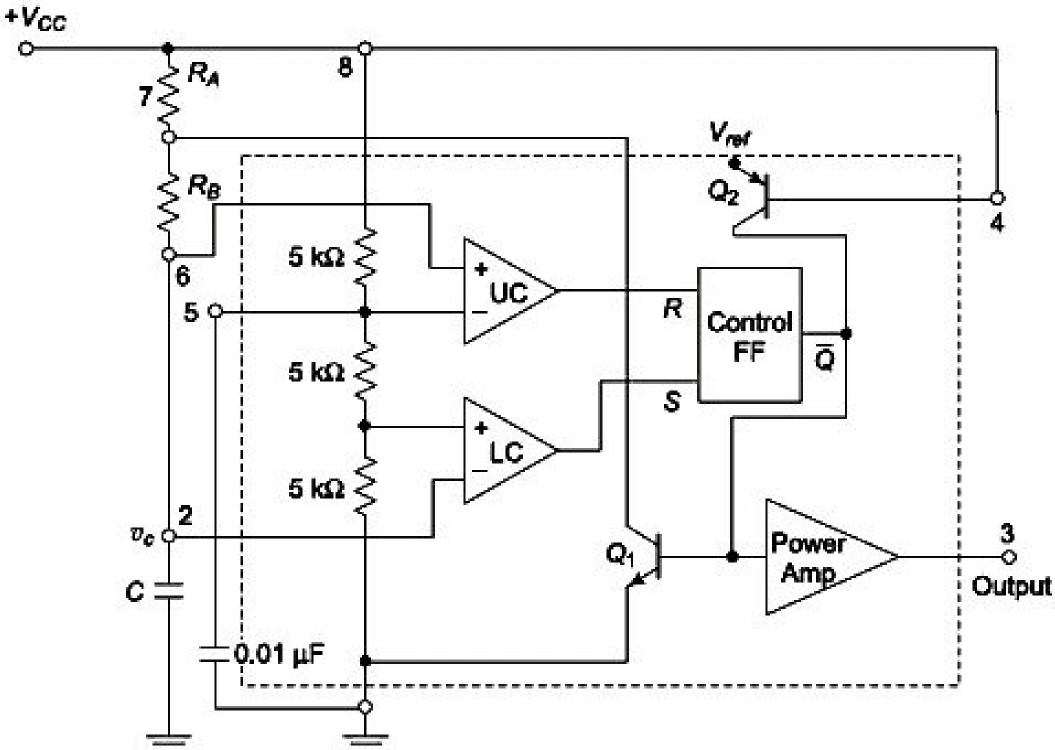


Fig. 7.27 *Functional diagram of astable multivibrator using IC 555 Timer*

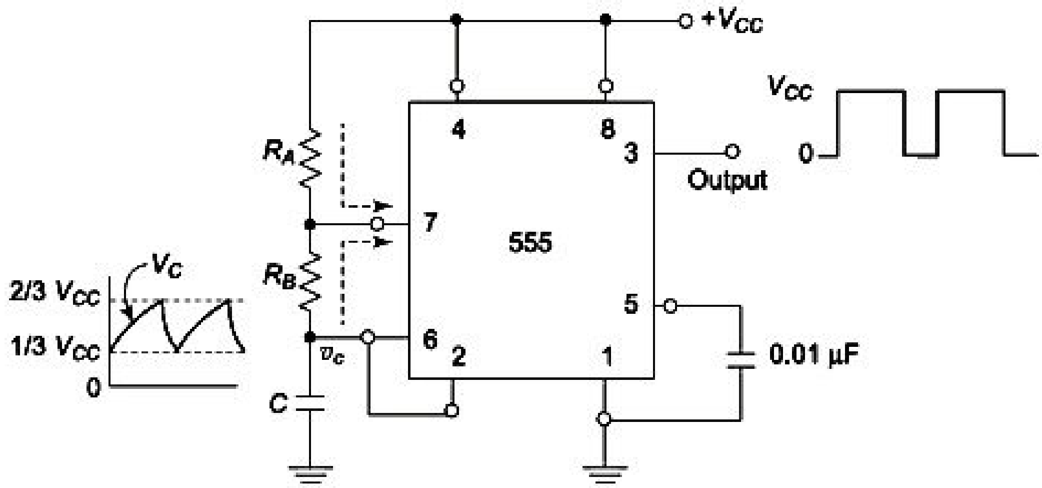


Fig. 7.28 *Connection diagram of astable multivibrator*

When the power supply V_{CC} is connected to the circuit, the capacitor C charges towards V_{CC}. The charging rate is determined by the time constant $(R_A + R_B)C$. During this period, the *output* (pin 3) is high, since $R = 0$, $S = 1$ and thus $Q = 0$. When the capacitor voltage reaches and rises just above $(2/3)V_{CC}$, the upper comparator triggers, and resets the flip-flop (FF). This makes internal *discharge* transistor Q_1 ON, resulting in the capacitor C discharging towards ground through the resistance R_B and Q_1. The time constant for discharging is R_BC. Since current can also flow through R_A into Q_1, the resistance R_A and R_B are to be made large enough to limit this current. A maximum current of 0.2A can be allowed to flow through the ON transistor Q_1.

When the timing capacitor C discharges, as it reaches and goes just less than $(1/3)V_{CC}$, the lower comparator gets triggered. This sets the flip-flop making $Q = 0$. This results in unclamping the external timing capacitor by switching Q_1 OFF. This cycle of charging to $(2/3)V_{CC}$, and discharging to $(1/3)V_{CC}$ repeats. Figure 7.29 shows the timing sequence and capacitor voltage waveform during the astable operation. The output (pin 3) is high during the internal charging of the capacitor from $(1/3)V_{CC}$ to $(2/3)V_{CC}$. The capacitor voltage v_c for a low pass RC circuit is given by

$$v_c = V_{CC}(1 - e^{t/RC})$$

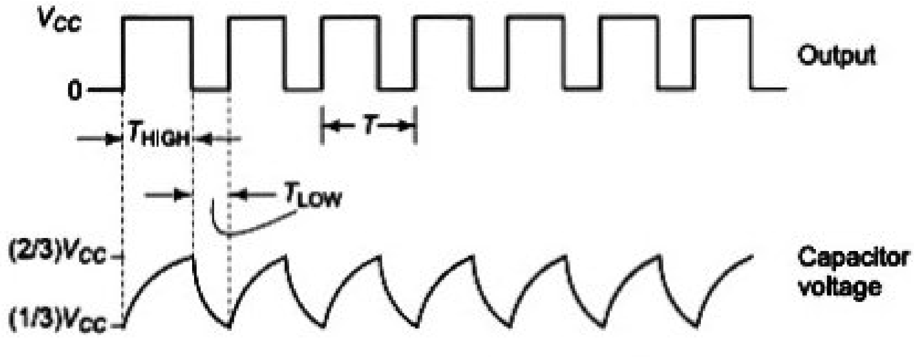


Fig. 7.29 *The timing waveform of astable multivibrator*

Here we assume a step input of V_{CC} Volts.

If t_1 is the time taken by the capacitor to charge from 0 to $(2/3)V_{CC}$, then

$$(2/3)V_{CC} = V_{CC}(1 - e^{-t_1/RC})$$

Therefore, $$t_1 = 1.098\ RC$$

If t_2 is the time taken by the capacitor to charge from 0 to $(1/3)V_{CC}$, then

$$(1/3)V_{CC} = V_{CC}(1 - e^{-t_2/RC})$$

Therefore, $$t_2 = 0.405\ RC$$

Then the time taken by the capacitor to charge from $(1/3)V_{CC}$ to $(2/3)V_{CC}$ is given by

$$\begin{aligned} t_{ON} &= t_1 - t_2 \\ &= 1.098\text{RC} - 0.405\ RC \\ &\approx 0.693\ \text{RC} \end{aligned}$$

Thus, t_{ON} for the circuit is

$$t_{ON} = 0.693(R_A + R_B)C$$

where R_A and R_B form the charging path. The output is in LOW level during the period of discharging from $(2/3)V_{CC}$ to $(1/3)V_{CC}$ and the voltage across the capacitor in such a condition is expressed by

$$(1/3)V_{CC} = (2/3)V_{CC}e^{-t/RC}$$

Hence, $t = 0.693\ RC$

Therefore, for the astable circuit,

$$t_{OFF} = 0.693\,R_B C$$

where R_B forms the discharging path for the current.

Thus, total time $T = T_{ON} + T_{OFF}$ or $T = 0.693(R_A + 2R_B)C$

and
$$f = \frac{1}{T} = \frac{1.45}{(R_A + 2R_B)C}$$

Figure 7.30 shows the various combinations of $(R_A + 2R_B)$ and C needed to obtain a stable output frequency. The *duty cycle D* of any pulse generator circuit is

$$D = \frac{\textit{high (ON) interval}}{\textit{period}} \times 100\% = \frac{T_{ON}}{T} \times 100\%$$

$$D = \frac{R_A + R_B}{R_A + 2R_B} \times 100\%$$

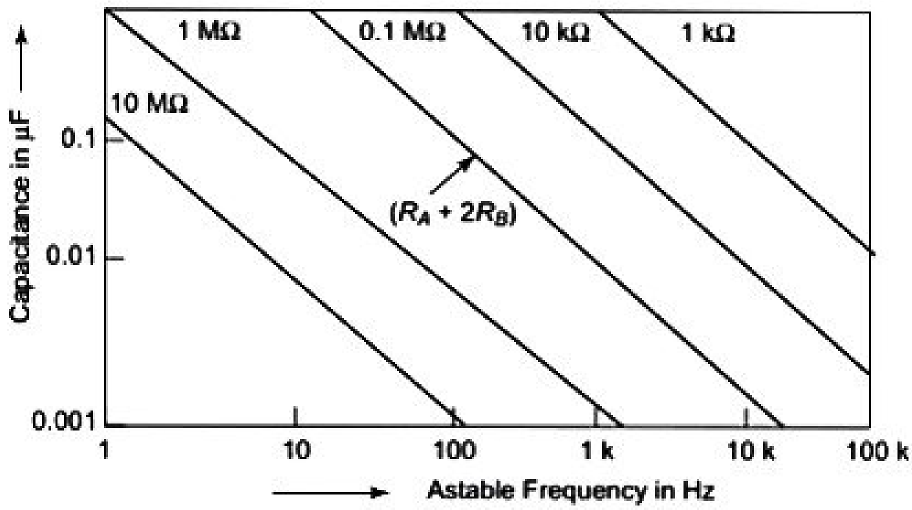


Fig. 7.30 *R_A, R_B and R_C combinations for different frequencies*

Defining the duty cycle in this manner always makes D greater than 50% since $T_{ON} > T_{OFF}$.
An alternative way of defining the duty cycle for this circuit is

$$D = \frac{T_{OFF}}{T} \times 100\% = \frac{R_B}{R_A + 2R_B} \times 100\%$$

It can be noted that the later definition takes into account the time when Q_1 is ON as T_{ON}. The resulting duty cycle value is less than 50%.

7.7.5 Applications of Astable Multivibrator

The important applications of the astable multivibrator are FSK generator, Pulse position modulator and Schmitt trigger.

FSK generator A timer IC 555 connected for FSK (Frequency-Shift-Keying) generation is shown in Fig. 7.31. The FSK method transmits data by identifying the logic 0 and 1 by means of two preset frequencies. The digital data input frequency is normally 150 Hz. Here, when the input is HIGH, Q_M is OFF. This makes the timer work in normal astable mode with the preset frequency defined by

$$f_1 = \frac{1.45}{(R_A + 2R_B)C}$$

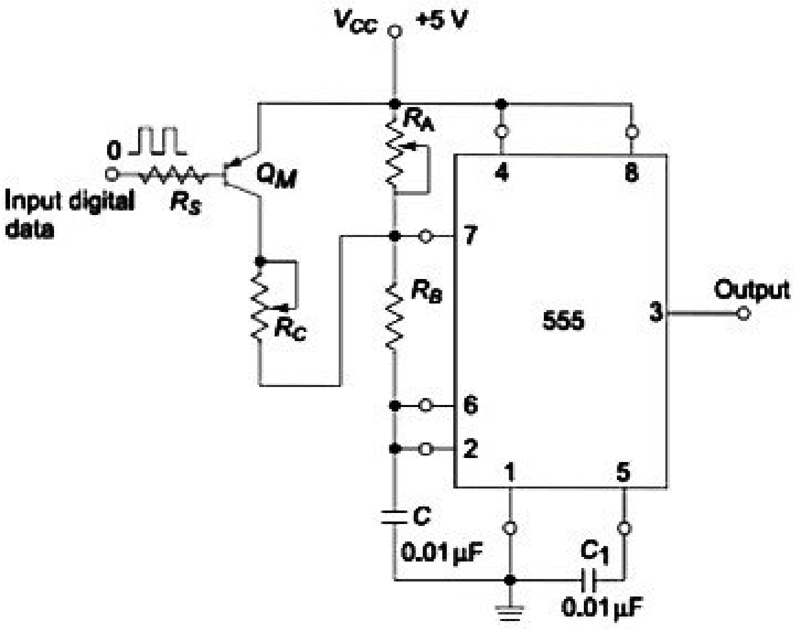


Fig. 7.31 *FSK generator connection diagram*

When the input goes LOW, Q_M is ON and it connects resistance R_C across R_A. The effective resistance $R_A \parallel R_C$ in series with R_B now forms the charging path. Therefore, the output frequency f_2 is

$$f_2 = \frac{1.45}{(R_A \| R_C + 2R_B)C}$$

Pulse-position modulator The timer IC 555 connected for pulse-position modulation is shown in Fig. 7.32(a). The modulating signal is applied to the modulating control input (pin 5). This changes the threshold voltage condition for the upper comparator. Therefore, the output pulse position varies due to the modulating signal. Figure 7.32(b) shows the output waveform obtained for an input triangular wave modulating signal. It can be seen that the output frequency changes with respect to the modulating signal leading to pulse position modulation.

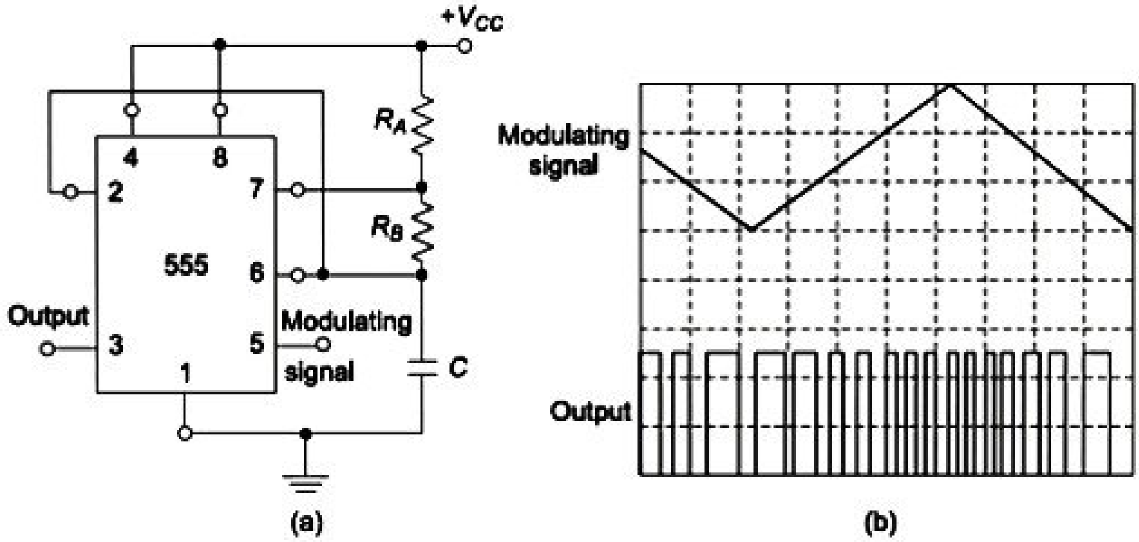


Fig. 7.32 *Pulse-position modulator using astable operation of IC Timer: (a) The connection diagram (b) Modulating signal and output waveforms*

Schmitt trigger using timer IC 555 The timer IC 555 can be used to function as a Schmitt trigger with variable threshold voltage levels.

The use of timer IC 555 connected as a Schmitt trigger circuit in astable mode of operation is shown in Fig. 7.33(a). The two internal comparator inputs (pins 2 and 6) are connected together and externally biased with a voltage $V_{CC}/2$ through R_1 and R_2 potential divider network. Since the voltage at pins 6 and 2 will trigger the upper comparator (UC) at $(2/3)V_{CC}$ and the lower comparator (LC) at $(1/3)V_{CC}$, the bias provided by R_1 and R_2 is centered within these two threshold levels. The input and output waveforms are shown in Fig. 7.33(b).

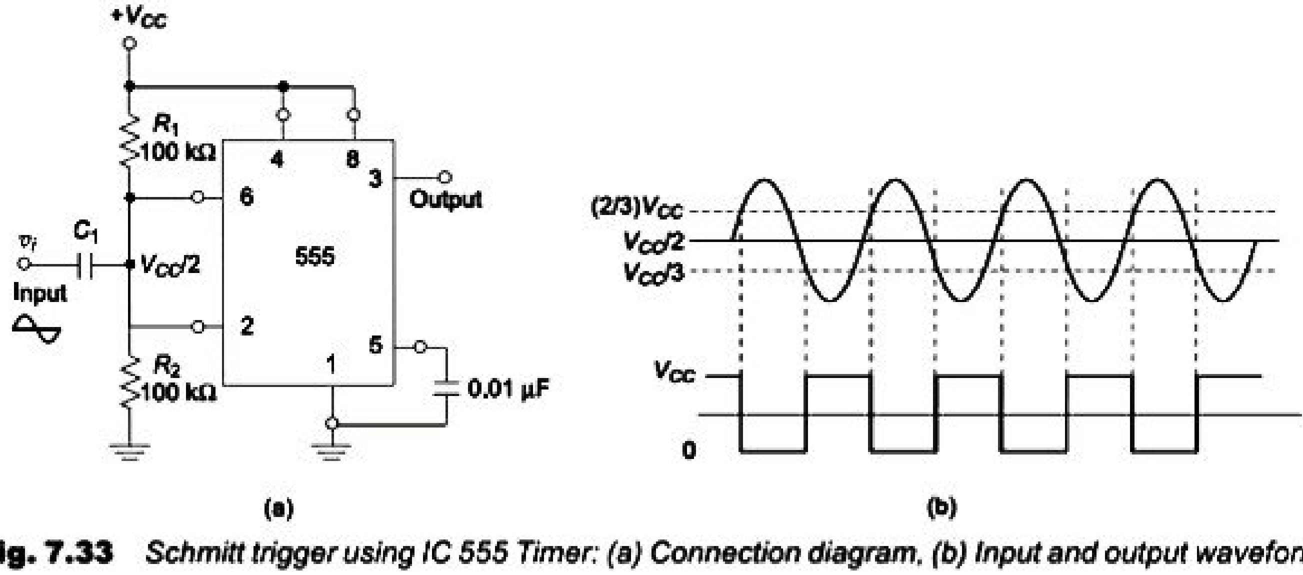


Fig. 7.33 *Schmitt trigger using IC 555 Timer: (a) Connection diagram, (b) Input and output waveforms*

8.4 IC VOLTAGE REGULATORS

Although voltage regulators can be designed using op-amps, it is quicker and easier to use IC voltage regulators. The IC voltage regulators are versatile, relatively inexpensive and are available with features such as programmable output, current/voltage boosting and floating operation for high voltage application. Some important types of linear IC voltage regulators are:

(i) Fixed positive/negative output voltage regulators
(ii) Adjustable output voltage regulators

Fixed voltage regulators 78XX series are three terminal positive fixed voltage regulators. There are seven voltage regulators with output voltages of 5V, 6V, 8V, 12V, 25V, 18V and 24V. The two digits XX of 78XX are used to identify the fixed output voltage of the regulator.

79XX series are negative fixed voltage regulators which are complements to the 78XX series devices. There are two additional voltage options of –2V and –5.2V available in 79XX series.

8.4.1 78XX and 79XX Voltage Regulators

The internal circuit diagram of a three terminal positive voltage regulator integrated circuit 7805 is shown in Fig. 8.7. This circuit provides over-current and thermal overload protections. The functional blocks of the regulator circuit are:

(i) The band gap voltage reference circuit is formed by the transistors Q_1 through Q_8. The band gap reference voltage V_R appears at the base of Q_6 that is across the resistor R_{18}. This voltage V_R is applied to the base of Q_1 as input.
(ii) Transistor Q_{10} along with resistor R_9 forms an emitter follower which provides the necessary base drive for the series pass output stage consisting of transistors Q_{15} and Q_{16}.
(iii) Transistor Q_{11} is a lateral PNP transistor with multiple collector terminals providing a multiple current source configuration. One of the collectors of Q_{11} acts as a current source active load for Q_{10}.
(iv) Transistor Q_{14} with current-limit resistor R_{16} of 0.3 Ω provides foldback current limiting for overload protection. Resistors R_3 and R_{14} with Zener diode D_2 provide the foldback dependence of the input-output voltage difference.
(v) Transistors Q_{12} and Q_{13} with Zener diode D_1, R_{11} and R_{12} provide thermal shutdown. Current source biasing for D_1 is provided by one of the collectors of Q_{11}.

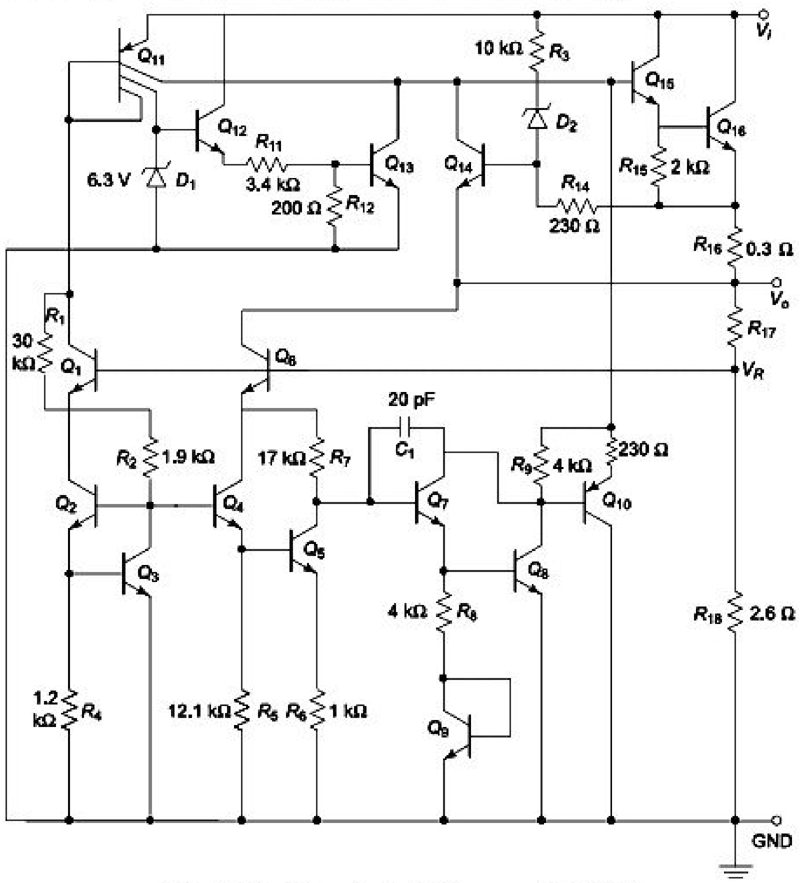


Fig. 8.7 *Internal circuit diagram of IC 7805*

The standard circuit connection of the 78XX monolithic voltage regulator is shown in Fig. 8.8. The input capacitor C_i is used to cancel the inductive effects due to long distribution leads and the output capacitor C_o improves the transient response and acts as a ripple filter also.

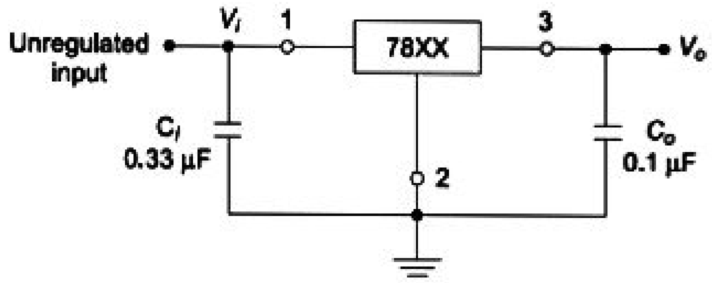


Fig. 8.8 *IC 78XX monolithic voltage regulator*

8.4.2 Dual Voltage Regulator

Figure 8.9 shows a dual voltage regulator. The circuit uses fixed positive (IC 7815) and fixed negative (IC 7915) voltage regulators to provide equal +15 V and −15 V respectively. The dual regulated voltage supplies as required for op-amps can be obtained from this circuit.

The advantage of this method is that it can supply a wide range of voltages at much higher currents with the use of heat sinks and external Metal Can package pass transistor.

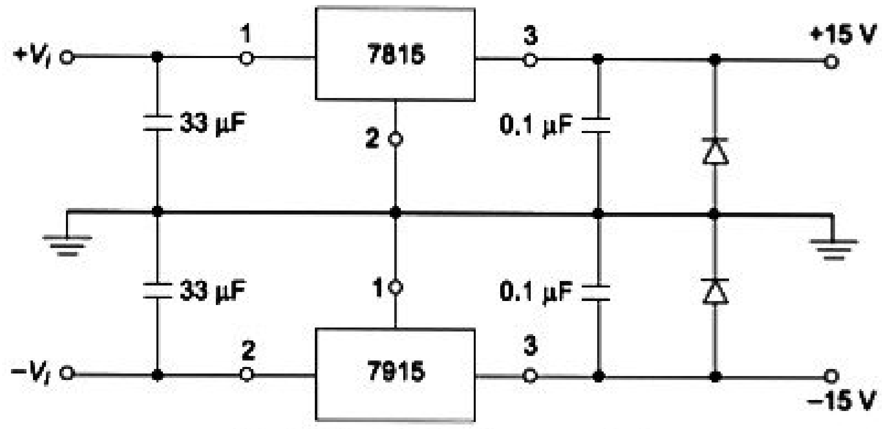


Fig. 8.9 *Dual voltage regulator*

8.4.3 Current Source using 7805 Regulation

The three terminal voltage regulator 7805 can be employed as a current source. This is achieved by connecting a current setting resistor R between the GND and OUT terminals of the regulator IC as shown in Fig. 8.10. Since the voltage at OUT terminal is always constant at 5V, the current through R is also maintained constant. Thus, by the use of KCL,

$$I_L = I_R + I_Q = \frac{5}{R} + I_Q$$

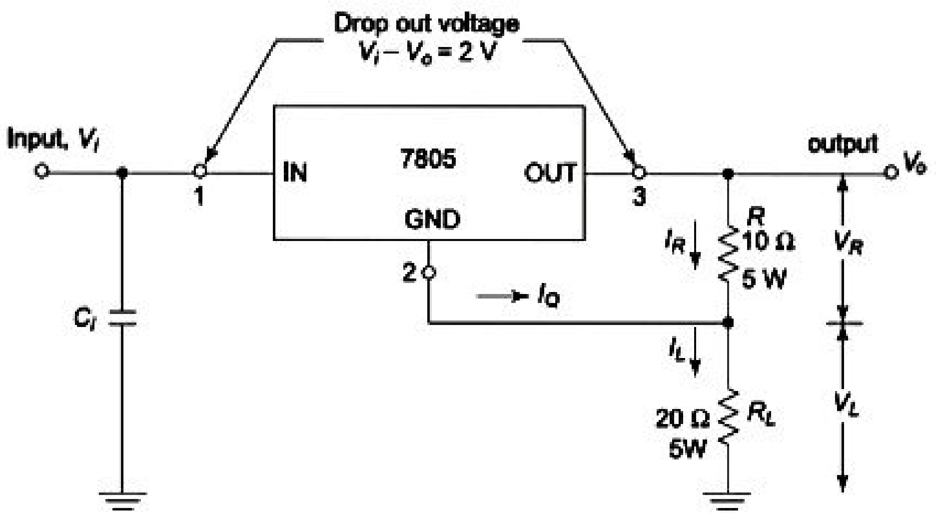


Fig. 8.10 *IC 7805 used as a current source*

8.4.4 Output Current Boosting

The output current of the three terminal regulator can be boosted or amplified by connecting an external pass-transistor Q_1 as shown in Fig. 8.11. The voltage drop across the resistor R_1 for low value of load currents is insufficient to make the transistor Q_1 ON, and therefore the regulator supplies the load current. When the load current I_L increases beyond a certain limit, the drop across R_1 increases and when it reaches 0.7 V, the transistor Q_1 turns ON, and supplies the additional current to the load. From transistor theory, the additional current is β times the base current.

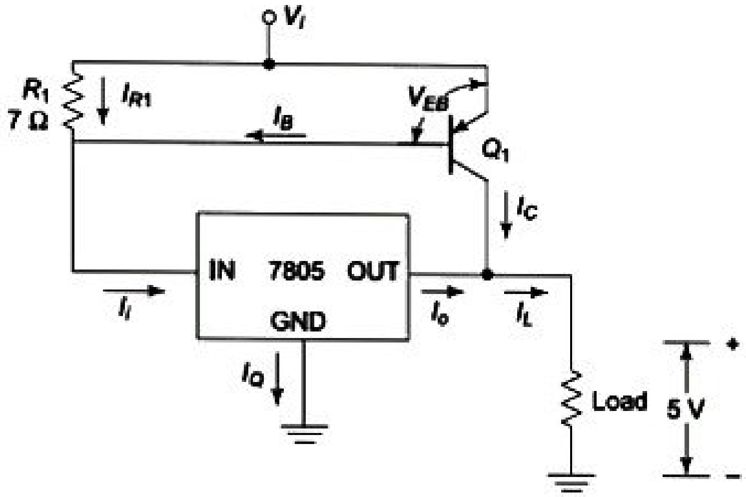


Fig. 8.11 *Boosting the output current of IC 7805*

Then, $$I_L = I_C + I_o \tag{8.2}$$

and $$I_C = \beta I_B$$

Thus $$I_L = \beta I_B + I_o \tag{8.3}$$

The output current I_o is given by

$$I_o = I_i - I_Q \approx I_i \text{ since } I_Q \text{ is negligibly small.}$$

Here, $$I_B = I_i - I_{R1}$$

$$\approx I_o - \frac{V_{EB(ON)}}{R_1} \tag{8.4}$$

Substituting Eq. (8.4) in Eq. (8.3)

$$I_L = I_o (\beta + 1) - \beta \frac{V_{EB(ON)}}{R_1} \tag{8.5}$$

The maximum current that can be obtained by using a 7805 regulator is 1A.

Assuming $V_{EB\,(ON)} = 0.7\text{V}$ and $\beta = 12$ for the transistor,

$$I_L = 1\,(12 + 1) - 12 \times \frac{0.7}{7} = 11.8\text{A}$$

5.9.1 Three-Terminal Fixed voltage regulators

78XX series are three-terminal positive fixed voltage regulators. There are seven voltage regulators with output voltages of 5V, 6V, 8V, 12V, 25V, 18V and 24V. The two digits XX of 78XX are used to identify the fixed output voltage of the regulator.

79XX series are negative fixed voltage regulators which are complements to the 78XX series devices. There are two additional voltage options of –2V and –5.2V available in 79XX series.

The internal circuits of three-terminal voltage regulator integrated circuits provide over-current and thermal overload protections.

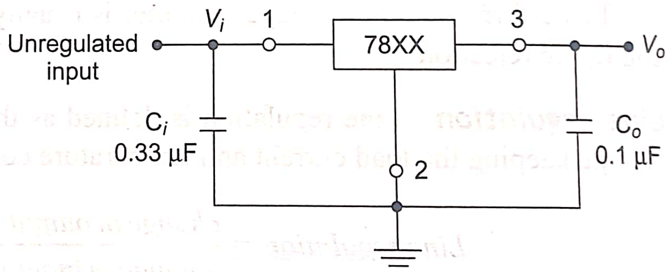


Fig. 5.37 *IC 78XX monolithic voltage regulator*

The standard circuit connection of the 78XX monolithic voltage regulator is shown in Fig. 5.37. The input capacitor C_i is used to cancel the inductive effects due to long distribution leads and the output capacitor C_o improves the transient response and acts as a ripple filter also.

5.10 THREE-TERMINAL ADJUSTABLE VOLTAGE REGULATORS

The fixed voltage regulators are designed and preset for a particular voltage of *positive/negative* polarities. There are applications which require

(i) regulated voltage sources which are precisely variable, and
(ii) some supply voltages which are not available from standard fixed voltage regulators.

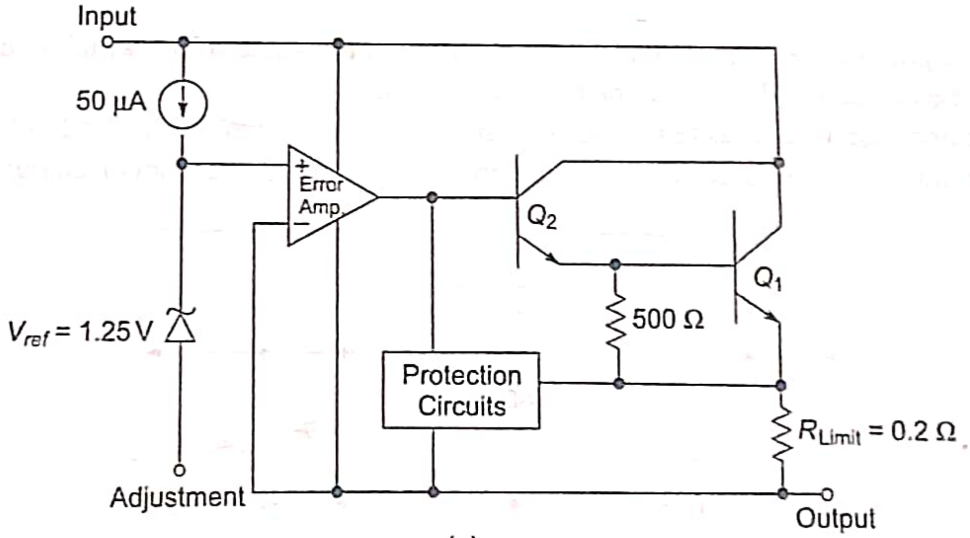
The most popular variable voltage regulators for such applications are LM117 / LM317 and IC723. The LM117/LM317 and LM137/LM337 families are adjustable three terminal positive and negative voltage regulators respectively. There are several variations of these ICs with different current ratings. The operating specifications are identified by their different suffixes and numbers.

5.10.1 LM117/LM317 Adjustable Positive Voltage Regulators

The LM117/LM317 series regulators are adjustable three terminal positive voltage regulators and they are capable of supplying output current of 0.1 A to 1.5 A, over a range of 1.2 V to 37 V output voltage range.

Circuit connection The internal functional diagram and the basic circuit connection for the LM317 regulator are shown in Fig. 5.41(a) and (b) respectively. The LM317 needs two resistors R_1 and R_2 for setting the output voltage. Normally capacitors are not required. However, when the LM317 is placed more than 15cm apart from the output filter capacitor C_2 of the power supply circuit, an input bypass capacitor, C_1 of 0.1 μF disc or 1 μF tantalum is needed to be connected as shown in Fig. 5.41(b). An additional aluminium or tantalum electrolytic capacitor of range 1 to 1000 μF may be connected at the output to improve the transient response characteristics. The unregulated voltage is applied at the terminal V_i of LM317 with respect to ground and it must be higher than the required regulated voltage V_o, by a value of at least 2V.

When the circuit is connected as shown in Fig. 5.41(b), the IC develops a typical value of *reference voltage* $V_{ref} = 1.25\text{V}$, between the output and adjustment terminals. This voltage appears across R_1 resulting in a current flow of $I_1 = \dfrac{V_{ref}}{R_1}$.



(a)

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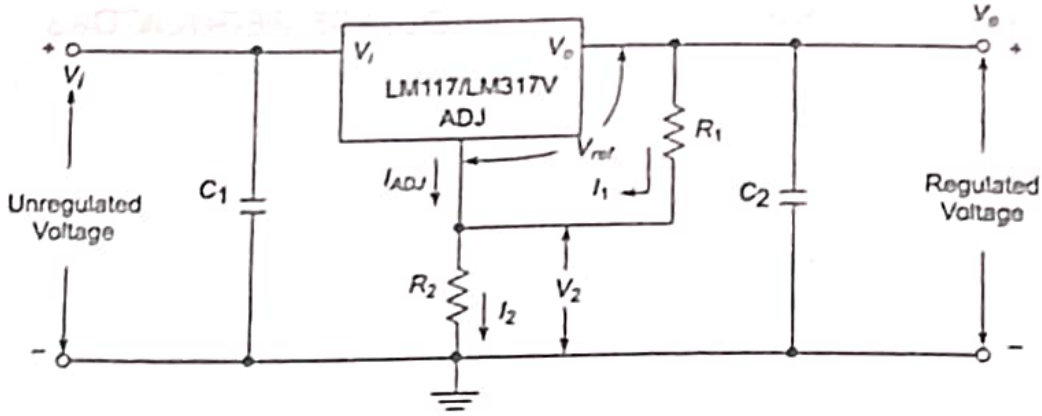


(b)

Fig. 5.41 *(a) LM317 voltage regulator functional diagram*
(b) Circuit connection for LM317 regulator

This current also flows through R_2 and an additional current of I_{ADJ} flows out of the adjustment terminal of the regulator through R_2. Thus the net current through R_2 is $I_2 = I_1 + I_{ADJ}$.

The voltage across R_2 is $V_2 = I_2 R_2 = (I_1 + I_{ADJ}) R_2$.

The net output voltage V_o is then given by

$$V_o = V_{ref} + V_2 = V_{ref} + R_2 (I_1 + I_{ADJ}) \tag{5.42}$$

Substituting $I_1 = \dfrac{V_{ref}}{R_1}$ in the above equation,

$$V_o = V_{ref}\left(1 + \frac{R_2}{R_1}\right) + I_{ADJ} R_2 \tag{5.43}$$

The last term $I_{ADJ} R_2$ indicates an error term of a typical value of 50 μA, and when R_2 is low, this term can be neglected.

Then,
$$V_o = 1.25\left(1 + \frac{R_2}{R_1}\right) \tag{5.44}$$

Hence, the output voltage V_o is a function of R_2 for a specific value of R_1, which is normally 240 Ω. The resistor R_1 is to be connected directly to the regulator output.

Protection diodes can be connected to the regulator circuit as shown in Fig. 5.42, when output capacitors of value greater than 25 μF are used. The diode D_1 prevents the capacitors from discharging into the regulator.

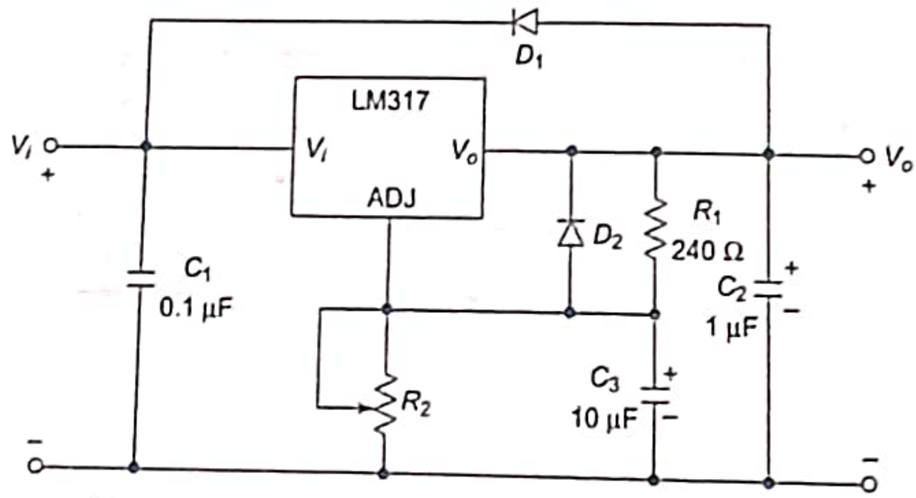


Fig. 5.42 *LM317 regulator with capacitors and protective diodes*

Figure 5.43 shows the circuit arrangement of an adjustable positive voltage regulator using LM317. The operation of the circuit is explained in Example 5.16.

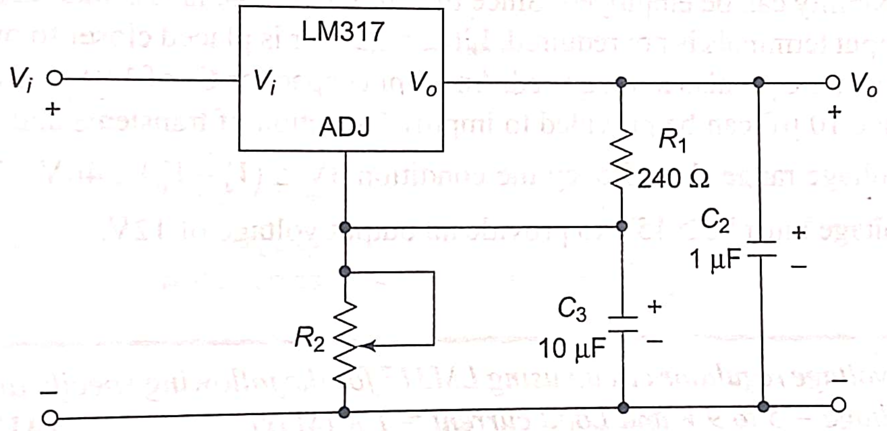


Fig 5.43 *LM317 adjustable positive voltage regulator*

5.10.2 LM137/LM337 Adjustable Negative Voltage Regulators

The LM137/LM337 series of adjustable three terminal negative voltage regulators are available for the corresponding types of LM117/LM317 positive voltage regulators. The circuit arrangement of LM337 connected for negative variable voltage regulation is shown in Fig. 5.45. Note that R_1 is of value 120 Ω and maximum input voltage which can be fed to V_i terminal is 50 V as against 60 V for LM317. The output voltage V_o is given by

$$V_o = 1.2\left(1+\frac{R_2}{R_1}\right) = 1.2\left(1+\frac{R_2}{120}\right)$$

$$V_o = 1.2\ \text{V} + 10\,\text{mA} \times R_2$$

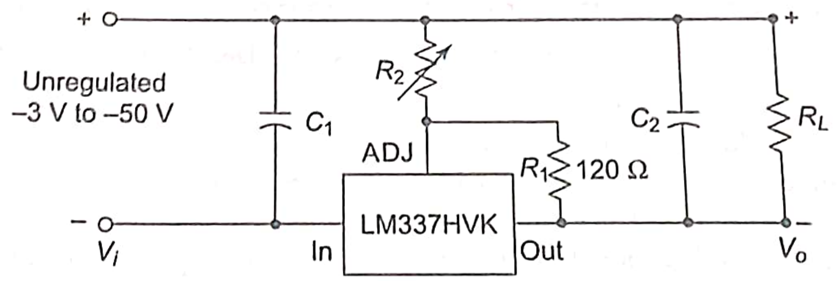


Fig. 5.45 *LM337 adjustable negative voltage regulator*

Table 5.2 shows the package types and grades of LM337 regulators. Figure 5.46 shows the schematic symbols and package types of the negative voltage regulators. The three terminals are input terminal V_i, output terminal V_o and adjustment terminal (ADJ).

Table 5.2 *Different grades and packages of LM337 regulators*

Device	*Available V_o (V)*	*Output Current (A)*	*V_i Max (V)*	*Ripple Rejection (dB)*	*Package Type*
LM337	−1.2 to −37	1.5	40	77	TO-39
LM337H	−1.2 to −37	0.5	40	77	TO-39
LM337HV	−1.2 to −37	1.5	50	77	TO-3
LM337HVH	−1.2 to −37	0.5	50	77	TO-39
LM337L	−1.2 to −37	0.1	40	65	TO-92
LM337M	−1.2 to −37	0.5	40	77	TO-202

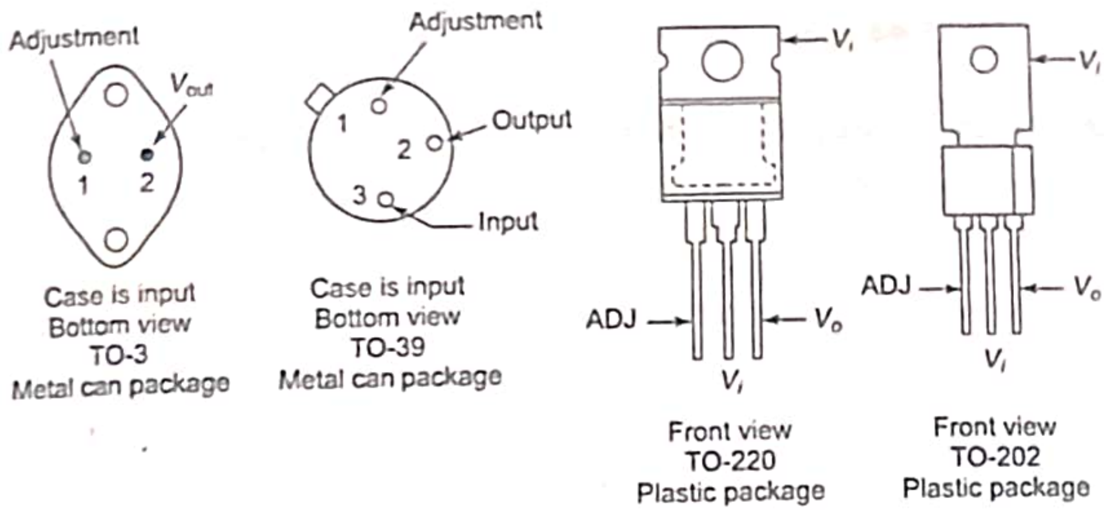


Fig. 5.46 *Schematic symbol and package types of LM137/LM337*

5.11 IC 723 GENERAL PURPOSE REGULATOR

[AU June'13, 8 marks, AU June'14, 16 marks, AU Dec'10, 8 marks, AU, Dec'13, 4 marks]

The three-terminal regulators such as 7805, 7815, 7905, 7915, etc. are capable of producing only fixed positive, or negative output voltages. Moreover, such regulators do not have short circuit protection. Therefore, these three terminal regulators evolved into dual polarity variable voltage regulators. They have provision for regulating positive and negative voltage inputs. The evolution further led to the monolithic linear voltage regulators and monolithic switching regulators. The monolithic linear voltage regulator type IC 723 is discussed in this section.

The IC 723 general purpose regulator overcomes the limitations of three terminal fixed voltage regulators. The IC 723 is a low current device, and can be employed for providing a load current up to 10 A or more by the addition of external transistors.

Figure 5.47 shows the detailed internal circuit diagram of IC 723. It consists of a temperature compensated Zener diode D_1, biased with a constant current source. The reference voltage V_{ref} is available from a buffer amplifier realised by transistors Q_1 through Q_6 and Zener diodes, D_1 and D_2. The V_{ref} has a typical value of 7.15 V. The series-pass element is realised by Darlington-connected transistors Q_{14} and Q_{15}. They boost the output current of the regulator to a value of 150 mA. The terminals + V and V_c are accessible externally. Current limit protection is provided by the transistor Q_{16}. *The sense voltage* is the voltage drop obtained across a suitable resistor connected between the terminals *current limit* (CL) and *current sense* (CS) of the IC 723. Transistors Q_7 through Q_{13} and resistors R_{10} and R_{11} form the error amplifier. The transistors Q_{11} and Q_{12} with their non-inverting and inverting input terminals produce a differential amplifier. The transistor Q_{13} acts as the current source for the differential amplifier.

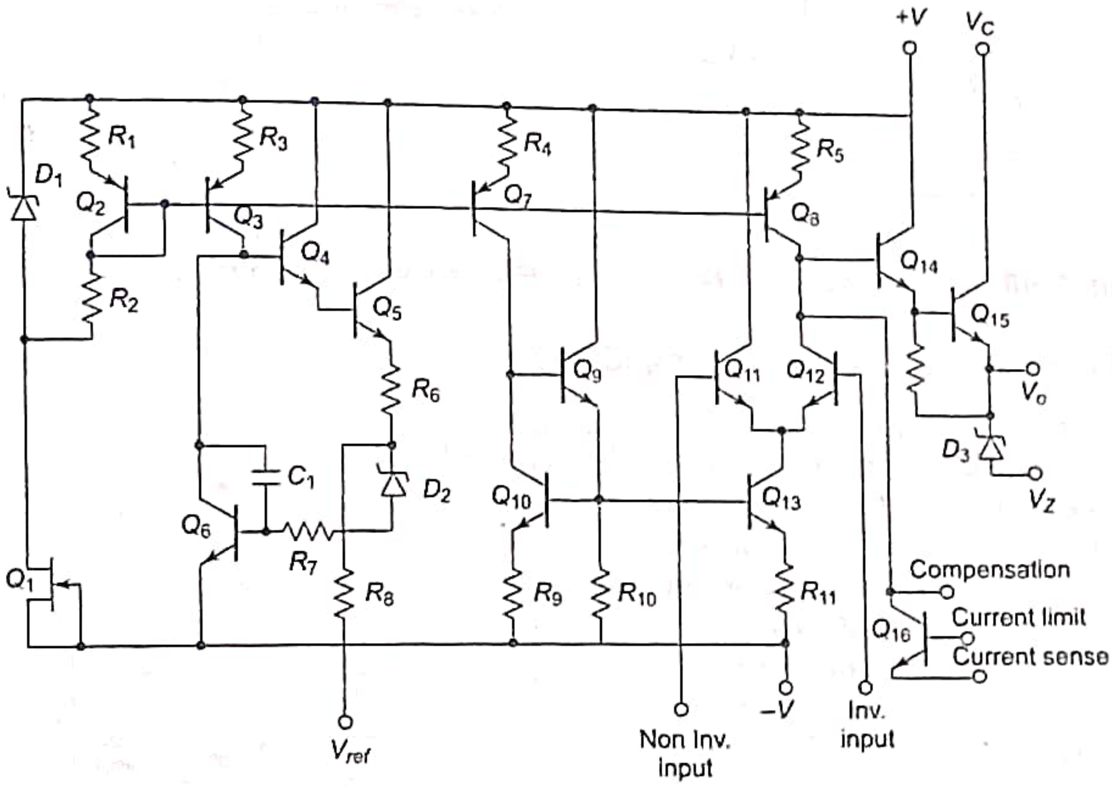


Fig. 5.47 *IC 723 Regulator internal circuit diagram*

The functional block diagram of IC 723 is shown in Fig. 5.48(a). Figures 5.48(b) and (c) show the pin diagrams for a 14-pin DIP and 10-pin Metal Can packages for the device. The Zener diode, the constant current source and reference amplifier form one section of the IC. The constant current source helps in maintaining a fixed output voltage from Zener diode D_2. The error amplifier, series pass element Q_1 and current limit transistor Q_2 form the second section. The error amplifier compares the input voltages applied at non-inverting (NI) and inverting (INV) input terminals. The error signal obtainable at the output of error amplifier drives the series pass element Q_1.

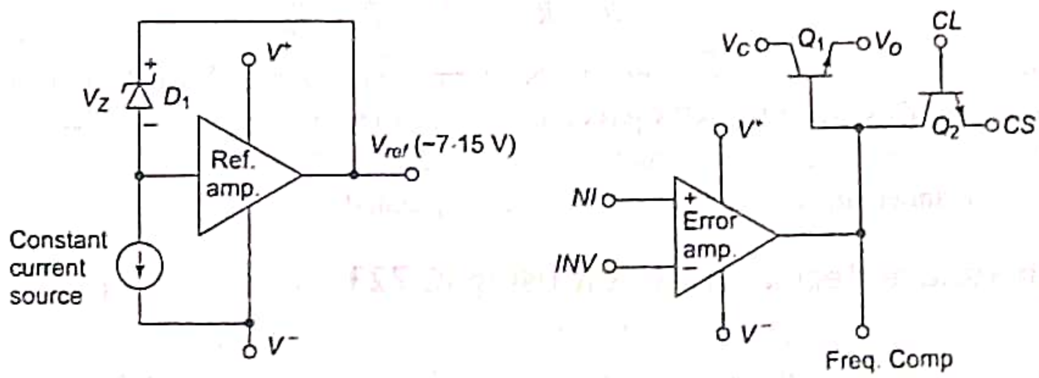


Fig. 5.48 (*a*) *Functional block diagram of IC 723*

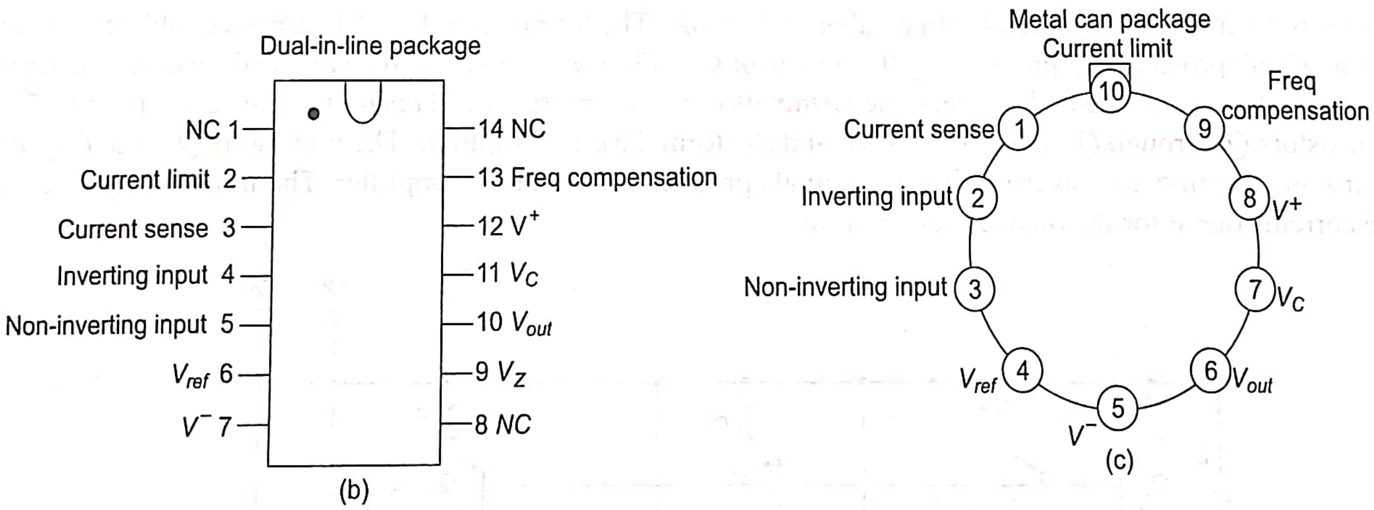
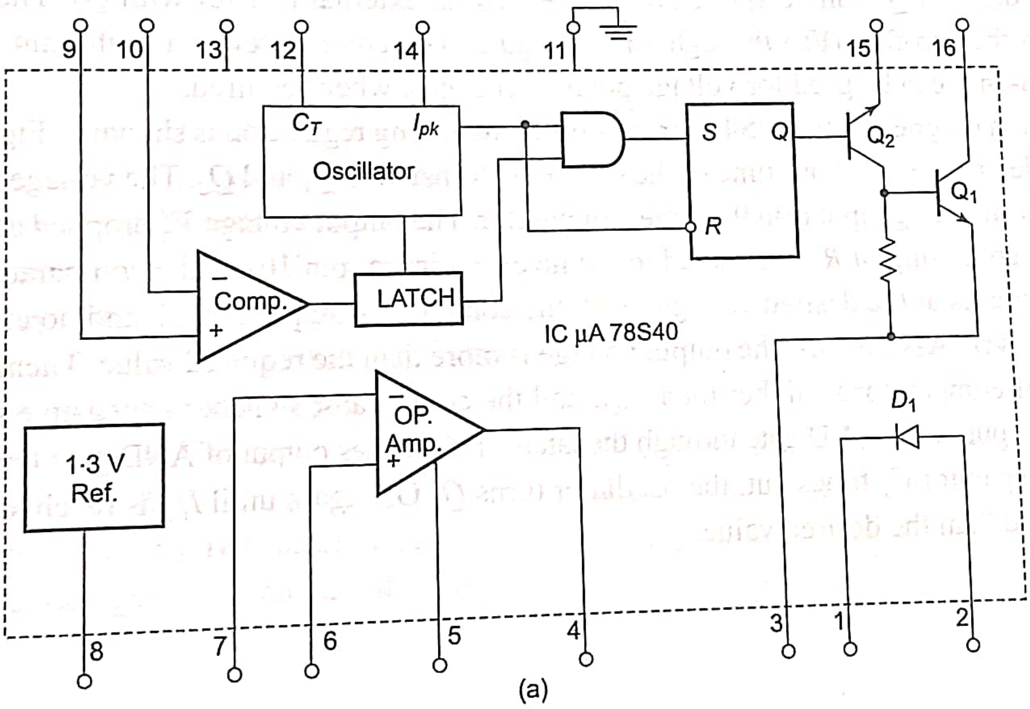


Fig. 5.48 *(b) Pin diagram of 14-pin DIP (c) Pin diagram of 10-pin Metal Can packages*

5.12.4 Monolithic Switching Regulator Circuit

[AU Dec'14, 16 marks, AU May'15, 10 marks]

Since the switching regulator circuit is complex, modern IC packages like Motorola MC3420/3520 or Silicon General SG1524 can be used instead of discrete components. The IC μA78S40 from Fairchild is a universal switching regulator. It is a pulse width modulated type of switching regulator. The internal block diagram and pin configuration of IC μA78S40 are shown in Fig. 5.59(a) and (b) respectively.



(a)

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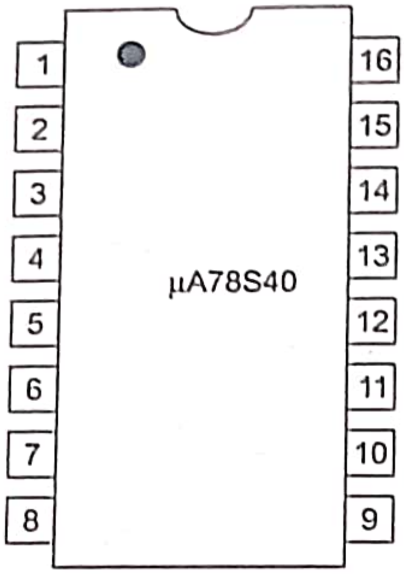


1. Diode cathode
2. Diode anode
3. Switch emitter
4. Op-Amp output
5. Op-Amp supply
6. Op-Amp non-inv Input
7. Op-Amp inv. Input
8. Reference
9. Comp. non-inv input
10. Comp. Inv. input
11. Ground
12. Timing capacitor
13. V_{CC}
14. I_{pk} sense
15. Drive collector
16. Switch collector

(b)

Fig. 5.59 *(a) Internal block diagram of IC µA78S40 (b) Pin configuration*

It consists of a temperature compensated voltage reference circuit, an oscillator, a high gain comparator, high current and high voltage output switch, a power switching diode D_1 and an op-amp. The reference voltage is available at pin 8. The oscillator is controllable for its duty cycle with the use of an active current-limit circuit. The internal switching frequency of oscillator is set by the timing capacitor C_T, connected between Pin 12 and ground (pin 11). The initial duty cycle (6 : 1) can be modified by the current limit circuiting which is activated at pin 14 (I_{PK}) and comparator outputs controlled by Pins 9 and 10. The typical frequency range of oscillator is between 100 Hz and 100 kHz.

The output transistor can operate at 40 V and supply current up to 1.5 A. The transistor Q_2 operated by the output of SR flip-flop drives the base of Q_1. The transistors Q_1 and Q_2 can be connected as a Darlington pair. Increased base drive to Q_1 can be applied by connecting an external resistor with Q_2. The oscillator output is connected to the flip-flop (FF) through an AND gate. The comparator is a high gain amplifier and the unconnected op-amp can be used for voltage polarity changes, when required.

The connection diagram of µA78S40 for step-down switching regulation is shown in Fig. 5.60. The output of comparator determines the OFF time of the switching transistors Q_1 and Q_2. The voltage reference of 1.3 V is applied to non-inverting input (pin 9) of the comparator. The output voltage V_o' dropped across the potential divider network consisting of $R_1 - R_2$ is fed to the inverting input (pin 10) of the comparator.

When the output is at the desired voltage level, the comparator output is high and no effect on the circuit operation is observed. Assume that the output voltage is more than the required value. Then, the voltage at the inverting input of comparator is higher than V_{ref}, and the comparator switches to negative saturation, and this is applied to one input of the AND gate through the latch. This makes output of AND gate low and Q_1 is turned OFF. When the capacitor C_T times out, the oscillator turns Q_1 ON again until I_{PK} is reached or output voltage is once again more than the desired value.

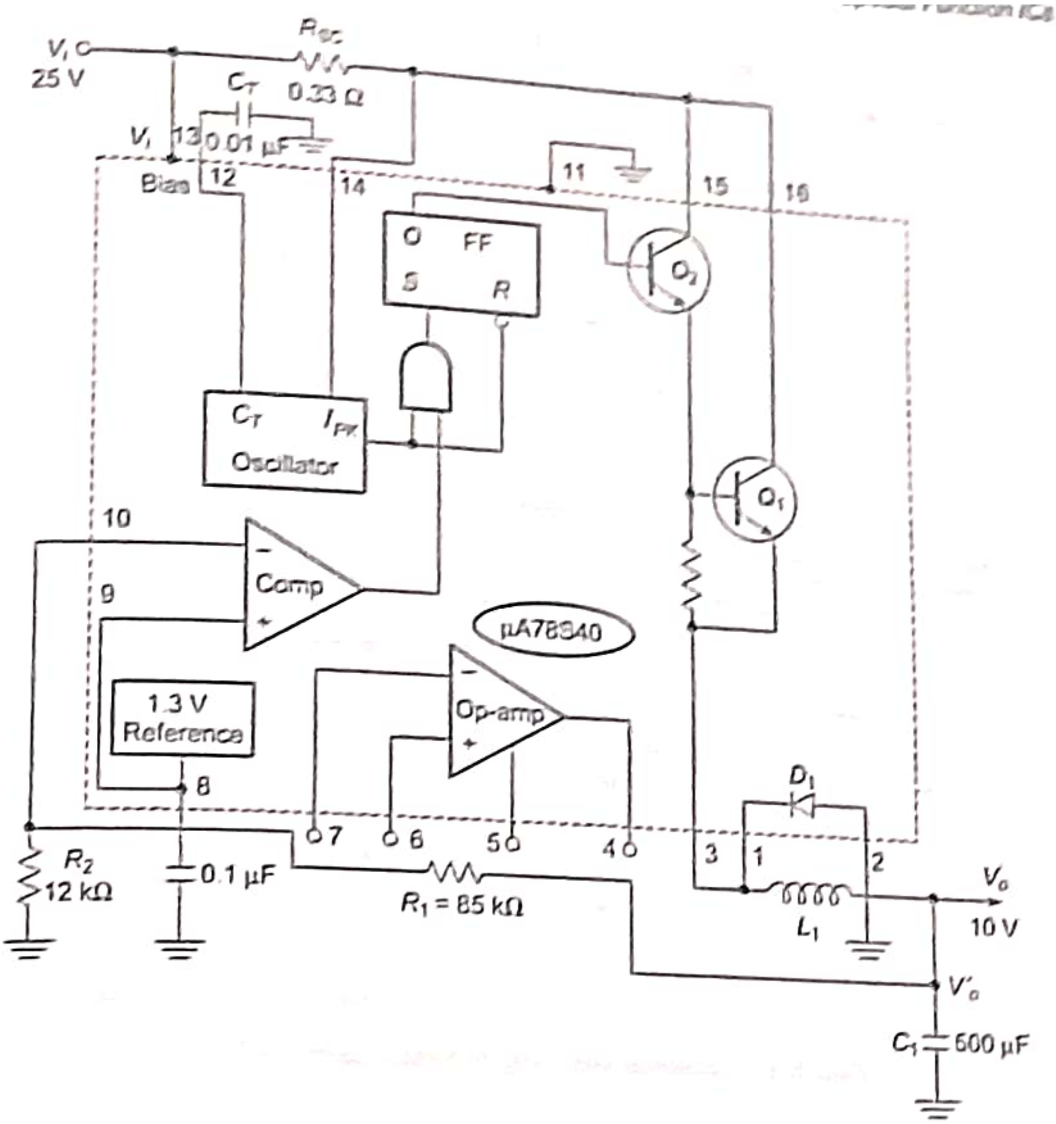


Fig. 5.60 *Step-down regulator using μA78S40*

When the output voltage drops lower than the required value, the average discharge current of inductor L_1 increases. This increases the time t_{ON}, since it needs more time to charge the inductor to I_{PK}. Increase in t_{ON} rises the voltage V_o to the desired value. When V_o increases, the inductor discharges less during t_{OFF}, and I_{PK} is reached faster when Q_1 is ON. This causes t_{OFF} to reduce.

Current limiting is achieved by connecting a sense resistor R_{SC} between I_{PK} sense (pin 14) and the V_{CC} (pin 13). It is activated when a voltage of 330 mV appears across R_{SC}. This terminates t_{ON} interval and starts t_{OFF} interval. The voltage drop V_{Df} of the power diode D_f determines the OFF time of inductor L_1 and the efficiency of the switching regulator. This regulator circuit is designed for stepping down from an input voltage of 25 V to an output voltage of 10 V.

The step-up switching regulator using μA78S40 along with the component values is shown in Fig. 5.61. This circuit generates an output voltage of 25 V when an input voltage of 10 V is applied.

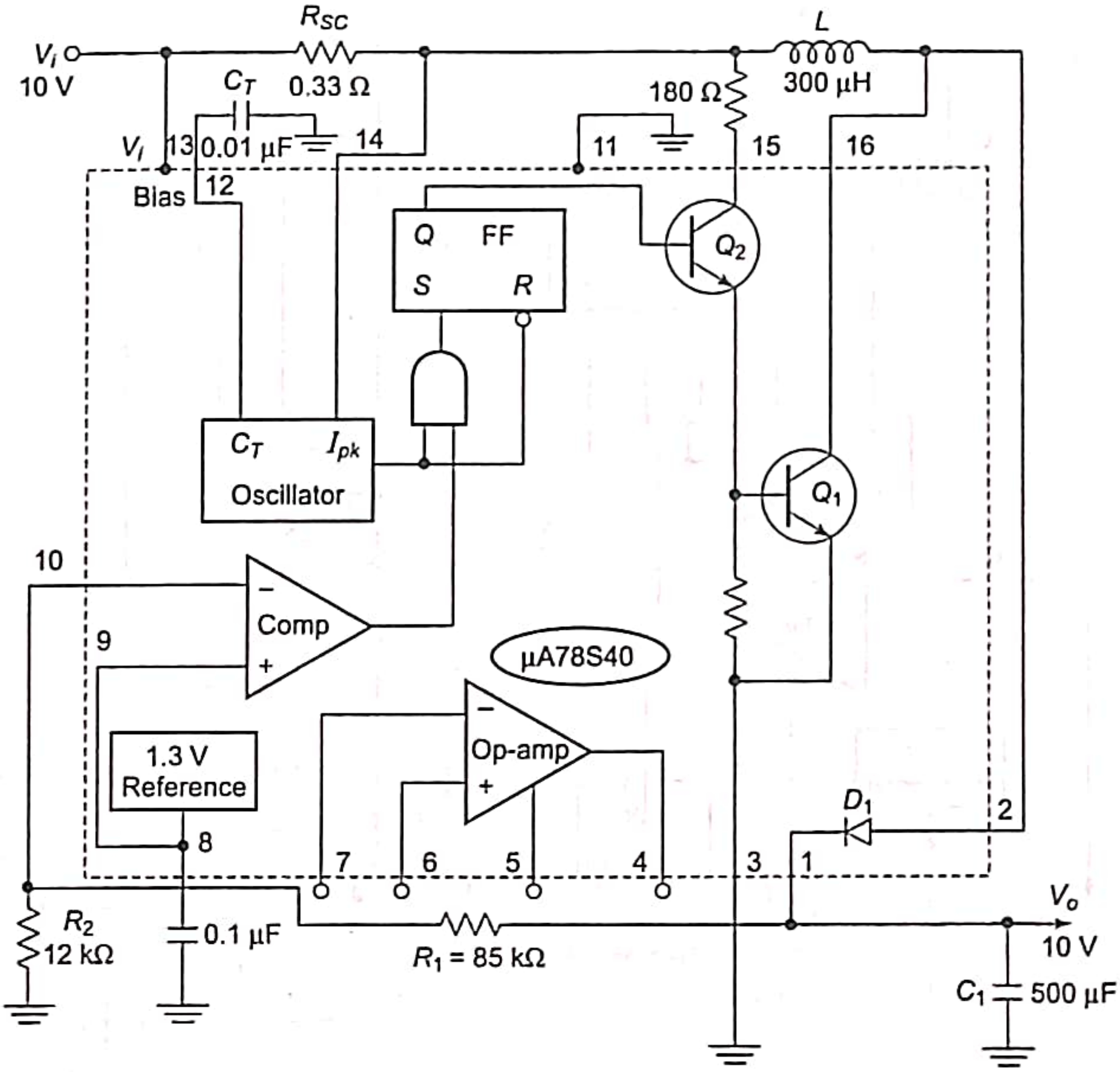


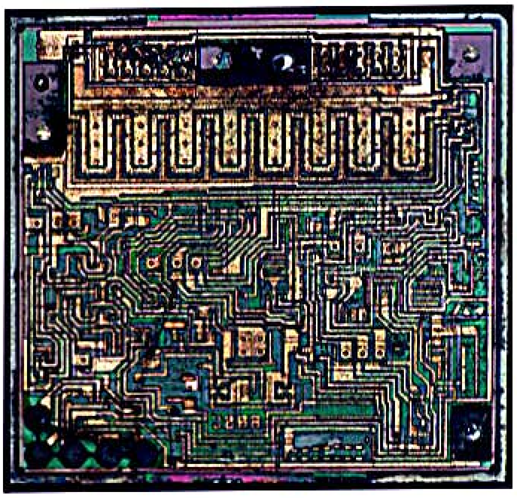
Fig. 5.61 *Step-up switching regulator using µA78S40*

Advantages of Switched Mode Power Supply (SMPS)

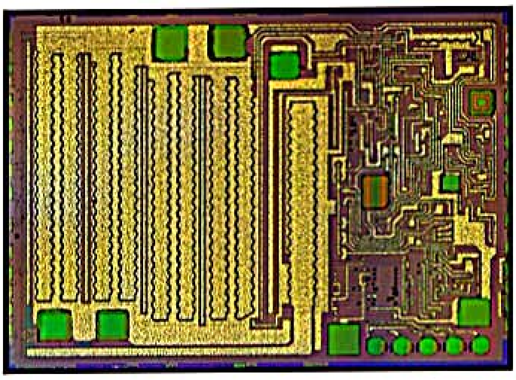
1. Efficiency is high because of less heat dissipation.
2. As the transformer size is very small, it will have a compact unit.
3. Protection against excessive output voltage by quick acting guard circuits.
4. Reduced harmonic feedback into the main supply.
5. Isolation from main supply without the need of large mains transformer.
6. Generation of low and medium voltage supplies is easy.
7. Switching supplies can change an unregulated input of 24V into a regulated dc output voltage of 1000V.
8. Though RF interference can be a problem in SMPS unless properly shielded, SMPS in TV sets is in synchronisation with the line frequency (15.625kHz) and thus switching effects are not visible on the screen.
9. SMPS are also used in personal computers, video projectors and measuring instruments.

Low-dropout regulator

A **low-dropout regulator (LDO regulator)** is a type of a DC linear voltage regulator circuit that can operate even when the supply voltage is very close to the output voltage.[1] The advantages of an LDO regulator over other DC-to-DC voltage regulators include: the absence of switching noise (in contrast to switching regulators); smaller device size (as neither large inductors nor transformers are needed); and greater design simplicity (usually consists of a reference, an amplifier, and a pass element). The disadvantage is that linear DC regulators must dissipate heat in order to operate.[2]



Die of the LM1117 low-dropout (LDO) linear voltage regulator.



Die of the LM2940L regulator

History

The adjustable low-dropout regulator debuted on April 12, 1977 in an *Electronic Design* article entitled "*Break Loose from Fixed IC Regulators*". The article was written by Robert Dobkin, an IC designer then working for National Semiconductor. Because of this, National Semiconductor claims the title of "*LDO inventor*".[3] Dobkin later left National Semiconductor in 1981 and founded Linear Technology where he was the chief technology officer.[4]

Components

The main components are a power FET and a differential amplifier (error amplifier). One input of the differential amplifier monitors the fraction of the output determined by the resistor ratio of R1 and R2. The second input to the differential amplifier is from a stable voltage reference (bandgap reference). If the output voltage rises too high relative to the reference voltage, the drive to the power FET changes to maintain a constant output voltage.

Regulation

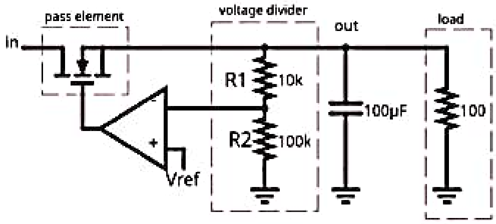


Fig. 1: Basic regulating loop of a low-dropout regulator feeds back its output voltage (Vout) through a voltage divider fed into the opamp's inverting input, so that Vout will be maintained at a fixed proportion above the reference voltage at the opamp's non-inverting input.

Low-dropout (LDO) regulators operate similarly to all linear voltage regulators. The main difference between LDO and non-LDO regulators is their schematic topology. Instead of an emitter follower topology, low-dropout regulators consist of an open collector or open drain topology, where the transistor may be easily driven into saturation with the voltages available to the regulator. This allows the voltage drop from the unregulated voltage to the regulated voltage to be as low as the saturation voltage across the transistor.[2]:Appendix A

In Fig. 1, the opamp's non-inverting input will have a voltage of:

$$V_{-} = \frac{R_2}{R_1 + R_2} V_{\text{out}} \ .$$

When this voltage is less than Vref, the opamp will turn on the pass element. The feedback loop keeps V_{-} about equal to V_{ref}. Solving for the output voltage yields:

$$V_{\text{out}} = \left(1 + \frac{R_1}{R_2}\right) V_{\text{ref}} \ .$$

If a bipolar transistor is used, as opposed to a field-effect transistor or JFET, significant additional power may be lost to control it, whereas non-LDO regulators take that power from voltage drop itself. For high voltages under very low Vin-Vout difference there will be significant power loss in the control circuit.[5]

Because the power control element is an inverter, another inverting amplifier is required to control it, increasing schematic complexity compared to simple linear regulator.

Power FETs may be preferable in order to reduce power consumption, yet this poses problems when the regulator is used for low input voltage, since FETs usually require 5 to 10 V to close completely. Power FETs may also increase the cost.

Efficiency and heat dissipation

The power dissipated in the pass element and internal circuitry (P_{LOSS}) of a typical LDO is calculated as follows:

$$P_{\text{LOSS}} = (V_{\text{IN}} - V_{\text{OUT}})\, I_{\text{OUT}} + V_{\text{IN}} I_{\text{Q}}$$

where I_{Q} is the quiescent current required by the LDO for its internal circuitry.

Therefore, one can calculate the efficiency as follows:

$$\eta = \frac{P_{\text{IN}} - P_{\text{LOSS}}}{P_{\text{IN}}} \quad \text{where} \quad P_{\text{IN}} = V_{\text{IN}} I_{\text{IN}}$$

However, when the LDO is in full operation (i.e., supplying current to the load) generally: $I_{\text{OUT}} \gg I_{\text{Q}}$. This allows us to reduce P_{LOSS} to the following:

$$P_{\text{LOSS}} = (V_{\text{IN}} - V_{\text{OUT}})\, I_{\text{OUT}}$$

which further reduces the efficiency equation to:

$$\eta = \frac{V_{\text{OUT}}}{V_{\text{IN}}}$$

It is important to keep thermal considerations in mind when using a low drop-out linear regulator. Having high current and/or a wide differential between input and output voltage could lead to large power dissipation. Additionally, efficiency will suffer as the differential widens. Depending on the package, excessive power dissipation could damage the LDO or cause it to go into thermal shutdown.

Quiescent current

Among other important characteristics of a linear regulator is the **quiescent current**, also known as ground current or supply current, which accounts for the difference, although small, between the input and output currents of the LDO, that is:

$$I_{\mathrm{Q}} = I_{\mathrm{IN}} - I_{\mathrm{OUT}}$$

Quiescent current is current drawn by the LDO in order to control its internal circuitry for proper operation. The series pass element, topologies, and ambient temperature are the primary contributors to quiescent current.[6]

Many applications do not require an LDO to be in full operation all of the time (i.e. supplying current to the load). In this idle state the LDO still draws a small amount of quiescent current in order to keep the internal circuitry ready in case a load is presented. When no current is being supplied to the load, P_{LOSS} can be found as follows:

$$P_{\mathrm{LOSS}} = V_{\mathrm{IN}} I_Q$$

Filtering

Torex XC6206 3.3 V LDO voltage regulator
in SOT23-3 package

In addition to regulating voltage, LDOs can also be used as filters. This is especially useful when a system is using switchers, which introduce a ripple in the output voltage occurring at the switching frequency. Left alone, this ripple has the potential to adversely affect the performance of oscillators,[7] data converters,[8] and RF systems[9] being powered by the switcher. However, any power source, not just switchers, can contain AC elements that may be undesirable for design.

Two specifications that should be considered when using an LDO as a filter are power supply rejection ratio (PSRR) and output noise.

Specifications

An LDO is characterized by its drop-out voltage, quiescent current, load regulation, line regulation, maximum current (which is decided by the size of the pass transistor), speed (how fast it can respond as the load varies), voltage variations in the output because of sudden transients in the load current, output capacitor and its equivalent series resistance.[10] Speed is indicated by the rise time of the current at the output as it varies from 0 mA load current (no load) to the maximum load current. This is basically decided by the bandwidth of the error amplifier. It is also expected from an LDO to provide a quiet and stable output in all circumstances (example of possible perturbation could be: sudden change of the input voltage or output current). Stability analysis put in place some performance metrics to get such a behaviour and involve placing poles and zeros appropriately. Most of the time, there is a dominant pole that arise at low frequencies while other poles and zeros are pushed at high frequencies.

Power supply rejection ratio

PSRR refers to the LDO's ability to reject ripple it sees at its input.[11] As part of its regulation, the error amplifier and bandgap attenuate any spikes in the input voltage that deviate from the internal reference to which it is compared.[12] In an ideal LDO, the output voltage would be solely composed of the DC frequency. However, the error amplifier is limited in its ability to gain small spikes at high frequencies. PSRR is expressed as follows:[11]

$$\mathbf{PSRR} = \frac{\Delta V_{\mathrm{IN}}^2}{\Delta V_{\mathrm{OUT}}^2} = 10 \log\left(\frac{\Delta V_{\mathrm{IN}}^2}{\Delta V_{\mathrm{OUT}}^2}\right) \mathbf{dB}$$

As an example, an LDO that has a PSRR of 55 dB at 1 MHz attenuates a 1 mV input ripple at this frequency to just 1.78 μV at the output. A 6 dB increase in PSRR roughly equates to an increase in attenuation by a factor of 2.

Most LDOs have relatively high PSRR at lower frequencies (10 Hz – 1 kHz). However, a Performance LDO is distinguished in having high PSRR over a broad frequency spectrum (10 Hz – 5 MHz). Having high PSRR over a wide band allows the LDO to reject high-frequency noise like that arising from a switcher. Similar to other specifications, PSRR fluctuates over frequency, temperature, current, output voltage, and the voltage differential.

Output noise

The noise from the LDO itself must also be considered in filter design. Like other electronic devices, LDOs are affected by thermal noise, bipolar shot noise, and flicker noise.[9] Each of these phenomena contribute noise to the output voltage, mostly concentrated over the lower end of the frequency spectrum. In order to properly filter AC frequencies, an LDO must both reject ripple at the input while introducing minimal noise at the output. Efforts to attenuate ripple from the input voltage could be in vain if a noisy LDO just adds that noise back again at the output.

Load regulation

Load regulation is a measure of the circuit's ability to maintain the specified output voltage under varying load conditions. Load regulation is defined as:

$$\textbf{Load Regulation} = \frac{\Delta V_{\mathrm{OUT}}}{\Delta I_{\mathrm{OUT}}}$$

The worst case of the output voltage variations occurs as the load current transitions from zero to its maximum rated value or vice versa.[6]

Line regulation

Line regulation is a measure of the circuit's ability to maintain the specified output voltage with varying input voltage. Line regulation is defined as:

$$\textbf{Line regulation} = \frac{\Delta V_{\mathrm{OUT}}}{\Delta V_{\mathrm{IN}}}$$

Like load regulation, line regulation is a steady state parameter—all frequency components are neglected. Increasing DC open-loop current gain improves the line regulation.[6]

Transient response

The transient response is the maximum allowable output voltage variation for a load current step change. The transient response is a function of the output capacitor value (C_{OUT}), the equivalent series resistance (ESR) of the output capacitor, the bypass capacitor (C_{BYP}) that is usually added to the output capacitor to improve the load transient response, and the maximum load-current ($I_{\mathrm{OUT,MAX}}$). The maximum transient voltage variation is defined as follows:

$$\Delta V_{\mathrm{TR,MAX}} = \frac{I_{\mathrm{OUT,MAX}}}{C_{\mathrm{OUT}} + C_{\mathrm{BYP}}} \Delta t_1 + \Delta V_{\mathrm{ESR}}$$ [6]

Where Δt_1 corresponds to the closed-loop bandwidth of an LDO regulator. ΔV_{ESR} is the voltage variation resulting from the presence of the ESR (R_{ESR}) of the output capacitor. The application determines how low this value should be.

5.18 ISOLATION AMPLIFIER

The isolation amplifier is an amplifier in which, there is no physical contact between the input and output sections. These amplifiers are used in applications requiring very large common-mode voltage difference between the input and output sections. Several thousand Volts can exist between the two sections while using such isolation amplifiers. Signal transmission (across the isolation barrier) is accomplished through an optical coupler, which acts as a 1:1 current translator. Current at the input of the device is replicated on the output side of the coupler. The isolated output current (I_o) is forced to flow through RF by the summing node action of the output op-amp.

They find use in medical instrumentation applications, where the patient must be isolated and protected from leakage currents. Most of the isolation amplifiers are hybrid ICs and consist of an input amplifier, a GaAs Light-Emitting Diode (LED), a silicon photo diode and an output amplifier. The input signal modulates the light output of the LED. The light emitted by the LED is detected by the photo-diode and converted to an electrical signal.

One of the main requirements of an isolation amplifier is its linearity of input-output characteristics. The inherent nonlinear current-input to light-output characteristics of the LED is the cause of the nonlinearity of isolation amplifier. Various opto-electronic devices and their circuits are being devised to obtain high degree of linearity in optically coupled analog signal transmission systems. Typical isolation amplifier chips are ISO100, 3450-3455 series from Burr-Brown, and AD293 and AD294 from Analog Devices.

5.18.1 IC ISO100 Isolation Amplifier

The IC ISO100 is an optically coupled isolation amplifier. It offers high accuracy, linearity and good stability against time and temperature. This is achieved by coupling light from an LED back to the input with negative feedback as well as forwarding the output. It needs careful directional matching of optical components.

The main features of the isolation amplifier ISO100 are as follows:

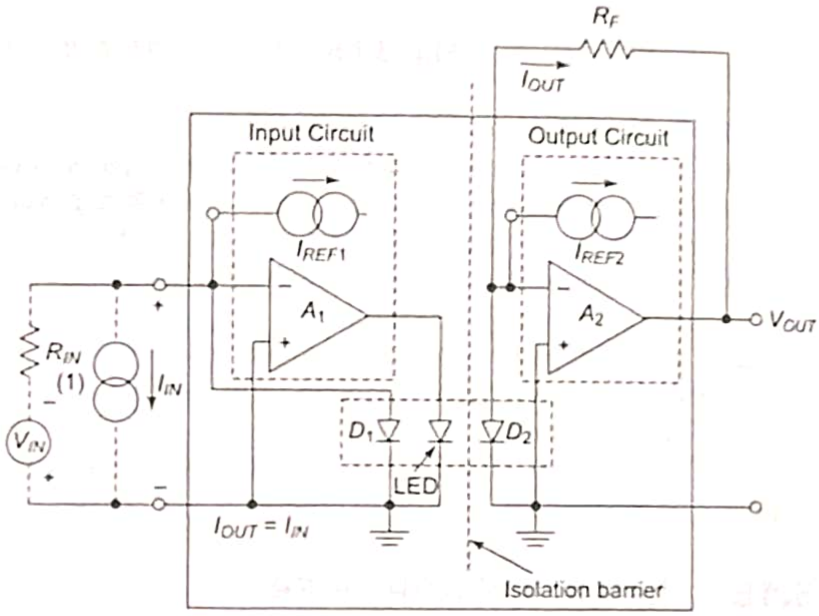
Supply voltage	:	±18 V
Isolation voltage	:	ac peak or dc 750 V

Input current	:	±1 mA
Storage temperature range	:	– 40°C to 100°C
Soldering lead temperature	:	+ 300°C
Output short-circuit duration	:	Continuous to ground
Leakage	:	0.3 μA (max) at 240 V/60 Hz
Bandwidth	:	60 kHz
18-pin DIP package		

Operation of ISO100 The ISO100 has several modes of operation, namely, unipolar or bipolar, voltage or current input, and inverting or non-inverting. Figure 5.86(a) and (b) show the simplified block diagram and pin diagram of ISO100. It can be considered as a unity gain current amplifier. Its output flows into a current-to-voltage converter. Hence, it is inherently a current input device. An isolation barrier exists between the input and output circuits of the IC. Therefore, signal transmission across the barrier is accomplished through an optical coupler. The current at the input of the device is replicated as I_{OUT} at the output side of the device. This output current I_{OUT} is forced to flow through the feedback resistor R_F. This is achieved by the action of the summing node of the output op-amp. The non-inverting input of the IC ISO100 is connected to the inverting input of the op-amp A_1. The two stages of inversion for the input signal compensate each other, thus producing a true output signal.

The transfer function of the IC ISO100 is given by $\frac{V_{OUT}}{V_{IN}} = \frac{R_F}{1 + A_{GE}}$ where A_{GE} is the gain error. It is defined as the deviation of the ratio, $\frac{I_{IN}}{I_{OUT}}$ from unity. It can thus be considered as the *coupling error*.

The optical coupler shown in Fig. 5.86(a) consists of a matched pair of photo diodes and an LED. This makes the signal transmission possible from input to output.



Note : (1) A negative input signal is required in the unipolar mode

Fig. 5.86 *(a) Block diagram of ISO100 Iso-amplifier*

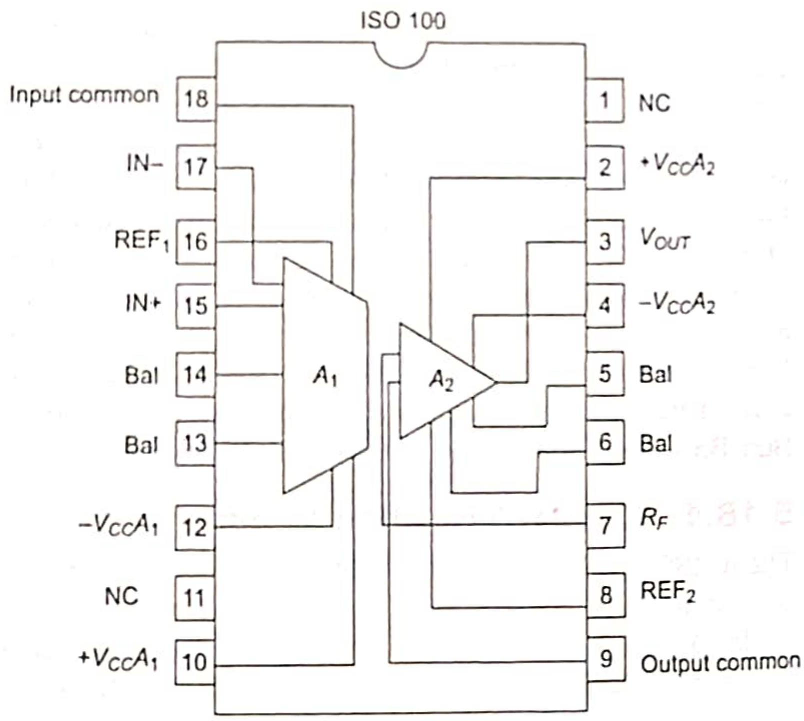


Fig. 5.86 *(b) Pin configuration of ISO100 Iso-amplifier*

The diode D_1 forms negative feedback path in combination with LED. The diode D_2 receives the optical signal across the isolation barrier.

Application Information

The advantages of ISO100 are its small size, low offset, low drift, wide bandwidth, ultra low leakage and low leakage. The precautions to be observed while designing the ISO are as follows:

(i) Input common and IN lines at Pins 17 and 18 must be grounded through separate lines. This is to avoid any large *dc* current flowing in input common feedback to signal input.

(ii) Shielded or twisted pair cables are preferred at the input for long lines.

(iii) The external capacitance across the isolation barrier is to be minimised.

(iv) Leakage and arcing are to be avoided. This can be done by spacing the isolation barrier, external components and conductor patterns far apart.

(v) The current I_1 should be greater than 200A to keep internal LED ON.

(vi) The maximum output voltage swing is determined by I_1 and R_F

$$V_{swing} = I_{1(MAX)} \times R_F$$

5.17 OPTO-COUPLERS

An opto-coupler is a solid state component in which the light emitter, the light path and the light detector are all enclosed within the component and cannot be changed from outside. As the opto-coupler provides electrical isolation between two circuits, it is also called an *opto-isolator*. An opto-isolator allows signal transfer without the use of coupling wires, capacitors or transformers. It can couple digital (logic 1 or 0) or analog (continuously variable) signals.

The schematic representation of an opto-coupler is shown in Fig. 5.80. An opto-coupler, also called an opto-electronic coupler, generally consists of an infrared LED and a photo-detector such as PIN photo diode for fast switching, photo transistor Darlington pair, or photo-SCR combined in a single package. Opto-isolators transduce input voltage to a proportional light intensity by using LEDs. The light is transduced back to output voltage using light sensitive devices. GaAs LEDs are used to provide spectral matching with the silicon sensors.

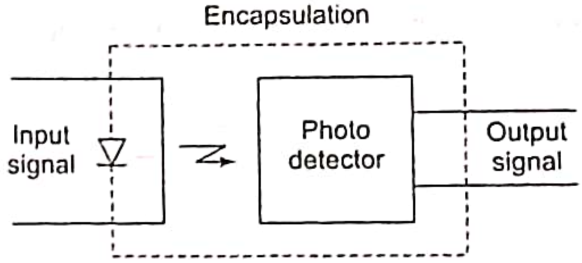


Fig. 5.80 *Schematic representation of opto-coupler*

The wavelength response of each device is made to be as identical as possible to permit the highest possible measure of coupling. There is a transparent insulating cap between each set of elements embedded in the structure to permit efficient passage of light. They are designed with very small response time such that they can be used to transmit data in MHz range of frequencies.

The rigid structure of this package permits one-way transfer of electrical signal from the LED to the photo-detector, without any electrical connection between the input and output circuitry. The extent of isolation between the input and output depends on the kind of material in the light path and on the distance between the light emitter and light detector. A significant advantage of the opto-isolator is its high isolation resistance, of the order of $10^{11}\ \Omega$, with the isolation voltages up to 2500 V achievable between the input and output signals. This feature allows it to be used as an interface between high voltage and low voltage systems. Application of this device includes the interfacing of different types of logic circuits and their use in *level-and-position-sensing* circuits.

In an opto-isolator, the power dissipations of LED and phototransistor are almost equal and I_{CEO} is measured in nano-amperes. The relative output current is almost constant when the case temperature varies from 25 to 75°C. The V_{CE} voltage of the photo transistor affects the resulting collector current only very slightly. The switching time of opto-isolator decreases with increased current, and for some devices it is exactly the reverse.

The schematic diagrams of the opto-couplers using a photodiode, photo Darlington pair and photo-SCR are illustrated in Fig. 5.81.

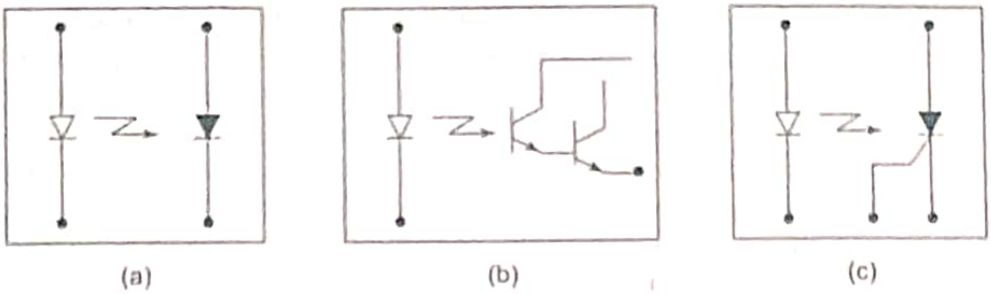


Fig. 5.81 *Schematic representation of opto-couplers using (a) Photodiode, (b) Photo-Darlington pair, and (c) Photo-SCR*

Figure 5.82 shows the basic circuit arrangement for the use of an opto-coupler. The source v_i and series resistor R_i determine the forward current I_i through the LED. Therefore, the light emitted from the LED depends on the signal v_i. This light incident on the photo diode generates a reverse current in the output circuit through the resistor R_o. Therefore the drop across the resistor R_o is proportional to the input voltage, and it varies proportionately with change in input signal source.

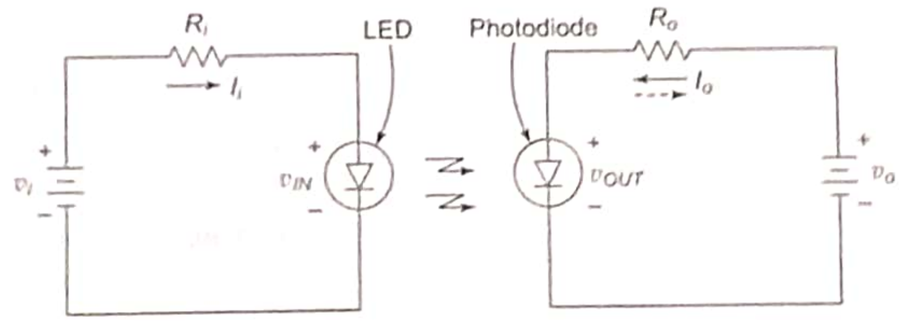


Fig. 5.82 *Basic circuit arrangement for the use of opto-coupler*

5.17.1 Characteristics of Opto-Couplers

The important characteristics of an opto-coupler are given below:

Collector-emitter voltage This is the maximum voltage that can be applied across collector and emitter of the receiving phototransistor (*no light* when it is turned OFF) before it may break-down.

Creepage distance This is physically a measure of how far a spark would have to travel around the outer side of the package to get from one side to the other. If the package has contaminants on it such as solder flux, or dampness, then a lower-resistance path will be created for noise signals to travel along the surface.

Forward current This is the current passing through the transmitting LED. Typically an opto-isolator will require about 5 mA to turn the output transistor ON.

Forward voltage This is the voltage that is dropped across the LED when it is turned ON by the input signal v_i. The LEDs drop typically 1 to 2 Volts, whereas the silicon diodes drop about 0.7 V.

Collector dark current This is the current that can flow through the output photo transistor when it is turned OFF.

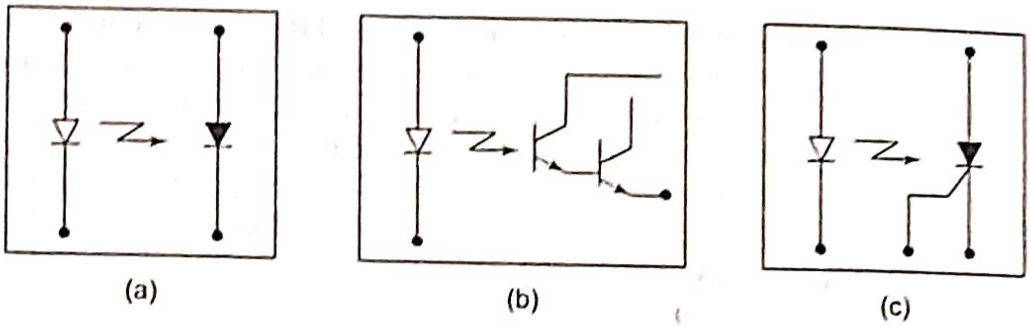


Fig. 5.81 *Schematic representation of opto-couplers using (a) Photodiode, (b) Photo-Darlington pair, and (c) Photo-SCR*

Figure 5.82 shows the basic circuit arrangement for the use of an opto-coupler. The source v_i and series resistor R_i determine the forward current I_i through the LED. Therefore, the light emitted from the LED depends on the signal v_i. This light incident on the photo diode generates a reverse current in the output circuit through the resistor R_o. Therefore the drop across the resistor R_o is proportional to the input voltage, and it varies proportionately with change in input signal source.

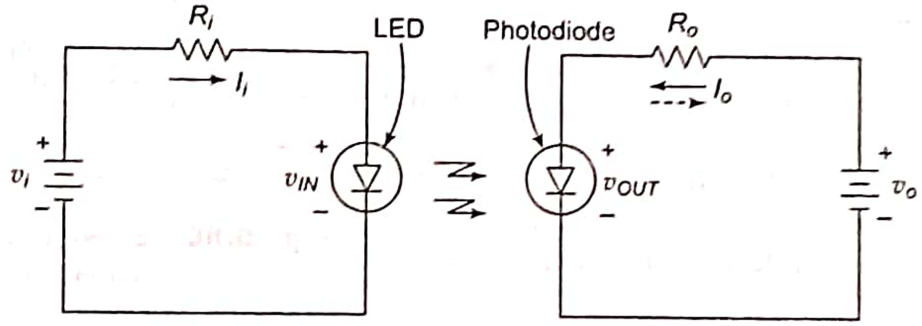


Fig. 5.82 *Basic circuit arrangement for the use of opto-coupler*

5.17.1 Characteristics of Opto-Couplers

The important characteristics of an opto-coupler are given below:

Collector-emitter voltage This is the maximum voltage that can be applied across collector and emitter of the receiving phototransistor (*no light* when it is turned OFF) before it may break-down.

Creepage distance This is physically a measure of how far a spark would have to travel around the outer side of the package to get from one side to the other. If the package has contaminants on it such as solder flux, or dampness, then a lower-resistance path will be created for noise signals to travel along the surface.

Forward current This is the current passing through the transmitting LED. Typically an opto-isolator will require about 5 mA to turn the output transistor ON.

Forward voltage This is the voltage that is dropped across the LED when it is turned ON by the input signal v_i. The LEDs drop typically 1 to 2 Volts, whereas the silicon diodes drop about 0.7 V.

Collector dark current This is the current that can flow through the output photo transistor when it is turned OFF.

Collector-emitter saturation voltage This is the voltage available between the collector and emitter when the output transistor is fully turned ON (saturated).

Isolation resistance This is the resistance from a pin on the input side to a pin on the output side. It must be very high to offer good isolation between the input and output sides of opto-coupler.

Response time The rise and fall times are the time durations that the output voltage requires to rise from zero to maximum and to fall from maximum to zero respectively. The rise time is dependent on the load resistor. Therefore, this value is always quoted for a fixed load resistor.

When the opto-couplers are used with pulse-width modulated (PWM) signals for applications such as speed control of motors, the response time of the opto-coupler, and the rise and fall time of the signals are the criteria for the selection of the opto-couplers.

Cut-off frequency This is effectively the highest frequency of square-wave that can be applied to the opto-isolator. It is actually the frequency at which the output voltage becomes half the maximum amplitude. This determines the operating bandwidth of the opto-coupler and it is related to the rise and fall times.

Current transfer ratio The input current is the forward current of LED that generates the emission of light, which is detected by photo diode, photo Darlington or photo-SCR to produce the output current. The ratio of output current I_o to the input current I_i is called the *current transfer ratio* (CTR). The different CTR ratings of the opto-couplers are shown in Table 5.3. This factor also depends on the proximity between LED and photo transistor and their efficiencies.

Table 5.3 *Current transfer ratios of opto-couplers*

Device	*Current Transfer Ratio*
LED-Photo diode	0.01 – 0.03
LED-Transistor	0. – 1
LED-Darlington	1 – 5
IRED-Transistor	10 – 15
GaAs IRED-Transistor	100 – 2000

5.17.2 Opto-Coupler ICs

The opto-coupler ICs are available in a variety of packages, with the most common being the six-pin mini DIP.

TLP112 opto-coupler The Type TLP112 from Toshiba is a mini-Flat coupler, which is suitable for surface mount assembly. It consists of a GaAlAs light emitting diode, optically coupled to a high speed detector of a single-chip photodiode-transistor assembly.

The important features are:

(i) An isolation voltage of 2500V rms (min) can be achieved
(ii) Switching speeds of $t_{pHL} = 0.8\ \mu s$, $t_{pLH} = 2\ \mu s(max)$ is possible
(iii) It is TTL compatible

The pin configuration and the input-output terminals are shown in Fig. 5.83(a) and (b) respectively.

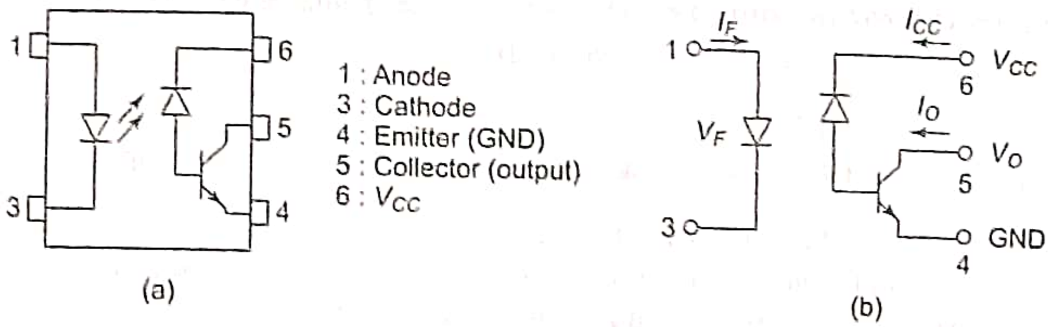


Fig. 5.83 *TLP112 Opto-coupler: (a) Pin configuration (b) Input-output terminals*

MOC3009-MOC3012 series opto-couplers The MOC3009 series opto-couplers consist of Gallium-Arsenide-Diode Infrared source and an optically coupled silicon Triac driver. It can provide 250 V driver output, with an electrical isolation of 7500 V peak. The IC is available in standard 6-pin Plastic DIP as shown in Fig. 5.84. It is directly interchangeable with MOC3009, MOC3010, MOC3011 and MOC3012.

The typical applications are Solenoid/Valve controls, Lamp ballasts, Interfacing microprocessors to 115 V ac peripherals, Motor controls and Incandescent lamp dimmers.

The pin configuration and the logic diagram of the MOC3009-MOC3012 Series ICs are shown in Figs. 5.84(a) and (b) respectively.

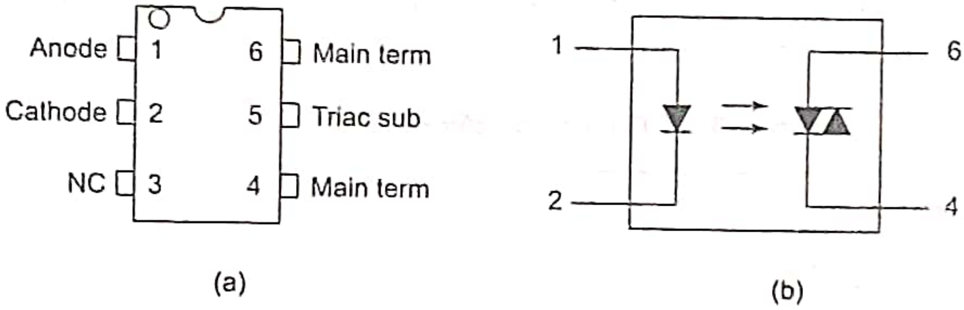


Fig. 5.84 *MOC3009-MOC3012 Series Opto-couplers: (a) Pin configuration (b) Logic diagram*

The typical application circuits for resistive load and inductive load using MOC3009 are shown in Figs. 5.85(a) and (b) respectively.

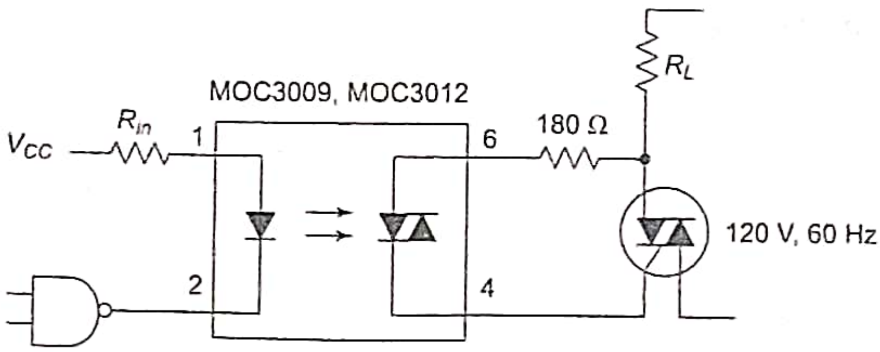


Fig. 5.85 *(a) Use of MOC3009 for resistive load*

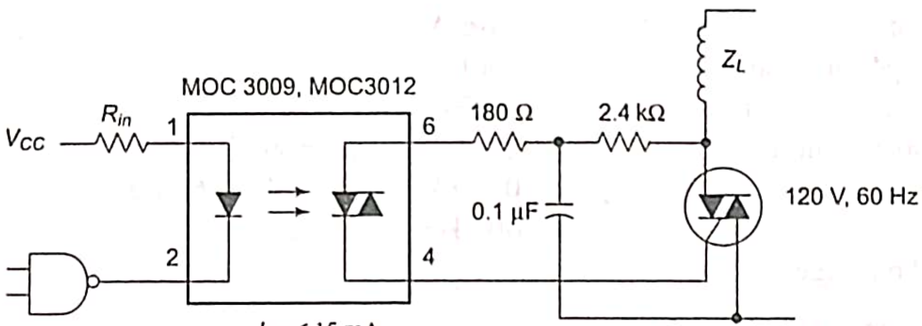


Fig. 5.85 *(b) Use of MOC3009 for inductive load*

The main advantages of opto-couplers are:

(i) The electrical isolation achieved between input and output is of the order of several Mega Ohms. This characteristic makes the device useful in high voltage applications where the voltages between the input and output are several thousand Volts.

(ii) The response time of opto-couplers is so small that they can be used for data applications involving Mega Hertz range of pulse frequencies.

(iii) They are capable of wideband signal transmission.

(iv) Easy interfacing with logic devices is possible.

(v) They are compact and portable.

(vi) They are more efficient than isolation transformers and relays.

(vii) The problems such as noise, transients and contact bounce etc. are completely eliminated.

5.19 FIBRE-OPTIC INTEGRATED CIRCUITS

[AU, June'13, 8 marks[

Opto-electronics and opto-electronic devices have undergone tremendous progress in recent years. Optics provides the advantages of large bandwidth, parallelism and reconfigurable characteristics. Electronic devices provide active components in information handling systems. Thus opto-electronic integrated circuits involve the integration of electronic and optical components, and optical interconnects. The fibre forms an optical interconnect medium. Such an interconnect medium provides a large bandwidth, high-speed data transmission, and immunity against mutual interference and cross-talk. They are unaffected by capacitive loading effects also. It results in size reduction, reduced power of system and increased fan-out capability.

Figure 5.87(a) shows the block diagram of an Opto-Electronic Integrated Circuit (OEIC). It combines the functions of optical detection, electronic functions such as switching and amplification and light transmission.

Figure 5.87(b) shows the essential elements and major components of a two-link fibre-optic communication facility. The transmitting end consists of an optical transmitter such as an LED or a laser diode source with driver,

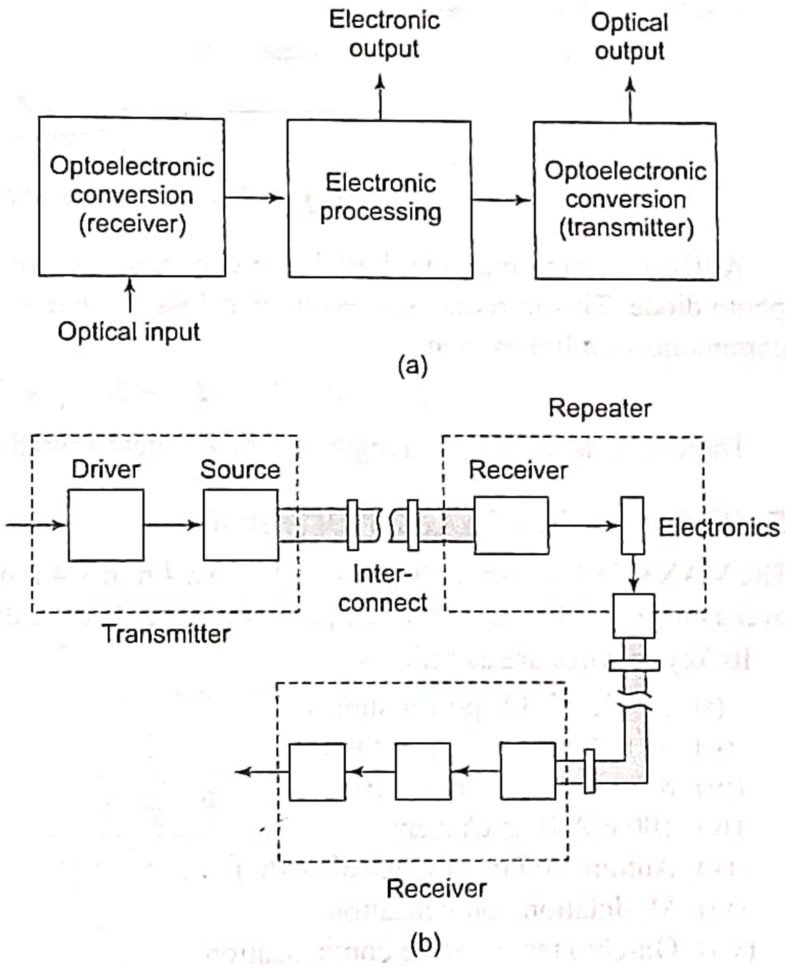


Fig. 5.87 *(a) Block diagram of Opto-Electronic-Integrated Circuit (OEIC) (b) Major components of a fibre-optic communication facility*

coupled to the fibre. The LED or laser diode turns ON and OFF according to the bit stream to be sent. The repeaters receive the signal from first link. The signal is amplified, reshaped, retimed and then retransmitted to the next link. The repeater consists of an avalanche photo-diode, or PIN diode, which senses the attenuated and dispersed train of light pulses. Then the pulses are processed before being sent to the second link. The receiver at the receiving end converts the incoming light pulses to electrical pulses. The electrical pulses are regenerated and distributed.

In a fibre-optic link, the total loss suffered by the fibre and all the connectors and splices along the length of the fibre optic communication path must not exceed a certain minimum value. The minimum acceptable received optical power P_r is dependent on the type of detector and allowable error rate. Assume a total light flux of power P_t is emitted from the optical source as shown in Fig. 5.88. Only part of the light is sent through the fibre, due to the inefficiency of coupling the diode to fibre end. This results in a part loss of L_{pt}. Similarly, the connector on the other end of the fibre introduces an insertion loss of L_C. The light passing through the fibre encounters losses due to absorption and leakage, which can be considered as ZL_f where L_f is the rate of loss and Z is the length of the fibre. The splice loss of L_S for each of N_S splices is also introduced.

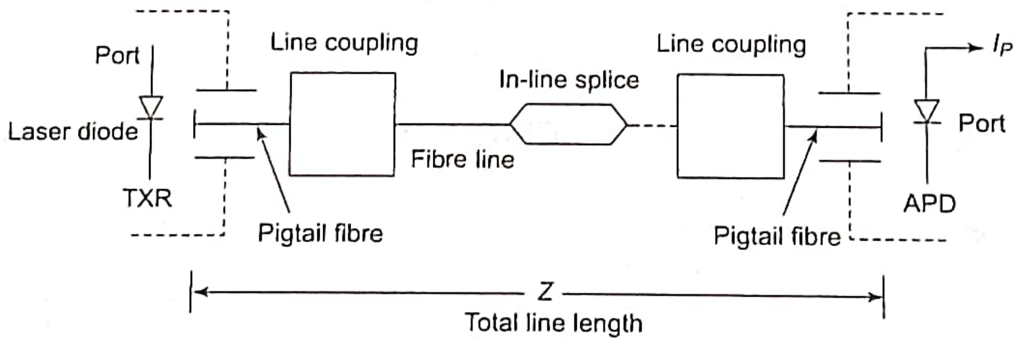


Fig. 5.88 *Elements in the fibre-optic link*

At the receiving end, the digital line connector shown feeds the light from the fibre into the avalanche photo diode. This introduces a second part loss L_{pr} and a second connector loss L_C. The *loss budget* for the communication link is then,

$$P_t - P_r = M + L_{pt} + L_{pr} + N_C L_C + N_S L_S + ZL_f$$

Therefore, the maximum length of link Z is determined and limited by the losses.